NexusEngine Guide

Core architecture, composition, integration, and migration reference

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Start Here

NexusEngine is the reusable Core beneath games, simulations, editors, and agents. It supplies deterministic runtime primitives and universal semantic capabilities. It does not supply every useful feature.

The shortest correct rule is:

```
NexusEngine                universal Core behavior
NexusEngine-Kits           reusable optional behavior
Experiments and games      authored product behavior
Hosts and adapters         platform and vendor implementations
```

Who This Guide Is For

- Application developers who need to assemble a runtime.
- Kit authors who need to choose an owner and dependency contract.
- Host authors who connect browsers, renderers, storage, networks, or SDKs.
- Agents that inspect and plan compositions through MCP.
- Maintainers migrating code to the `0.0.4` semantic Domain model.

Reading Paths

Build something: read First Runtime, Domains And Atomic Kits, then Composition And Recipes.

Integrate a platform: read Registries And Security, then Hosts, Providers, And Adapters.

Use an agent: read Composition And Recipes, then MCP Agent Workflow.

Maintain Core: read State, Lifecycle, And Idempotence, Testing And Proof, and Migrating To 0.0.4.

The Non-Negotiable Boundary

A Core atom has one responsibility, is product-neutral, installs idempotently, owns explicit state, supports snapshot and reset, isolates nondeterminism, and has direct proof. If any requirement is unknown, the behavior remains outside Core until proved.

This fail-closed rule keeps Core small enough to understand and flexible enough to compose.

Sources Of Truth

1. Domain manifest v2 records and the source they reference.
2. Generated catalog, package exports, ownership ledger, and registry hash.
3. This modular guide.
4. Migration and extraction ledgers.

Historical plans, generated run packets, and the retired ProtoKit repository are evidence. They do not override the active manifest catalog.

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The NexusEngine Mental Model

NexusEngine separates meaning from implementation. A Domain names a semantic owner. An atomic Kit installs one capability. A provider performs work behind a contract. An adapter translates between boundaries. A recipe selects and configures atoms without becoming a new runtime owner.

The Vocabulary

Term	Meaning	Example
Engine	Runtime container for state, systems, events, resources, and installed Kits	`createEngine()`
Domain	Semantic ownership boundary	`n:object`, `n:simulation:motion`
Atomic Kit	One installable, idempotent responsibility	Object registry or placement
Provider	Replaceable implementation of a Domain contract	Physics backend or speech synthesis
Adapter	Translation across two public contracts	Motion plan to physics drive request
Recipe	Data selecting Domains, Kits, and settings	Object plus Placement composition
Registry	Non-executable metadata describing exact sources	Package, version, commit, integrity
Host	Application-owned lifecycle and platform boundary	Editor, browser application, server

State Has One Owner

Every durable concept has one state owner. Other Domains refer to that state through descriptors, identifiers, commands, events, or public APIs. They do not keep shadow copies that can drift.

For example, Object owns object identity and intrinsic geometry meaning. Placement owns where an object is placed. A renderer may project both into a scene, but renderer objects are not Core state.

Composition Is Additive

Atoms declare `requires` and `provides` tokens. Composition resolves those edges into an ordered plan. Local idempotence makes repeated installation safe; the composition receipt makes the complete accepted plan exactly-once across process restarts.

Core Does Not Mean Default

A capability may be universal and still opt-in. Core means the behavior satisfies the ownership contract. It does not mean every engine instance installs it automatically.

Data Is Not A Runtime Owner

Authored presets, balance values, levels, quests, materials, and complete game loops remain recipe or product data. Core can define the schema and deterministic evaluator while the product owns the authored values.

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Build A First Runtime

Install the package, import only the semantic surfaces you need, construct the Engine, then advance it with explicit time.

Minimal Engine

The runnable source for this example is `examples/guide/basic-engine.mjs`.

```
import { createEngine } from "nexusengine";

const engine = createEngine({ kits: [] });
engine.tick(1 / 60);
console.log(engine.clock.frame);
```

The root API is intentionally small. Domain factories and atoms are imported from generated semantic subpaths.

Add Object And Placement

The runnable source is `examples/guide/object-placement.mjs`.

```
import { createEngine } from "nexusengine";
import { createObjectRegistryKit } from "nexusengine/domains/object/registry";
import { createObjectPlacementKit } from "nexusengine/domains/object/placement";

const engine = createEngine({
  kits: [createObjectRegistryKit(), createObjectPlacementKit()]
});
```

Placement requires the Object descriptor contract. Composition can resolve that provider automatically; direct construction requires you to install both.

Inspect Installed Surfaces

Domain Service Kits register semantic paths and named APIs. Use the engine's addressability APIs instead of reaching into private state.

```
engine.n.path("n:object");
engine.n.path("n:object:placement");
engine.n.ownerOf("n:object");
engine.n.paths();
engine.n.apis();
```

Keep Time Explicit

Pass elapsed time into `tick`. Do not read wall-clock time inside deterministic Core systems. A host may sample a clock, but it converts that sample into explicit runtime input before Core processes it.

Save And Restore

Each stateful atom exposes its manifest-declared snapshot and reset behavior. Applications should persist the complete host composition receipt beside application state so the accepted runtime can be reconstructed after restart.

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Domains And Atomic Kits

The semantic path tells you where a responsibility belongs. The manifest tells tools and humans what that owner may and may not do.

Domain Manifest V2

Every top-level semantic Domain declares:

- stable identity, path, immediate parent, label, version status
- one responsibility and explicit owned meaning
- forbidden responsibilities
- owned state and schema
- inputs, systems, outputs, and lifecycle
- required and optional dependencies
- public atomic Kits, providers, and adapters
- settings schema and proof references

Generation fails when a public atom lacks source or proof. Compliance is never inferred from a folder name.

Semantic Families

Family	Main paths
Runtime	`n:runtime`, realtime, data, transaction, persistence, sequence, startup
Composition	`n:composition`, `n:mcp`, `n:policy`
World	`n:spatial`, scale, `n:object`, `n:world`, navigation, generation, terrain, water, scene, weather, asset
Simulation	`n:simulation`, physics, recovery, motion, economy, operations, hazard, pursuit, progression, `n:compute`, model
Actors	`n:actor`, creature, character, player, `n:agent`
Interaction	`n:interaction`, input
Visual semantics	`n:material`, `n:lighting`, `n:camera`
Rendering	`n:render`: resources, contracts, pipelines, batches, provider lifecycle
Presentation	output, animation, audio, UI, speech, capture, sky
Infrastructure	`n:network`, `n:diagnostics`, contract-only `n:host`

The generated Domain Index appendix contains every active path.

Canonical Physics Foundation

`n:physics` owns the canonical Physics identity. Its `n:physics:contracts` subdomain owns the portable provider, state, command, event, and query boundaries. Each boundary is an independently installable atomic Kit with one generated public subpath.

Contract records are strict and JSON-portable. They reject unknown top-level fields, non-finite numbers, functions, platform handles, cycles, and non-plain objects, then normalize accepted records into deterministic key order. The contract package does not claim body ownership, collision detection, solving, or provider execution; those capabilities promote only with their own proof.

The existing `n:simulation:physics` surface remains separate during the staged `0.0.5` implementation. Its eventual cutover is not implied by the presence of the new contract package.

`n:physics:lifecycle` is the next canonical subdomain. It separates six reasons to change:

- Installation owns aggregate phase and provider identity.
- Startup owns readiness requests and provider receipts.
- Step owns strict step ordering and provider-neutral frame receipts.
- Shutdown owns stop requests and completion receipts.
- Reset restores the composed lifecycle through public APIs.
- Snapshot captures and restores those APIs atomically.

The lifecycle does not call a solver or install a concrete provider. A later provider package consumes its portable requests and supplies the reviewed receipts. Failed coordinated mutations restore every touched lifecycle state.

`n:physics:material` separates six physical material responsibilities:

- Friction normalizes static, dynamic, rolling, spinning, and anisotropic data.
- Restitution normalizes bounce coefficient and activation threshold.
- Density uses positive kilograms-per-cubic-meter records.
- Surface owns physical classification and portable tags only.
- Combine Policy resolves two materials symmetrically without solving contact.
- Physics Material owns immutable IDs and exact-once registry mutations.

The material registry depends on the five specialist capabilities through public tokens. A material record contains no shader, texture, sound, particle, collider, solver impulse, or provider handle. Those concerns remain separate even when a product maps one physical surface to visual and audio effects.

`n:physics:world` separates seven solver-facing environment responsibilities:

- World Settings normalizes coordinate handedness, length units, up axis, optional bounds, and out-of-bounds policy.
- Gravity Field owns uniform and point-gravity acceleration records.
- Force Field owns non-gravity force or acceleration records.
- Wind Field owns uniform, deterministic gust, and corridor flow velocity.
- Time Scale combines explicit Physics-only scale records deterministically.
- Simulation Region resolves physical simulate, sleep, or disable behavior.
- Physics World owns immutable world IDs and references the six capabilities.

World sampling is read-only and provider-neutral. It returns portable acceleration, force, wind velocity, region, bounds, and scaled-delta records. Authored weather stays under `n:world:weather`; atmosphere corridors and game routes stay with World or the product; Runtime owns clocks and schedules; later Body and Solver packages consume the Physics output.

Restored Behavior Shape

Twenty-six historical source modules become 27 atomic behaviors because the old World Physics module mixed contact and recovery ownership. Those behaviors are now separate World Contact and Soft Respawn atoms. None of the historical root aliases return.

The restoration ledger validates each historical source checksum, every old export, its corrected semantic owner, and its proof. A behavior with a game-adjacent name can remain Core only when its manifest describes a universal primitive without authored rules, presets, rendering ownership, or hidden cross-Domain effects.

Atomicity Test

An atom should have one reason to change. If a module owns inventory, combat, rendering, and progression together, it is a composition or product feature, not an atom.

Splitting is semantic, not file-count driven. Several source files may implement one indivisible lifecycle. Conversely, one legacy file may contain several owners and must be separated.

Dependency Tokens

Use capability tokens for behavior dependencies and Domain paths for semantic presence. Requirements must describe what the atom consumes; providers list what they supply. Composition rejects missing providers, cycles, identity collisions, and changed content.

Public Imports

Only generated package exports are public. Production code does not import private files from sibling Domains. Cross-Domain collaboration goes through a public subpath or an explicit adapter.

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Authoring

`n:authoring` owns editable source for scripts, agents and editing applications. Its 19 kits install together through `createAuthoringDomain()`. Project is the single document authority; the other kits register typed source and operations. Use this guide to start editing. The general Engine README remains the runtime entry point.

Start editing in a local browser

The Editor repository provides the real Engine host, filesystem persistence, Canvas controls, a Three viewport and GLB/PNG adapters. In that repository:

```
npm ci
npm run authoring -- create --project /absolute/path/to/my-project
npm run authoring -- open --project /absolute/path/to/my-project
```

Open the printed localhost URL. Create a cube, select its object in the outliner, use the transform gizmo or the box face edit, undo/redo, save and export GLB. Open/New project controls save the current project before switching directories. The command panel accepts an array of typed operations. The CLI also exposes JSON-line stdio for agents:

```
npm run authoring -- stdio --project /absolute/path/to/my-project
```

The Authoring host requires the pinned Engine package. It never uses the Editor's legacy fallback runtime. The optional WebGPU implementations live in NexusEngine-Kits; Authoring source has no renderer, filesystem, PNG codec, browser object or GPU handle.

Edit directly with portable JavaScript

```
import { createEngine } from "nexusengine";
import { createAuthoringDomain } from "nexusengine/domains/authoring";

const engine = createEngine({ kits: createAuthoringDomain() });
const project = engine.n.authoringProject;
const command = (requestId, operations) => project.execute({
  requestId,
  epoch: project.context().epoch,
  operations,
});

command("create-box", [{ id: "mesh.cube", args: { id: "box", size: 2 } }]);
const box = project.getDocument("box");
const preview = project.preview({
  requestId: "raise-top",
  epoch: project.context().epoch,
  operations: [{
    id: "mesh.transform",
    args: {
      id: box.id,
      expectedRevision: box.revision,
      selection: { mode: "face", ids: ["f4"] },
      translation: [0, 0.25, 0],
    },
  }],
});
project.acceptPreview(preview);
project.undo({ requestId: "undo-top", epoch: project.context().epoch });
project.redo({ requestId: "redo-top", epoch: project.context().epoch });
const savedSource = project.getSnapshot();
```

The same workflow is executable in `examples/authoring/first-edit.mjs`. Pure Core scripts need no window or background process. An application host is needed for persistence, transport, background jobs and preview. Documents do not start a second runtime.

Discover paths, kits and commands

```
engine.n.ownerOf("n:authoring:mesh"); // authoring-mesh-document-kit
engine.n.api("authoringMesh");       // installed API metadata and owner
project.kinds();                     // installed source schemas
project.tools();                     // commands, paths, inputs, effects and limits
project.tools().filter(tool => tool.domainPath === "n:authoring:mesh");
```

Commands are addressed by their discovered operation ID and carry their owning Domain path. API aliases such as `engine.n.authoringMesh` are registered by the same manifests. Do not turn a path into a private filesystem import. All public factory subpaths are under `nexusengine/domains/authoring`; the generated catalog contains their exact mappings.

`tools()` includes an input schema, accepted field list, profile, document reads and writes, cancellation boundary, common errors and receipt schema. Complex source schemas and geometric preconditions also require `project.preview()`; a JSON-schema check alone does not validate prospective topology or references. Kit installation is trusted application code. Project files never deserialize functions or authorize remote registry execution.

Document and transaction rules

- A document has `id`, `kind`, `schemaVersion`, monotonic `revision`, immutable `content`, typed `dependencies` and a canonical SHA-256 content hash.
- A request has a nonempty `requestId`, current `epoch` and 1–256 operations. Edits and deletions require the target revision from before the transaction. `expectedRevision: 0` is only for editing a document created earlier in that same staged transaction.
- Repeating a retained request with identical content returns its original receipt. Reusing the ID with different content fails. A stale revision or unknown old epoch fails before commit.
- Operations stage together. Schema, dependency kinds, reference cycles, element references and cross-document invariants are checked against the complete prospective state. Failure changes neither source nor receipts/history.
- Built-in source dependencies use `latest` references. Exact revision references in extension schemas require an explicit typed rebase; restoring a snapshot must not silently rewrite such references.
- Undo/redo retains changed document versions and shares unchanged immutable mesh elements. A new edit clears redo; a no-op preserves redo. Snapshots copy portable source by default; `getSnapshot({immutable:true})` shares validated read-only document versions for serialization. Default history retention is 128 entries.
- Reset destroys source and starts a new epoch. Restore validates the full snapshot and history, allocates fresh current revisions/epoch, and retains historical receipts. Default receipt capacity is 10,000; overflow is explicit.
- Restored source requires its kind implementations to be installed. Unknown schema versions fail explicitly. There is no earlier published Authoring format that is silently migrated into this format.

Child service snapshots identify their installed registration; reset disposes transient evaluation caches or Runtime executions and preserves source. Project alone snapshots/restores/resets editable documents and history.

A receipt distinguishes the committed mutation from subsequent export or visual review. It contains request identity, before/after clocks, changed document identities, operation results, validation and recovery information. Successful transport delivery alone does not establish valid geometry or visual quality.

Capability profiles

Every row below names implemented behavior. Algorithm limits are intentional, and unsupported inputs fail before replacing source. This is a source editing system with a minimal client; it does not claim Blender feature parity.

Child path under `n:authoring`	Source and supported operations	Boundary
Root / Project	Contract, schema registration, documents, discovery, transactions, previews, receipts, history, snapshots, recovery	One Project writer; portable source only
`workspace`	Open document references, active document, named view references, mode and tool; open/close/set/delete	Closing a view never deletes its source; no new root Workspace domain
`editing`	Vertex/edge/face selection; assembly-instance selection; add/remove/toggle, expand/contract, connected, quad loops/rings	Selected elements and objects must remain valid after a transaction
`mesh`	Box, plane/grid, circle/disc, cylinder/cone, sphere, torus; translation, quaternion rotation, scale, pivot, coordinate frame, proportional falloff	100,000 vertices / 200,000 faces; finite coordinates within ±1e12; Y-up source
`mesh` topology	Delete, extract, duplicate, split, triangulate, explicit weld, fill, extrude, inset, edge subdivision, bridge, loop cut, knife chord, dissolve, convex bevel	Details below; attribute preservation or explicit rejection
`curve`	Polyline/cubic Bézier points, adaptive evaluation, parallel-transport tube sweep, caps and closed sweeps	Up to 10,000 controls / 100,000 evaluated samples; 180-degree cusps reject

Child path under `n:authoring`	Source and supported operations	Boundary
`modifier`	Ordered enabled stacks; mirror, array, Catmull–Clark subdivision, solidify, smooth, shrinkwrap, twist/taper/bend, Boolean, decimation, remesh; reorder/remove/apply	Evaluated output is separate until an atomic Apply
`brush`	Distance-based sampling, pressure, constant/linear/smooth falloff, axis symmetry	Finite explicit stroke and sample budgets
`sculpt`	Grab, inflate, smooth, flatten, mask and layer strength; preview and history	Fixed topology; no dynamic-topology sculpting
`uv`	Planar, cylindrical, spherical, toroidal and face projections; seam flags; per-face unwrap, island detection, packing and transforms	Per-face unwrap is not LSCM; padding must fit the requested resolution
`uv` diagnostics	Degenerate/mirrored triangles, area ratios, conformal stretch and positive-area overlap	Bounded overlap candidate checks; boundary contacts are excluded
`material`	Metallic-roughness PBR, five texture roles, samplers, alpha/cutoff, double sided, emissive; constant/checker/noise/multiply/mix graphs	Unsupported shader nodes reject; procedural color must bake before delivery
`paint`	RGBA8 tiled layers, linear-space compositing, fill, brush strokes, clone offsets, vertex colors, procedural color and normal baking	64×64 tiles; 1–4096 image dimensions, 1–16 layers; synchronous bake ≤1M pixels
`rig`	Bone hierarchy, rest transforms, parent/cycle validation, add/update/remove; pose limits, copy, look-at and CCD IK evaluation	≤512 bones; bounded IK; look-at/IK require positive uniform scales
`skin`	Explicit bind matrices, inverse-distance-to-segment binding and diffusion; normalized weights, locked influences, brush painting, smoothing, mirroring and LBS evaluation	1–32 source influences; topology/rest changes require a same-transaction rebind/remap
`animation`	Seconds-based clips, STEP/LINEAR/CUBICSPLINE TRS tracks, quaternion interpolation, key editing, poses, relative shape keys, clip arrangements	Later arrangement entries override the same property; no implicit timeline clock
`animation` conversion	Explicit one-to-one rest-aligned retarget; constraint bake; common-timeline morph bake	Checks interval quarter/midpoints against 0.001 source-unit, 0.1-degree and 0.0001 scale tolerances; this is sampled evidence, not a continuous mathematical bound
`assembly`	Shared assets, nested transforms, collections, visibility/export flags, variants, cameras/lights, instance duplicate/remove/update, surface scatter	Static scatter prototypes, area-weighted seeded placement, density mask, normal alignment, spacing, scale/orientation variation
`domain-composition`	Discover trusted Core registry, author/validate trees, add/remove nodes, plan composition	Reuses public Composition contracts; no remote code execution
`sequence`	Source definitions with Before/During/After, bounded steps and named operation arrays; current Runtime execution integration	Finite Linear manual advancement; no automatic agent planning loop
`publishing`	Immutable evaluated packet, source hashes/revisions, object/asset/presentation descriptors, geometry, PBR images, rigs/skins/clips/shapes, dependency closure, bounded cache	External adapters encode files; four influences by default, explicit measured reduction if requested

Mesh topology and attributes

Vertices, faces, corners and deterministic edges have stable identities. The mesh validator rejects nonmanifold edges, disconnected vertex fans, degenerate edges, self-intersecting polygon boundaries and invalid references. Concave planar polygons use ear clipping. Nonplanar quads have deterministic projected triangulation; nonplanar polygons with more than four corners must be explicitly triangulated before deformation. Global triangle-to-triangle self-intersection is not a general validity claim.

Numeric vertex/corner attributes interpolate through declared remapping. Parent face values copy. Unparented new edges have no invented seam/crease values. Reserved attributes are `uv0` (corner/2), `color` (vertex/4), `mask` (vertex/1), `material` (face/1), and `seam`, `sharp`, `crease` (edge/1). Bound skin and shape source reject incompatible topology changes until explicitly rebuilt together.

- Inset: one strictly convex face. Extrude: selected face region and boundary walls.
- Weld: explicit removed/retained pairs within tolerance; no silent face collapse.
- Bridge: two matching ordered loops. Loop cut: a noncrossing quad strip, 2–32 segments.
- Knife: chord between two existing nonadjacent vertices of one face.
- Dissolve: one interior edge with compatible face/corner attributes; discontinuities reject.
- Bevel: flat chamfer on selected edges of a closed convex solid, at most 64 original/chamfer planes. Run before attribute authoring; this profile rejects meshes with authored attributes rather than inventing UV/weight transfer.

Evaluated geometry splits source corners as needed, generates smooth/hard-edge normals, UVs, tangent handedness, colors and material groups. These arrays are portable; they are not renderer-owned buffers.

Modifier limits

Mirror seam merging requires a half-space source. Subdivision uses binary crease and boundary rules, up to four levels within the mesh budget. Shrinkwrap uses an AABB tree with exact triangle-distance leaves and a 10M test budget. Boolean is BSP CSG on closed outward manifolds, ≤4,000 faces per operand, bounded depth and edge conformity. Decimation uses validated edge collapse on ≤5,000 vertices and protects supported boundaries/seams/creases. Remesh is subdivision, triangulation and relaxation; it is not a voxel remesher.

Animation and delivery

Retargeting requires explicit bone mappings. Constraint-bearing rigs require clips baked against the current rig identity before publishing. Editing a baked key invalidates that bake receipt. Morph tracks must share key times and interpolation for GLB; continuous tracks can be baked to a common timeline. STEP discontinuities are preserved or explicitly rejected by conversion profiles.

The Editor GLB adapter writes actual glTF 2.0 binary buffers, PBR material/texture bindings, embedded PNGs, standalone PNGs, hierarchy, joints, inverse bind matrices, weights, clips, shape position/normal deltas, cameras and punctual lights. It validates with Khronos before atomically publishing a content-addressed directory. An independent Three GLTFLoader renders the resulting bytes and samples animation. A validator warning remains visible in the receipt; it is not discarded.

The packet/source revision check runs again at publication. Stale output is not committed as current output. Cancellation cleans staging directories. FBX and arbitrary Blender node compatibility are not implemented format profiles.

Sequences and durable hosts

`engine.n.authoringSequence.start(id, { runId })` creates a finite execution owned by existing `n:runtime:sequence`. `request(stepId)` prepares the active Project request; `acknowledge(stepId, receipt)` requires the actual retained Project receipt. The Editor host executes and journals that request before acknowledging it. `advance(stepId)` is the direct synchronous Core convenience path.

Ticks do not advance manually driven Authoring leaves. Wrong-step targeting, changed request content and invalid operations fail. Successful or failed runs release Runtime nodes; receipts remain available. Cancellation stops the bounded run. A caller or agent inspects results and plans a later run. The declared attempt budget is source metadata, not an automatic retry/replanning controller.

Persistence and background work

Filesystem projects contain `project.json`, content-addressed `documents/`, `blobs/`, `checkpoints/`, and an ordered hash-chained `journal.jsonl`. Checkpoints include source/history and reference image tiles as blobs. An exclusive local writer session and generation checks prevent accidental competing saves. Files and parent directories are synced before manifest replacement. A failed manifest write preserves the previous checkpoint. Missing/corrupt records fail explicitly.

Browser IndexedDB storage is a separate explicit-save profile with atomic transactions and generation checks. It does not claim the Node journal/lease profile or cross-device collaboration.

The Editor worker adapter bounds concurrency, queued work, time, V8 heap and transfer size. It reports stages, supports cancellation/termination/restart, and checks source revisions before committing a derived mesh or image. Large derived image edits checkpoint directly when they exceed the journal frame profile. Closing a host cancels its workers. These are resource budgets, not an operating system security sandbox.

Proofs and reproducible examples

Core tests run normally through `npm test`:

- `tests/core-domains/core-authoring-public.mjs`
- `tests/core-domains/core-authoring-foundation.mjs`
- `tests/core-domains/core-authoring-geometry.mjs`
- `tests/core-domains/core-authoring-surfaces-rig.mjs`
- `tests/core-domains/core-authoring-integration.mjs`
- `tests/core-domains/core-authoring-modeling.mjs`

Editor recipes and tests exercise the actual packed Engine dependency. The donut recipe creates geometry, icing, UVs, color/normal maps and seeded sprinkles through installed commands. Mechanical and organic recipes cover bevels, shared parts, rigging, weights, clips and shape keys. The headless development harness records source hashes, bounded commands, failures and all nine verification stages.

The Engine capability/command inventory and Editor evidence record distinguish contract proof, rendered output and measured workload limits. Batch checks at 1, 10 and 100 jobs establish bounded isolation/retry/resume behavior. They do not establish throughput or quality at hundreds of thousands of scenes. Multiple users, distributed asset services, larger meshes, 8K images, full Blender workspaces and production-scale deployment require additional measured profiles.

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State, Lifecycle, And Idempotence

Idempotence means repeating an accepted operation does not create a second effect. It is enforced locally by each atom and globally by the composition controller.

Install

An atomic Kit has a stable identity and manifest fingerprint. Installing matching content again returns the installed API without adding state, systems, or subscriptions. Reusing the same identity with different content is a hard conflict.

Commands

State-changing commands need a stable idempotency key when they may be retried. The state owner records the accepted key and result. Repeating the same command returns the original result. Reusing the key with changed content fails before mutation.

Snapshot

A snapshot contains portable semantic state. It excludes renderer objects, functions, sockets, file handles, GPU resources, and vendor SDK instances. Providers snapshot their own implementation state behind their contract when necessary.

Reset

Reset restores the configured baseline. Calling reset twice returns the same snapshot and must not duplicate resources or lifecycle hooks.

Physics Lifecycle

The canonical Physics lifecycle uses explicit stages rather than treating installation as proof that a provider is ready:

```
uninstalled -> installed -> starting -> ready -> stopping -> installed
                                  \-> failed <-/
```

Every mutation carries an `operationId`. Exact replay returns the original receipt; changed content under the same ID fails before mutation. Installation owns the phase. Startup, Step, Shutdown, Reset, and Snapshot own separate state resources and coordinate only through public capability APIs. Multi-API reset and restore operations capture pre-call snapshots and roll every component back when any validation or load fails.

Physics Materials

Physical material descriptors are immutable under one material ID. Defining a new record requires an `operationId`; exact replay returns the original receipt, changed command content fails before mutation, and a different command cannot replace an existing ID with different material content. Removal follows the same exact-once rule.

Friction, restitution, density, surface, and combine-policy normalization are read-only. Pair resolution does not mutate either material or policy state and returns byte-equivalent output when the two material arguments are reversed. Snapshots contain only portable records, sorted IDs, revisions, and receipts.

Physics Worlds

Gravity, force, wind, time-scale, simulation-region, and Physics world records use immutable IDs and exact-once mutation receipts. Repeating an accepted definition returns the original receipt. Reusing its operation ID with changed content, or redefining a record ID with different content, fails before state changes.

Field, region, scale, and aggregate world sampling are read-only. A world snapshot stores only normalized records, sorted IDs, revisions, and receipts; it never stores provider worlds, native handles, clocks, weather objects, or query diagnostics. Loading a world snapshot validates every capability reference before replacing state.

Replay

Deterministic replay starts from an accepted snapshot and applies the same ordered inputs. Any nondeterministic source is injected by the host and recorded as input. Random behavior uses explicit seeds or provider receipts.

Composition-Level Exactly-Once

The controller hashes the normalized request, dependency plan, exact Kit source descriptors, configuration, and resolved executable fingerprints. The resulting plan ID is reviewed by a human. After apply, its receipt is persisted. Repeating the same plan returns that receipt without invoking the host again.

Transactional Failure

Before apply, the host supplies a snapshot. If host mutation or receipt persistence fails, the controller restores both host and controller state. A failed plan has no success receipt.

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Composition And Recipes

Composition turns semantic intent into a deterministic, reviewable installation order.

Registry First

Core discovery starts from merged registry metadata generated from manifests. It never scans the filesystem at runtime. Imported registries may add identities but may not replace Core records or collide with an existing Domain path.

Selection Inputs

A request may select:

- individual atomic Kit IDs
- Domain IDs, which select the Domain's public atoms
- recipe IDs, which select declared Domains and Kits
- a scoped composition tree
- per-Kit configuration and allowed statuses

The retired `bundles` request key is rejected. Use recipes for declarative composition.

Validate Before Plan

Validation checks identity, dependencies, provider availability, status policy, cycles, and ordering without resolving executable code.

The runnable inspection example is `examples/guide/composition-inspection.mjs`.

```
const validation = engine.n.composition.planning.validate({
  kits: ["object-placement-kit"]
});

if (!validation.ok) throw new Error("Composition is invalid");
```

Plan

Planning adds required providers and returns a stable dependency order. MCP planning also asks the application host to preflight exact executable sources. Preflight is read-only.

Recipe Ownership

A recipe is data. It can express a useful combination without creating a new state owner. Reusable optional recipes belong in a trusted Kit registry. Complete game recipes belong with the game.

Core includes six data-only recipes for proving the restored universal graph:

- `procedural-navigation`
- `terrain-character-traversal`
- `management-operations`
- `vehicle-rescue-logistics`
- `spatial-guidance`
- `hazard-pursuit`

These recipes select atoms and adapters; they do not contain authored levels, economies, objectives, or tuning. Applications provide that data themselves.

Apply

Apply requires the exact reviewed plan ID and explicit authorization. The controller resolves the same metadata, rejects drift, applies through the host, persists a receipt, and returns the original receipt on replay.

GUIDE CHAPTER

MCP Agent Workflow

MCP makes the semantic catalog inspectable by an agent. It does not make the agent the runtime owner and does not authorize mutation by itself.

Opt-In Boundary

An application installs the MCP registry Domain, registers the Composition provider, supplies a transactional host, and chooses a transport. Without those steps, the package exposes no live MCP server.

Read Tools

Tool	Purpose
`domains_list`, `domain_get`	Discover semantic ownership
`kits_list`, `kit_explain`	Preserve the existing Kit-oriented inspection surface
`atoms_list`, `atom_get`	Page through atomic capabilities and dependency evidence
`recipes_list`, `recipe_get`	Inspect declarative compositions
`registry_sources_list`	Review package, version, commit, integrity, environment, permissions, and status
`composition_validate`	Check a request without resolving code
`composition_plan`	Resolve exact sources and produce a stable plan ID

Mutation Tool

`composition_apply` is the only mutation tool in the provider. It requires explicit authorization for the exact plan ID. A transport or application may impose additional policy.

Review Sequence

```
inspect relevant Domains and atoms
-> validate the request
-> plan exact sources
-> show commit, integrity, permissions, and capability changes
-> human approves the exact plan ID
-> apply once
-> persist receipt
-> runtime continues after MCP disconnects
```

Prompts

The provider exposes exactly two short prompts: `inspect-and-plan` and `review-and-apply`. They route the workflow; they do not embed the full guide.

Resources

Registry records and guide chapters are individual MCP resources. An agent reads only the records and chapters relevant to its current decision. The complete PDF is for humans, not prompt context.

GUIDE CHAPTER

Registries And Security

A registry record is metadata. Reading, searching, merging, or validating it must not execute package code.

Exact Source Identity

An executable Kit record includes:

- package name and exact version
- canonical package subpath and export name
- immutable 40-character source commit
- SHA-256 integrity
- environments and requested permissions
- requires and provides tokens
- status and settings schema

Moving branches, tags, and unpinned URLs are not executable identities.

Resolution Rules

The host verifies that paths stay inside the resolved package root, imports the declared export only, compares integrity, and computes an executable fingerprint. Browser modules require immutable commit URLs. Local and package paths cannot escape their package root.

No Runtime Installation

Core never runs a package manager during composition. When required code is absent, planning returns a structured installation receipt with the exact package and version. Installation is a separate human-controlled operation; planning is repeated afterward.

Collision Rules

- An imported registry cannot replace a Core identity.
- Two Domains cannot claim the same semantic path.
- One Kit ID cannot refer to changed content.
- A repeated accepted plan returns its original receipt.
- Integrity mismatch, wrong export, path escape, or missing source fails before mutation.

Trust Boundary

Integrity proves identity, not benevolence. Approved JavaScript still runs with the permissions of its host. Review source ownership, commit, permissions, and environment before approval. Use process or platform isolation when untrusted code must run.

GUIDE CHAPTER

Hosts, Providers, And Adapters

Core owns semantic contracts. Concrete platform work belongs at a leaf boundary.

Host

The host owns application lifecycle, package availability, executable resolution, runtime construction, persistence, and rollback. Core retains only portable host capability contracts under `n:host`.

Browser, Node, native, editor, terminal, repository, filesystem, storage, transport, and vendor SDK implementations are outside Core.

Provider

A provider implements one Domain contract. Examples include physics solving, model inference, speech synthesis, asset retrieval, or capture rendering. Core can own request and result schemas while the provider owns vendor handles, caches, and external effects.

For Physics, `nexusengine/domains/physics/provider-contract` validates a provider's lifecycle and execution methods without installing or executing the provider. Provider capability metadata must be portable; backend worlds, native handles, caches, and solver state remain provider-owned. State, commands, events, and query records cross the boundary through the sibling contract Kits under `n:physics:contracts`.

The provider does not become ready merely because it was selected. `n:physics:lifecycle` emits explicit startup, step, shutdown, reset, and snapshot requests. The provider executes the relevant operation and returns a portable receipt. Core owns sequencing and replay; the provider owns backend worlds, handles, solver state, and platform effects.

Physical material records cross the provider boundary as portable input. Core owns their identity, validated coefficients, surface classification, and deterministic pair policy. A provider may translate those records into native backend materials, but native handles and caches never enter Core snapshots. The provider executes the resulting contact response; `n:physics:material` does not calculate impulses or silently map physical surfaces to renderer or audio assets.

Physics world records also cross the provider boundary as portable input. Core resolves world settings, gravity acceleration, generic fields, physical flow, time scale, and simulation-region policy. A provider applies those records to its native world, bodies, broad phase, and solver. Provider handles and cached native field objects never enter Core snapshots.

`n:world:weather` remains the authored weather authority. A product or explicit adapter may translate accepted weather or atmosphere descriptors into a physical wind-field command. That translation does not transfer weather, rendering, or gameplay ownership into `n:physics:world`.

For Render, `nexusengine/domains/render/provider-contract` validates resource and frame execution methods without installing or invoking the provider. Portable resource, frame, resolved-pass, shader-interface, and event records cross the sibling contract Kits under `n:render:contracts`. Native devices, GPU handles, compiled shaders, command encoders, caches, and platform surfaces remain provider- or host-owned.

`n:render:lifecycle` records which provider composition is selected and tracks its startup, shutdown, reset, snapshot, and recovery receipts. It never retains the provider object or invokes its methods. A ready recovery receipt may return the lifecycle to `ready`; a nonready receipt returns it to `installed`, after which the provider must complete a new startup. Runtime lifecycle remains the owner of engine ticking and generic Kit installation.

`n:render:device` records the portable identity and accepted capabilities of a device returned by that provider. Separate atoms own feature declarations, numeric limits, capability profiles, semantic memory reservations, logical queue submissions, acquisition state, loss records, and read-only diagnostics. Provider receipts are explicit command inputs. GPU handles, real allocation, command encoding, queue execution, and repair never enter these Core records.

`n:render:resource` records exact execution-resource identities, references, portable residency, integrity evidence, accounting, and provider operation receipts. `n:render:buffer` builds portable logical Buffer descriptors, explicit layouts, and typed views over those exact identities. Buffer updates refer to Device queue submissions and accept explicit provider

receipts; Core never retains source bytes or performs allocation, mapping, transfer, command submission, or repair.

`n:render:texture` builds portable Texture formats, exact Texture records, typed subresource views, mip plans, stream requests, and residency evidence on those Resource and Buffer contracts. A stream names the exact source-provided mip content, a bounded copy-source staging Buffer range, and one Device queue submission. Completion records a matching provider receipt; only completed streams may establish resident subresources. Asset still owns source content and decoding, Presentation owns visual and authored shadow meaning, Pipeline and Frame own attachment execution, and providers own allocation, upload, mip generation, eviction, repair, and backend format mapping.

`n:render:shader` records portable language capabilities, immutable source and include revisions, module and program topology, deterministic variants, bounded permutations, logical compile state, normalized reflection, and cache links to resident `shader-program` Resources. Compile requests name an exact SHA-256 module/source/include closure and a matching Device queue submission; completion accepts only an explicit provider receipt for the same device, submission, and compile identity. The existing Shader interface schema remains under `n:render:contracts`. Core does not preprocess, parse, compile, link, retain binaries or GPU programs, execute backend reflection, or repair source.

`n:material` associates exact portable Material slots with Shader interfaces, typed parameter values, resident Texture views and subresources, sampler descriptors, complete instances, and Shader variants. Validation names one completed compile and matching reflection record, while cache records point to resident `material` Resources with the same composition hash. Current resolution always rechecks Texture lifecycle and subresource residency, so an eviction makes dependent validation and cache use stale without changing stored Material state. Presentation still owns authored visual meaning, and providers own GPU bind groups, sampler objects, uploads, binding commands, execution, and repair.

A Presentation render-layer graph expresses visual meaning and ordering policy. It is not a GPU command graph. A later explicit bridge resolves an accepted Presentation graph into strict Render pass records; the Render Pass contract does not silently take ownership of semantic layers or reorder them.

Adapter

An adapter translates two public contracts without becoming the state owner of either. Its manifest names both requirements and the capability it provides. Adapters import public semantic subpaths, never private sibling files.

The restored behavior graph uses nine optional adapters:

- locomotion contact response
- camera world occlusion
- vehicle water response
- lifecycle economy
- lifecycle facility
- facility economy
- occupant request
- transport request
- request economy

Each atom remains useful without its adapters. Installing an adapter never installs another atom and never turns a read-only query into a hidden mutation.

Presentation Boundary

Presentation Core contains renderer-neutral descriptors and policies. Three.js objects, WebGL resources, browser audio, native widgets, and authored presets belong in renderer, host, Kit, or product repositories.

Example Ownership Decisions

- Generic sky descriptors: `n:presentation:sky`.
- Authored sunset preset: recipe or product data.
- Speech request lifecycle: `n:presentation:speech`.

- PocketTTS implementation: external provider Kit.
- Motion trajectory: `n:simulation:motion`.
- Animation clip descriptor: `n:presentation:animation`.
- Two-bone pose solving: `n:simulation:motion`.

Bundled Graphics Language Kit

The optional `nexusengine/domains/render/graphics-language` entry composes existing Render contracts, resource identity and lifecycle services with NexusGL native/WASM execution and platform presentation. Core owns portable installation and handover status; the optional kit owns executable provider instances and buffers. Applications keep scene state and select their pipeline. See `docs/GRAPHICS-LANGUAGE.md` for one-call installation, the shared baseline and failure behavior.

GUIDE CHAPTER

Build Projects And Targets

`n:build` is NexusEngine's isolated build-time Domain. It is physically shipped with the Engine so projects do not own compiler, toolchain, platform host, packaging, artifact, or receipt logic. It is not installed by `createEngine()` and cannot be imported by runtime Domains.

Boundary

```
project (read only)              ~/.nexusengine (Build owned)
├── src                          ├── sources/<sha256>
├── content                      ├── toolchains/<identity>
├── assets                       ├── builds/<plan-hash>/<target>
├── tests                        ├── artifacts/<project>/<plan-hash>
└── package.json                 └── receipts/<plan-hash>.json
```

Build fingerprints every project file before planning and immediately before execution. It fingerprints the project again after every aggregate run. Any changed, added, or removed source path fails project immutability proof.

Large repositories may declare exact target entrypoints and an analysis include set in `package.json`. Paths are literal repository-relative paths, not globs. Excluded files remain part of the whole-project immutability fingerprint but do not enter target compilation graphs.

```
{
  "nexusengineBuild": {
    "include": ["index.html", "src"],
    "entries": {
      "web-static": "src/main.js",
      "android-xr": "src/native-main.js",
      "pcvr": "src/native-main.js"
    }
  }
}
```

Commands

```
nexusengine inspect ./project

nexusengine plan ./project \
  --profile native-preferred \
  --target web-live \
  --target web-static \
  --target android-xr \
  --target pcvr

nexusengine build ./project \
  --profile native-preferred \
  --out /absolute/artifact/root \
  --approve-plan sha256:<exact-plan-hash> \
  --target web-static
```

At least one repeated `--target` is required. Order and duplicates normalize to one sorted target set, so they do not change the plan hash. An interactive terminal may confirm the displayed exact plan. Noninteractive execution always requires `--approve-plan`.

Pipeline

```
read-only project inventory
-> SHA-256 source fingerprint
-> TypeScript compiler AST and typed diagnostics
-> AST-derived module, effect, and dependency analysis
-> Kit IR
-> deterministic Execution IR
-> whole-Kit portability classification
-> target capability and fallback selection
-> exact plan approval
-> isolated target stages
-> artifact integrity and project immutability proof
-> persistent receipt
```

The four whole-Kit execution modes are `native`, `native-adapter`, `javascript`, and `unsupported`. JavaScript is not silently translated into native behavior. A module must explicitly declare portable-native intent and pass semantic parity before native lowering can be accepted.

Targets

- `web-live` copies the local immutable module closure and generates a service worker that verifies every SHA-256 before populating a content cache.
- `web-static` emits a self-contained local directory and rejects unresolved external browser packages.
- `android-xr` shares OpenXR runtime, action, view, and submission contracts, then creates an external Android ARM64 Gradle/NDK stage and validates the APK closure without installing it on a device.
- `pcvr` shares the same OpenXR contracts, then adds Windows x64 host, generated-runtime linking, PE packaging, and no-runtime validation.

Native planning is not native proof. Supported numeric Kit IR lowers into deterministic Rust; other whole Kits may use the QuickJS-NG sandbox only when their requested capabilities are supported. OpenXR and QuickJS-NG source comes from exact commits with verified archive SHA-256 values. A native target is `package-proven` only after its target compiler and package validator succeed. The release requires Linux Android XR and Windows PCVR package jobs; runtime and headset execution remain separate `hardware: false` evidence for `0.0.4`. Every Web artifact also contains `nexusengine-build-diagnostics.json` with its exact plan, registry, target, closure, source, Engine dependency, and linker identity for a product diagnostic view.

Source Security

Build accepts npm registry integrity, crates.io checksums, exact Git commits, versioned vendor installers, and immutable HTTPS ESM URLs. Metadata discovery does not execute source. Moving references, absent licenses, integrity mismatches, path escapes, incomplete dependency closure, and duplicate source identities fail before target execution.

Exact `git+https` lock entries and npm's exact `git+ssh://git@github.com/...#<commit>` lock spelling resolve to one canonical GitHub HTTPS source identity inside isolated npm stages. Credentials, non-GitHub SSH sources, and moving refs remain rejected. Build retains any caller-provided rewrites used by a controlled local source mirror.

The base npm install performs no Build downloads. Network provisioning is a separate approved operation and caches verified bytes by content hash.

MCP

The opt-in Build provider exposes:

```
build_targets_list
build_inspect
build_plan
build_apply
build_receipt_get
```

`build_apply` requires MCP authorization and the exact reviewed plan hash. Repeated successful apply returns the existing receipt without rebuilding. During a partial failure, successful target stages remain cached while blocked or failed targets remain visible in the aggregate failed receipt.

GUIDE CHAPTER

Testing And Proof

Tests prove contracts at the level where failure would matter. A green unit test is not enough when a change affects package exports, composition, consumers, or rendered output.

Manifest Proof

Validate unique IDs, normalized paths, immediate parents, one owner per public module, source paths, export names, schemas, lifecycle declarations, dependencies, and proof references.

Atomic Lifecycle Proof

For each stateful atom:

1. Install twice and prove no duplicate state or systems.
2. Apply the same command twice and compare state and result.
3. Reuse an idempotency key with changed content and require failure.
4. Snapshot, load, and compare equality.
5. Reset twice and compare equality.
6. Replay ordered inputs from the same snapshot.

Composition Proof

Test missing providers, cycles, collisions, status rejection, stable ordering, repeated apply, process restart, receipt persistence, host rollback, and continued runtime operation after MCP disconnect.

Restoration Proof

The restoration generator checks 26 historical source checksums against the last complete Git snapshot, requires exactly 27 replacement atoms, nine adapters, and six recipes, and confirms every proof path exists. Regression tests cover each named historical defect, JSON-portable snapshot boundaries, exact-once operations, deterministic ties, large-delta advancement, and side-effect-free queries.

The root disposition ledger must classify those exact 26 sources as `core-restored`. A source returning to `external-kit`, a missing old export, or a replacement that is no longer manifest-reachable fails documentation and migration checks.

Combination Coverage

Test each atom alone, every declared dependency edge, pairwise cross-Domain combinations, official recipes, high-risk three-way stacks, and installation-order permutations. Do not attempt every mathematical combination.

Registry Security Proof

Prove that metadata reads cannot execute code. Reject moving refs, integrity mismatch, path escape, wrong export, missing package, duplicate identity, and changed-content replay before mutation.

Consumer Proof

Install the packed Engine tarball in clean temporary directories. Then prove Kits, Editor, and Simulator against that exact artifact without local symlinks. Browser-facing workflows require Playwright and human-view inspection in addition to DOM assertions.

Documentation Proof

Regenerate all derived files, run drift checks, execute guide examples, resolve links, render the PDF, verify every page contains content, and visually inspect rendered page images for clipping or overlap.

GUIDE CHAPTER

Migrating To 0.0.4

`0.0.4` is a hard semantic cutover. Old paths and root symbols are removed in the same change as their replacements. There are no runtime forwarding exports.

Import Strategy

Keep root imports for bootstrap and runtime contracts. Import Domain factories and atoms from generated semantic subpaths.

```
import { createEngine } from "nexusengine";
import { createMotionKit } from "nexusengine/domains/simulation/motion";
import { createPhysicsKit } from "nexusengine/domains/simulation/physics";
```

Do not use `n:core-*`, `nexusengine/core-kits/*`, or private source paths.

Runtime Paths

Use semantic paths such as `n:object`, `n:runtime:transaction`, and `n:material`. Immediate parent relationships are validated by manifests.

Removed Implementations

Concrete hosts, renderers, shaders, platform storage, model mocks, speech engines, authored sky presets, genre rules, and complete games moved outside Core or became caller-owned data. Universal behavior from 26 removed modules is restored through 27 corrected semantic atoms. The Restored Behavior Migration appendix lists every old export, configuration change, snapshot change, event change, optional adapter, and proof.

Physics Name Collision

The historical `createPhysicsKit` from `src/world-physics-kit.js` and the current provider-neutral `createPhysicsKit` do not provide the same behavior. Migrate the historical behavior to `createWorldContactKit` plus `createSoftRespawnKit`. Use the current factory only for Physics contracts and providers. There is no alias or legacy snapshot coercion between them.

Consumer Migration Order

1. Replace old imports with generated semantic package subpaths.
2. Replace old runtime paths with semantic paths.
3. Install explicit atoms that were formerly implicit defaults.
4. Move platform implementations into the host or an external Kit.
5. Replace compatibility aliases with the owning Domain API.
6. Run from a packed Engine artifact in a clean directory.

Consumer changes are intentionally separate from the Engine-local restoration wave. A migration row marked `outside-restoration-wave` does not mean the Core replacement is missing; it means the consumer has not yet been changed.

Changelog Contract

The changelog and migration map explain replacements. They are not compatibility code. A removed symbol stays removed.

GUIDE CHAPTER

ProtoKit Extraction

ProtoKits is retired as an active development destination. Its history remains evidence for reconstructing useful behavior with Core atoms, external Kits, recipes, and game-owned code.

Frozen Source

Extraction uses a clean read-only worktree at one immutable remote default-branch commit. The inventory records the remote, commit, file count, and source snapshot SHA-256. A stale or dirty local checkout is never canonical evidence.

Unit Of Classification

Folder count is not capability count. The extractor records folders, files, exports, factories, manifests, tests, dependencies, state, lifecycle signals, commands, events, providers, adapters, and consumers. Every discovered source surface receives exactly one disposition.

Dispositions

Disposition	Meaning
`core-reuse`	Existing Core atom already owns the behavior
`core-composition`	Existing atoms reproduce it through composition
`core-new-atom`	A missing universal atom passed every promotion gate
`external-kit`	Reusable but optional, niche, platform, or vendor behavior
`recipe-data`	Authored data or declarative composition
`game-owned`	Complete product or gameplay behavior
`duplicate`	Semantically identical to a named canonical owner
`rejected-unproven`	Insufficient evidence for a safe owner

New Atom Gate

A candidate must have one indivisible responsibility, deterministic behavior or isolated nondeterminism, product neutrality, snapshot and reset support, no concrete presentation ownership, and two semantically different consumers. Failure of one condition keeps it outside Core.

Reconstruction

Useful behavior is preserved as a recipe: source lineage, required Core atoms, external owner, data, policy, and proof status. Preserving behavior does not require preserving the old package shape.

Archive Boundary

Writers and automations remain disabled. Archiving the remote repository is a separate external mutation that requires explicit approval after extraction coverage is complete.

GUIDE CHAPTER

Release And Operating Model

Release proof is performed against committed, reproducible sources. Local-green evidence is necessary but is not the same as a release.

Generated Truth

The Core catalog, package exports, ownership ledger, API reference, guide indexes, MCP chapter resources, and guide identify the same registry SHA-256. Generate mode writes them; check mode rejects drift.

Required Gates

```
npm ci --ignore-scripts --no-audit --no-fund
npm run core:check
npm run ownership:generate
npm run migrations:check
npm run protokits:check
npm run guide:check
npm run guide:examples
npm test
npm run test:release
npm run boundaries:check
npm run docs:check
npm pack --dry-run --json
```

Consumer repositories must install the exact packed or committed Engine source. Local symlinks and sibling private imports invalidate clean-consumer proof.

Human Approval

After every local and hosted gate passes, present the repository identity, previous numeric branch, derived next version, exact default-branch SHA, checks, and warnings. Approval applies only to that tuple.

Immediately before branch creation, refetch the remote and confirm the default branch still points to the approved SHA. Create the immutable numeric branch with a non-force exact-SHA push, then read the remote ref back and compare it byte for byte.

Outside This Guide Build

This documentation build does not push, publish, archive ProtoKits, mutate Google Drive, create a GitHub Release, publish npm, or create the numeric version branch. Each action has its own approval gate.

GENERATED REFERENCE

Domain Index

Registry SHA-256: `147e7302e0563cdf246cc33c197ba32fec606f783817d345244544e3ce486c76`

- `n:actor`: Own neutral embodied actor identity and shared actor references.
- `n:actor:creature`: Own neutral creature embodiment definitions and references.
- `n:actor:character`: Own active embodied character identity and neutral runtime bindings.
- `n:actor:player`: Own neutral player identity, possession, control authority, and spawn generations.
- `n:agent`: Own product-neutral observation, proposal, decision-cycle, execution receipt, and replay evidence contracts.
- `n:asset`: Own asset identity, manifests, bundles, content-addressed jobs, readiness, and provider contracts.
- `n:authoring`: Own canonical headless creation, import, editing, validation, persistence and publishing with default format implementations.
- `n:authoring:create`: Own canonical create operations, default providers, diagnostics and source-safe lifecycle.
- `n:authoring:importing`: Own canonical importing operations, default providers, diagnostics and source-safe lifecycle.
- `n:authoring:persistence`: Own canonical persistence operations, default providers, diagnostics and source-safe lifecycle.
- `n:authoring:publishing:export`: Own canonical publishing export operations, default providers, diagnostics and source-safe lifecycle.
- `n:authoring:validation`: Own canonical validation operations, default providers, diagnostics and source-safe lifecycle.
- `n:authoring:modifier`: Own modifier authoring contracts and operations.
- `n:authoring:publishing`: Own publishing authoring contracts and operations.
- `n:authoring:sequence`: Own sequence authoring contracts and operations.
- `n:authoring:domain-composition`: Own domain-composition authoring contracts and operations.
- `n:authoring:assembly`: Own assembly authoring contracts and operations.
- `n:authoring:animation`: Own animation authoring contracts and operations.
- `n:authoring:skin`: Own skin authoring contracts and operations.
- `n:authoring:rig`: Own rig authoring contracts and operations.
- `n:authoring:paint`: Own paint authoring contracts and operations.
- `n:authoring:material`: Own material authoring contracts and operations.
- `n:authoring:uv`: Own uv authoring contracts and operations.
- `n:authoring:curve`: Own curve authoring contracts and operations.
- `n:authoring:sculpt`: Own sculpt authoring contracts and operations.
- `n:authoring:brush`: Own brush authoring contracts and operations.
- `n:authoring:workspace`: Own workspace authoring contracts and operations.
- `n:authoring:project`: Own editable project contracts and operations.
- `n:authoring:mesh`: Own editable mesh contracts and operations.
- `n:authoring:editing`: Own editable editing contracts and operations.
- `n:build`: Own isolated build-time source analysis, compilation, toolchains, targets, artifacts, receipts, and proof without entering application runtime composition.
- `n:build:source`: Own read-only project source, immutable dependency identities, content caches, fingerprints, and module graphs.
- `n:build:analysis`: Own real syntax, type, effect, and dependency analysis for build inputs.
- `n:build:ir`: Own serializable Kit IR, Execution IR, validation, and source lineage maps.
- `n:build:classification`: Own whole-Kit portability, capability resolution, and fallback selection.
- `n:build:orchestration`: Own normalized requests, target sets, deterministic plans, approvals, execution, and receipts.
- `n:build:compile`: Own deterministic Rust lowering, JavaScript fallback descriptors, runtime ABI, and native link plans.

- `n:build:toolchain` : Own immutable toolchain sources, discovery, approved provisioning, isolated stages, and process execution.
- `n:build:target` : Own target registration and target-specific build providers.
- `n:build:artifact` : Own content-addressed artifact caches, manifests, integrity, and external output.
- `n:build:proof` : Own project immutability, runtime parity, and target validation evidence.
- `n:build:target:web-live` : Own verified live ESM loading and content-hash browser caching.
- `n:build:target:web-static` : Own self-contained static Web artifact materialization.
- `n:build:target:openxr` : Own shared native OpenXR session, input, frame, view, swapchain, and submission contracts.
- `n:build:target:android-xr` : Own Android ARM64 OpenXR host generation, Gradle packaging, and APK validation.
- `n:build:target:pcvr` : Own Windows x64 OpenXR host generation, executable packaging, and validation.
- `n:build:target:host-native` : Own local native and embedded Node contract proof on the build host.
- `n:camel` : Own bundled Camel execution and browser/native binding to the installed public Nexus API graph.
- `n:camera` : Own camera targets, smoothing, framing, views, projections and portable camera execution records.
- `n:camera:third-person` : Own renderer-neutral third-person camera follow descriptors.
- `n:composition` : Own deterministic Domain and Kit discovery, dependency planning, plan identity, and exactly-once apply receipts.
- `n:compute` : Own portable compute graphs, resource requirements, dispatch intent, executor hosting, and execution-family contracts.
- `n:compute:graph` : Own portable compute topology, dependencies, and deterministic plans.
- `n:compute:graph:node` : Own portable node identity, ports, operations, and requirements.
- `n:compute:graph:dependency` : Own data, control, and barrier dependencies.
- `n:compute:graph:plan` : Own deterministic validation, ordering, partitioning, and batching.
- `n:compute:resource` : Own portable compute resource requirements, access intent, and receipts.
- `n:compute:resource:buffer` : Own portable compute buffer size, usage, and access semantics.
- `n:compute:resource:image` : Own portable compute image format, view, and access semantics.
- `n:compute:resource:binding` : Own portable compute binding slots, layouts, and sets.
- `n:compute:dispatch` : Own provider-neutral execution intent, workgroups, submissions, and receipts.
- `n:compute:dispatch:workgroup` : Own portable workgroup shape and count semantics.
- `n:compute:host` : Own compute executor compatibility, deterministic selection, and lifecycle.
- `n:compute:host:capability` : Own compute-specific features, limits, requirements, and profiles.
- `n:compute:host:selection` : Own deterministic compatibility, preferences, and executor selection.
- `n:compute:host:lifecycle` : Own acquisition, readiness, recovery, and release contracts for compute executors.
- `n:compute:host:execution` : Own execution-family classification and realization of portable compute work.
- `n:compute:host:execution:gpu` : Own GPU-class compute execution semantics over Host GPU resources.
- `n:compute:host:execution:gpu:vulkan` : Own portable Vulkan compute contracts over a Host GPU environment without pretending a browser runtime exists.
- `n:compute:host:execution:gpu:opengl` : Own portable OpenGL compute contracts over a Host GPU environment without pretending a browser runtime exists.
- `n:compute:host:execution:cpu` : Own processor-class compute execution and deterministic CPU fallback.
- `n:compute:host:execution:cpu:javascript` : Own JavaScript and Worker execution of portable Compute graphs.
- `n:compute:host:execution:cpu:wasm` : Own WebAssembly compute execution, memory, SIMD, and threading contracts.
- `n:compute:host:execution:cpu:native` : Own native CPU extension contracts for threads, vector execution, and synchronization.
- `n:compute:model` : Own model descriptors, registries, inference requests/results, and provider contracts.
- `n:compute:model:inference` : Own provider-neutral inference requests, results, and compute requirements.
- `n:compute:model:inference:provider` : Own inference provider capability and contract semantics without owning runtimes.

- `n:diagnostics` : Own renderer-neutral telemetry, health, determinism, performance, replay, and debug evidence descriptors.
- `n:host` : Own host capability descriptors, fallback contracts, and shared physical execution-environment ownership while keeping backend handles provider-private.
- `n:host:gpu` : Own the shared GPU environment, portable physical-resource identity, device lifecycle, cross-consumer readiness, and recovery used by Compute and Render.
- `n:host:gpu:capability` : Own backend-neutral GPU features, limits, profiles, and compatibility requirements.
- `n:host:gpu:device` : Own portable logical GPU device identity and the device lifecycle shared by GPU consumers.
- `n:host:gpu:device:adapter` : Own portable adapter discovery and selection semantics.
- `n:host:gpu:device:logical-device` : Own logical-device acquisition, identity, generation, and state.
- `n:host:gpu:device:queue` : Own portable shared-queue submission ordering and completion receipts.
- `n:host:gpu:device:lifecycle` : Own readiness, loss, release, and generation transitions for the shared logical GPU device.
- `n:host:gpu:resource` : Own portable shared GPU-resource identity, usage, revision, residency, references, and lifetime.
- `n:host:gpu:synchronization` : Own engine-level cross-consumer ownership, readiness, transitions, and completion state for shared GPU resources.
- `n:host:gpu:recovery` : Own shared device-loss records, resource invalidation, and restoration coordination.
- `n:interaction` : Own targets, affordances, activation progress, semantic requirements, prompts, and completion results.
- `n:interaction:input` : Own semantic input actions, axes, contexts, bindings, dead zones, and adapter contracts.
- `n:interaction:assistance-target` : Own assistance target urgency, attachment, completion, loss, and deterministic selection.
- `n:interaction:environmental-affordance` : Own portable affordance proximity and activation state.
- `n:interaction:request` : Own portable request queue and fulfillment boundaries.
- `n:interaction:request:queue` : Own queued request patience, fulfillment, expiry, and effect descriptors.
- `n:interaction:request:fulfillment` : Own spatial request destination, deadline, completion, expiry, and reward state.
- `n:interaction:transfer-zone` : Own portable transfer-zone acceptance, dwell, capacity, occupancy, and completion state.
- `n:lighting` : Own portable light definitions, reflection policy, negotiation and observation records.
- `n:material` : Own visual material descriptions, parameters, instances, portable execution bindings and validation.
- `n:mcp` : Own opt-in transport-neutral MCP contracts, provider registration, authorization, and protocol dispatch.
- `n:network` : Own portable transport, session, authority, synchronization, and replication contracts.
- `n:network:transport` : Own provider-neutral transport capabilities and portable message channels.
- `n:network:multiplayer` : Own the portable multiplayer protocol boundary and deterministic inbound queue contract.
- `n:network:multiplayer:session` : Own match identity, peer readiness, and connection phases.
- `n:network:multiplayer:authority` : Own host and client roles plus state ownership declarations.
- `n:network:multiplayer:tick-sync` : Own deterministic RTT, clock-offset, drift, and remote-tick mapping records.
- `n:network:multiplayer:replication` : Own sequence numbers, acknowledgements, input frames, and snapshot envelopes.
- `n:object` : Own renderer-neutral object identity, intrinsic geometry meaning, fidelity, vegetation identity, and placement.
- `n:object:shape` : Own source and derived geometric shapes, provider jobs, qualification, and fallback.
- `n:object:fidelity` : Own valid object forms, fidelity packages, readiness, and contextual adaptation.
- `n:object:vegetation` : Own rooted plant species, instances, lifecycle, and deterministic variation.
- `n:object:vegetation:tree` : Own deterministic tree structure, canopy, growth, and fidelity descriptors.
- `n:object:vegetation:foliage` : Own deterministic foliage structure and descriptors.
- `n:object:vegetation:ecology` : Own deterministic vegetation suitability scoring and species selection.
- `n:object:placement` : Own deterministic placement transforms, grounding, alignment, fit, and validation receipts.
- `n:physics` : Own the canonical backend-neutral Physics boundary and compose its atomic capability subdomains.
- `n:physics:contracts` : Own portable Physics provider, state, command, event, and query boundary schemas.

- `n:physics:lifecycle`: Own deterministic installation, startup, stepping, shutdown, reset, and snapshot orchestration contracts.
- `n:physics:body`: Own portable provider-neutral body identity, state, mass properties, sleep state, lifecycle, and exact-once registry mutations.
- `n:physics:shape`: Own portable provider-neutral collision-shape identity, geometry descriptors, validation, and exact-once registration.
- `n:physics:material`: Own portable physical material identity, coefficients, surface classification, and deterministic pair-combine policy.
- `n:physics:collider`: Own portable collider identity, attachment, filtering, sensor semantics, lifecycle, and exact-once records.
- `n:physics:detection`: Own provider-neutral broad-phase and narrow-phase collision classification without contact or solver behavior.
- `n:physics:constraints`: Own portable constraint descriptors, exact records, lifecycle status, revisions, and break policy semantics.
- `n:physics:world`: Own portable solver-facing Physics world records, physical fields, Physics time scales, and physical simulation regions.
- `n:policy`: Own product-neutral permission, guard, sandbox, and runtime safety decisions.
- `n:presentation`: Own renderer-neutral presentation descriptors and output policy contracts.
- `n:presentation:output`: Own surface, safe-area, viewport, aspect, bar, and render-resolution policy.
- `n:presentation:animation`: Own animation, pose, blend, rig, and timeline descriptors.
- `n:presentation:audio`: Own audio cue, music, ambience, mix, volume, and spatial audio descriptors.
- `n:presentation:ui`: Own renderer-neutral HUD, menu, prompt, notification, focus, selection, accessibility, and scale descriptors.
- `n:presentation:speech`: Own provider-neutral speech requests, voices, utterance lifecycle, and synthesis result contracts.
- `n:presentation:capture`: Own observation requests, view sets, framing, capture jobs, progress, and result contracts.
- `n:presentation:sky`: Own generic sky, atmosphere, cloud, horizon, and celestial descriptors.
- `n:render`: Own the canonical backend-neutral render-execution boundary and compose its atomic capability subdomains.
- `n:render:contracts`: Own portable Render provider, resource, frame, resolved-pass, shader-interface, and event boundary schemas.
- `n:render:lifecycle`: Own provider-neutral Render composition installation, startup, shutdown, reset, snapshot, and recovery state.
- `n:render:device`: Own portable Render device contracts, capability negotiation, semantic accounting, lifecycle, loss, and diagnostics.
- `n:render:surface`: Own portable output-surface descriptors, logical regions, format choices, and deterministic transition intents.
- `n:render:resource`: Own portable Render execution-resource identity, references, semantic residency, accounting, operation receipts, and lifecycle state.
- `n:render:buffer`: Own portable logical Buffer descriptors, explicit layouts, semantic typed views, and bounded provider update receipts.
- `n:render:texture`: Own portable logical Texture descriptors, typed views, formats, mip plans, streaming records, and proven subresource residency.
- `n:render:shader`: Own provider-neutral Shader source lineage, module and program composition, variants, compile state, reflection observations, and semantic cache links.
- `n:render:pipeline`: Own portable rendering pipeline state and validation.
- `n:render:batch`: Own portable rendering batch state and validation.
- `n:render:execution`: Own realization-family semantics for portable Render work.
- `n:render:execution:gpu`: Own semantics common to GPU-class rendering over Host GPU resources.
- `n:runtime`: Own deterministic engine lifecycle, ticks, state mutation contracts, and runtime service installation.

- `n:runtime:realtime` : Own deterministic frame context and realtime phase execution.
- `n:runtime:data` : Own schemas, snapshots, selectors, migrations, deterministic random streams, and portable data envelopes.
- `n:runtime:transaction` : Own portable repeat-safe operation and transaction receipts.
- `n:runtime:persistence` : Own save/load targets, save slots, recovery records, and adapter contracts.
- `n:runtime:sequence` : Own deterministic sequence nodes, ordered execution, and frame-driven sequence state.
- `n:runtime:startup` : Own launch truth, preparation facts, continuation choice, structured failure, and readiness receipts.
- `n:runtime:sequence:schedule` : Own deterministic elapsed-time schedules and occurrence records.
- `n:runtime:data:observation` : Own bounded portable committed observation history and provenance.
- `n:simulation` : Own deterministic simulation objectives, resources, hazards, pressure, checkpoints, timers, and resolution contracts.
- `n:simulation:physics` : Own backend-neutral physical bodies, colliders, contacts, constraints, queries, and provider contracts.
- `n:simulation:physics:articulated` : Own articulated body topology, joint dynamics inputs, and backend-neutral articulation state.
- `n:simulation:motion` : Own intent-to-motion descriptors, movement modes, trajectories, velocity state, movement policies, and deterministic pose solving.
- `n:simulation:motion:articulated` : Own target poses, joint limits, articulation motion plans, and drive requests.
- `n:simulation:motion:locomotion` : Own deterministic action-to-motion intent and locomotion frame calculation.
- `n:simulation:motion:vehicle` : Own deterministic vehicle movement, boost, bounds, and impact frames.
- `n:simulation:physics:world-contact` : Own portable world-contact resolution and correction records.
- `n:simulation:recovery` : Own portable subject recovery records and deterministic recovery state.
- `n:simulation:recovery:soft-respawn` : Own exact-once portable subject recovery records.
- `n:simulation:economy` : Own portable economic account and cargo state primitives.
- `n:simulation:economy:accounts` : Own finite account balances and economy transaction records.
- `n:simulation:economy:cargo` : Own portable cargo inventory, carrying, condition, deposits, and quota state.
- `n:simulation:operations` : Own portable facility, occupant, and transport operation primitives.
- `n:simulation:operations:facility` : Own deterministic facility capacity, condition, status, cycles, and operation receipts.
- `n:simulation:operations:occupant-flow` : Own deterministic occupant spawning, patience, service, and abandonment state.
- `n:simulation:operations:transport-route` : Own deterministic transport stops, carriers, capacity, travel, and arrival receipts.
- `n:simulation:hazard-field` : Own deterministic bounded hazard state, spawning, motion, and collision queries.
- `n:simulation:pursuit-pressure` : Own coherent pursuit distance, warning bands, caught state, recovery, and transition history.
- `n:simulation:progression` : Own portable progression capability and lifecycle ownership boundaries.
- `n:simulation:progression:lifecycle` : Own prerequisite-gated lifecycle timing, completion, and portable effect descriptors.
- `n:spatial` : Own renderer-neutral transforms, coordinate spaces, bounds, zones, distance queries, and deterministic spatial math.
- `n:spatial:scale` : Own subject scale, scale anchors, proximity bands, and deterministic scale queries.
- `n:world` : Own world identity, cells, partitions, surfaces, deterministic assembly, and world state receipts.
- `n:world:scene` : Own host-neutral scene identity, lifecycle, transition, binding descriptors, and scene snapshots.
- `n:world:weather` : Own weather conditions, tendencies, regions, layers, sampling, and deterministic evolution.
- `n:world:foundation` : Own deterministic foundation contributions, composition, sampling, and cell resolution.
- `n:world:feature` : Own semantic world feature definitions, registries, lifecycle, queries, and composition.
- `n:world:feature:landform` : Own semantic elevation and landform feature descriptors.
- `n:world:feature:hydrology` : Own semantic watershed, water path, water body, and wetland feature descriptors.

- `n:world:feature:ecology` : Own semantic biome, habitat, vegetation-region, and ecotone feature descriptors.
- `n:world:feature:settlement` : Own semantic settlement, route, structure, and infrastructure feature descriptors.
- `n:world:feature:atmosphere` : Own semantic cloud, fog, wind, thermal, precipitation, and visibility feature descriptors.
- `n:world:navigation` : Own renderer-neutral navigation graphs, path queries, route fields, and landmark guidance.
- `n:world:navigation:navmesh` : Own deterministic 2D navigation cells, portals, and 3D waypoint graphs derived from walkability.
- `n:world:navigation:pathfinding` : Own deterministic path requests, A* resolution, results, and graph adapters.
- `n:world:navigation:route-field` : Own generic route marker and corridor descriptors plus deterministic proximity queries.
- `n:world:navigation:landmark-guidance` : Own reusable landmark discovery, reach, completion, priority, and proximity state.
- `n:world:generation` : Own deterministic generic region, connector, point, graph, and walkability generation.
- `n:world:terrain` : Own deterministic terrain layer evaluation, sampling, cell preparation, streaming state, and portable cell evidence.
- `n:world:water-surface` : Own renderer-neutral water zones, currents, drag, depth, wave phase, hazards, and spatial queries.

GENERATED REFERENCE

Core Dependency Table

Registry SHA-256: `147e7302e0563cdf246cc33c197ba32fec606f783817d345244544e3ce486c76`

Owner	Requires	Optional
`n:actor`	-	-
`n:actor:creature`	`n:actor`	-
`n:actor:character`	`n:actor`	-
`n:actor:player`	`n:actor:character`	-
`n:agent`	-	-
`n:asset`	-	-
`n:authoring`	`n:runtime`	-
`n:authoring:create`	`n:authoring:project`	-
`n:authoring:importing`	`n:authoring:project`	-
`n:authoring:persistence`	`n:authoring:project`	-
`n:authoring:publishing:export`	`n:authoring:project`	-
`n:authoring:validation`	`n:authoring:project`	-
`n:authoring:modifier`	`n:authoring:mesh`, `n:authoring:project`	-
`n:authoring:publishing`	`n:authoring:assembly`, `n:authoring:project`	-
`n:authoring:sequence`	`n:authoring:project`	-
`n:authoring:domain-composition`	`n:authoring:project`	-
`n:authoring:assembly`	`n:authoring:animation`, `n:authoring:material`, `n:authoring:mesh`, `n:authoring:project`, `n:authoring:rig`, `n:authoring:skin`	-
`n:authoring:animation`	`n:authoring:mesh`, `n:authoring:project`, `n:authoring:rig`, `n:authoring:skin`	-
`n:authoring:skin`	`n:authoring:brush`, `n:authoring:mesh`, `n:authoring:project`, `n:authoring:rig`	-
`n:authoring:rig`	`n:authoring:project`	-
`n:authoring:paint`	`n:authoring:brush`, `n:authoring:material`, `n:authoring:mesh`, `n:authoring:project`	-
`n:authoring:material`	`n:authoring:project`	-
`n:authoring:uv`	`n:authoring:mesh`, `n:authoring:project`	-
`n:authoring:curve`	`n:authoring:mesh`, `n:authoring:project`	-
`n:authoring:sculpt`	`n:authoring:brush`, `n:authoring:mesh`, `n:authoring:project`	-
`n:authoring:brush`	`n:authoring:project`	-
`n:authoring:workspace`	`n:authoring:project`	-
`n:authoring:project`	`n:authoring`	-

Owner	Requires	Optional
`n:authoring:mesh`	`n:authoring:project`	-
`n:authoring:editing`	`n:authoring:mesh`, `n:authoring:project`	-
`n:build`	-	-
`n:build:source`	`n:build`	-
`n:build:analysis`	`n:build`	-
`n:build:ir`	`n:build`	-
`n:build:classification`	`n:build`	-
`n:build:orchestration`	`n:build`	-
`n:build:compile`	`n:build`	-
`n:build:toolchain`	`n:build`	-
`n:build:target`	`n:build`	-
`n:build:artifact`	`n:build`	-
`n:build:proof`	`n:build`	-
`n:build:target:web-live`	`n:build:target`	-
`n:build:target:web-static`	`n:build:target`	-
`n:build:target:openxr`	`n:build:target`	-
`n:build:target:android-xr`	`n:build:target`	-
`n:build:target:pcvr`	`n:build:target`	-
`n:build:target:host-native`	`n:build:target`	-
`n:camel`	-	-
`n:camera`	-	-
`n:camera:third-person`	`character:resolution`, `motion:velocity`, `n:camera`	-
`n:composition`	-	`n:mcp`
`n:compute`	-	-
`n:compute:graph`	`n:compute`	-
`n:compute:graph:node`	`n:compute:graph`	-
`n:compute:graph:dependency`	`n:compute:graph`	-
`n:compute:graph:plan`	`n:compute:graph`	-
`n:compute:resource`	`n:compute`	-
`n:compute:resource:buffer`	`n:compute:resource`	-
`n:compute:resource:image`	`n:compute:resource`	-
`n:compute:resource:binding`	`n:compute:resource`	-
`n:compute:dispatch`	`n:compute`	-
`n:compute:dispatch:workgroup`	`n:compute:dispatch`	-
`n:compute:host`	`n:compute`, `n:host`	-
`n:compute:host:capability`	`n:compute:host`	-
`n:compute:host:selection`	`n:compute:host`	-

Owner	Requires	Optional
`n:compute:host:lifecycle`	`n:compute:host`	-
`n:compute:host:execution`	`n:compute:host`	-
`n:compute:host:execution:gpu`	`n:compute:host:execution`, `n:host:gpu`	-
`n:compute:host:execution:gpu:vulkan`	`n:compute:host:execution:gpu`, `n:host:gpu`	-
`n:compute:host:execution:gpu:opengl`	`n:compute:host:execution:gpu`, `n:host:gpu`	-
`n:compute:host:execution:cpu`	`n:compute:host:execution`	-
`n:compute:host:execution:cpu:javascript`	`n:compute:host:execution:cpu`	-
`n:compute:host:execution:cpu:wasm`	`n:compute:host:execution:cpu`	-
`n:compute:host:execution:cpu:native`	`n:compute:host:execution:cpu`	-
`n:compute:model`	`n:compute`	-
`n:compute:model:inference`	`n:compute:model`	-
`n:compute:model:inference:provider`	`n:compute:model:inference`	-
`n:diagnostics`	-	-
`n:host`	-	-
`n:host:gpu`	`n:host`	-
`n:host:gpu:capability`	`n:host:gpu`	-
`n:host:gpu:device`	`n:host:gpu`	-
`n:host:gpu:device:adapter`	`n:host:gpu:device`	-
`n:host:gpu:device:logical-device`	`n:host:gpu:device`	-
`n:host:gpu:device:queue`	`n:host:gpu:device`	-
`n:host:gpu:device:lifecycle`	`n:host:gpu:device`	-
`n:host:gpu:resource`	`n:host:gpu`	-
`n:host:gpu:synchronization`	`n:host:gpu`	-
`n:host:gpu:recovery`	`n:host:gpu`	-
`n:interaction`	-	-
`n:interaction:input`	-	-
`n:interaction:assistance-target`	`n:interaction`	-
`n:interaction:environmental-affordance`	`n:interaction`	-
`n:interaction:request`	`n:interaction`	-
`n:interaction:request:queue`	`n:interaction`	-
`n:interaction:request:fulfillment`	`n:interaction`	-
`n:interaction:transfer-zone`	`n:interaction`	-
`n:lighting`	-	-
`n:material`	`n:render:shader`, `n:render:texture`	-
`n:mcp`	-	`n:composition`
`n:network`	-	-

Owner	Requires	Optional
`n:network:transport`	`n:network`	-
`n:network:multiplayer`	`n:network`, `n:network:transport`	-
`n:network:multiplayer:session`	`n:network:multiplayer`	-
`n:network:multiplayer:authority`	`n:network:multiplayer`	-
`n:network:multiplayer:tick-sync`	`n:network:multiplayer`	-
`n:network:multiplayer:replication`	`n:network:multiplayer`	-
`n:object`	-	`n:asset`, `n:simulation:physics`
`n:object:shape`	`object:descriptor-contract`	-
`n:object:fidelity`	`object:descriptor-contract`	-
`n:object:vegetation`	`n:object`	-
`n:object:vegetation:tree`	`n:object:vegetation`	-
`n:object:vegetation:foliage`	`n:object:vegetation`	-
`n:object:vegetation:ecology`	`n:object:vegetation`	-
`n:object:placement`	`object:descriptor-contract`	-
`n:physics`	`n:runtime`	-
`n:physics:contracts`	`n:physics`	-
`n:physics:lifecycle`	`n:physics`, `physics:command-schema`, `physics:event-schema`, `physics:provider-contract`, `physics:state-schema`	-
`n:physics:body`	`n:physics`, `physics:command-schema`, `physics:event-schema`, `physics:state-schema`	-
`n:physics:shape`	`n:physics`, `physics:command-schema`, `physics:state-schema`	-
`n:physics:material`	`n:physics`, `physics:command-schema`, `physics:event-schema`, `physics:state-schema`	-
`n:physics:collider`	`n:physics`, `physics:body-registry`, `physics:command-schema`, `physics:event-schema`, `physics:material-registry`, `physics:shape-registry`, `physics:state-schema`	-
`n:physics:detection`	`n:physics`, `n:physics:collider`, `n:physics:shape`	-
`n:physics:constraints`	`n:physics`, `physics:body-registry`, `physics:command-schema`, `physics:event-schema`, `physics:state-schema`	-
`n:physics:world`	`n:physics`, `physics:command-schema`, `physics:event-schema`, `physics:state-schema`	-
`n:policy`	-	-
`n:presentation`	-	-
`n:presentation:output`	`n:presentation`	-
`n:presentation:animation`	`n:presentation`	-

Owner	Requires	Optional
`n:presentation:audio`	`n:presentation`	-
`n:presentation:ui`	`n:presentation`	-
`n:presentation:speech`	`n:presentation`	-
`n:presentation:capture`	`n:presentation`	-
`n:presentation:sky`	`n:presentation`	-
`n:render`	`n:runtime`	-
`n:render:contracts`	`n:render`	-
`n:render:lifecycle`	`n:render`, `render:provider-contract`	-
`n:render:device`	`n:render`, `render:installation`, `render:provider-contract`	-
`n:render:surface`	`n:render`, `render:device-contract`, `render:device-lifecycle`, `render:provider-contract`	-
`n:render:resource`	`n:render`, `render:device-contract`, `render:device-lifecycle`, `render:device-memory`, `render:device-queue`, `render:resource-schema`	-
`n:render:buffer`	`n:render`, `n:render:resource`, `render:device-queue`, `render:resource-identity`, `render:resource-lifecycle`	-
`n:render:texture`	`n:render`, `n:render:buffer`, `n:render:resource`, `render:buffer-resource`, `render:device-queue`, `render:resource-identity`, `render:resource-lifecycle`	-
`n:render:shader`	`n:render`, `n:render:contracts`, `n:render:device`, `n:render:resource`, `render:device-capability`, `render:device-queue`, `render:resource-identity`, `render:resource-lifecycle`, `render:shader-schema`	-
`n:render:pipeline`	`n:render`	-
`n:render:batch`	`n:render`	-
`n:render:execution`	`n:render`	-
`n:render:execution:gpu`	`n:host:gpu`, `n:render:execution`	-
`n:runtime`	-	-
`n:runtime:realtime`	`n:runtime`	-
`n:runtime:data`	`n:runtime`	-
`n:runtime:transaction`	`n:runtime`	-
`n:runtime:persistence`	`n:runtime:data`	-
`n:runtime:sequence`	`n:runtime`	-
`n:runtime:startup`	`n:runtime`	`n:asset`
`n:runtime:sequence:schedule`	`n:runtime:sequence`	-
`n:runtime:data:observation`	`n:runtime:data`	-
`n:simulation`	`n:runtime:realtime`	-

Owner	Requires	Optional
`n:simulation:physics`	`n:simulation`	-
`n:simulation:physics:articulated`	`n:simulation:physics`	-
`n:simulation:motion`	`n:simulation`	-
`n:simulation:motion:articulated`	`n:simulation:motion`	-
`n:simulation:motion:locomotion`	`n:simulation:motion`	-
`n:simulation:motion:vehicle`	`n:simulation:motion`	-
`n:simulation:physics:world-contact`	`n:simulation:physics`	-
`n:simulation:recovery`	`n:simulation`	-
`n:simulation:recovery:soft-respawn`	`n:simulation`	-
`n:simulation:economy`	`n:simulation`	-
`n:simulation:economy:accounts`	`n:simulation`, `transaction:idempotency`	-
`n:simulation:economy:cargo`	`n:simulation`	-
`n:simulation:operations`	`n:simulation`	-
`n:simulation:operations:facility`	`n:simulation`	-
`n:simulation:operations:occupant-flow`	`n:simulation`	-
`n:simulation:operations:transport-route`	`n:simulation`	-
`n:simulation:hazard-field`	`n:simulation`	-
`n:simulation:pursuit-pressure`	`n:simulation`	-
`n:simulation:progression`	`n:simulation`	-
`n:simulation:progression:lifecycle`	`n:simulation`	-
`n:spatial`	-	-
`n:spatial:scale`	`n:spatial`	-
`n:world`	`n:spatial`	-
`n:world:scene`	`n:world`	-
`n:world:weather`	`n:world`	-
`n:world:foundation`	`n:world`	-
`n:world:feature`	`n:world`	-
`n:world:feature:landform`	`n:world:feature`	-
`n:world:feature:hydrology`	`n:world:feature`	-
`n:world:feature:ecology`	`n:world:feature`	-
`n:world:feature:settlement`	`n:world:feature`	-
`n:world:feature:atmosphere`	`n:world:feature`	-
`n:world:navigation`	`n:world`	-
`n:world:navigation:navmesh`	`navigation:walkability-source`	-
`n:world:navigation:pathfinding`	`navigation:navmesh`	-
`n:world:navigation:route-field`	`n:world`	-
`n:world:navigation:landmark-guidance`	`n:world`	-

Owner	Requires	Optional
`n:world:generation`	`n:world`	-
`n:world:terrain`	`n:world`	-
`n:world:water-surface`	`n:world`	-

GENERATED REFERENCE

Generated Core API Reference

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Domains

Domain	Parent	Responsibility	Status
`n:actor`	-	Own neutral embodied actor identity and shared actor references.	stable-candidate
`n:actor:creature`	`n:actor`	Own neutral creature embodiment definitions and references.	stable-candidate
`n:actor:character`	`n:actor`	Own active embodied character identity and neutral runtime bindings.	stable-candidate
`n:actor:player`	`n:actor`	Own neutral player identity, possession, control authority, and spawn generations.	stable-candidate
`n:agent`	-	Own product-neutral observation, proposal, decision-cycle, execution receipt, and replay evidence contracts.	stable-candidate
`n:asset`	-	Own asset identity, manifests, bundles, content-addressed jobs, readiness, and provider contracts.	stable-candidate
`n:authoring`	-	Own canonical headless creation, import, editing, validation, persistence and publishing with default format implementations.	stable-candidate
`n:authoring:create`	`n:authoring`	Own canonical create operations, default providers, diagnostics and source-safe lifecycle.	stable-candidate
`n:authoring:importing`	`n:authoring`	Own canonical importing operations, default providers, diagnostics and source-safe lifecycle.	stable-candidate
`n:authoring:persistence`	`n:authoring`	Own canonical persistence operations, default providers, diagnostics and source-safe lifecycle.	stable-candidate
`n:authoring:publishing:export`	`n:authoring:publishing`	Own canonical publishing export operations, default providers, diagnostics and source-safe lifecycle.	stable-candidate
`n:authoring:validation`	`n:authoring`	Own canonical validation operations, default providers, diagnostics and source-safe lifecycle.	stable-candidate
`n:authoring:modifier`	`n:authoring`	Own modifier authoring contracts and operations.	stable-candidate

Domain	Parent	Responsibility	Status
`n:authoring:publishing`	`n:authoring`	Own publishing authoring contracts and operations.	stable-candidate
`n:authoring:sequence`	`n:authoring`	Own sequence authoring contracts and operations.	stable-candidate
`n:authoring:domain-composition`	`n:authoring`	Own domain-composition authoring contracts and operations.	stable-candidate
`n:authoring:assembly`	`n:authoring`	Own assembly authoring contracts and operations.	stable-candidate
`n:authoring:animation`	`n:authoring`	Own animation authoring contracts and operations.	stable-candidate
`n:authoring:skin`	`n:authoring`	Own skin authoring contracts and operations.	stable-candidate
`n:authoring:rig`	`n:authoring`	Own rig authoring contracts and operations.	stable-candidate
`n:authoring:paint`	`n:authoring`	Own paint authoring contracts and operations.	stable-candidate
`n:authoring:material`	`n:authoring`	Own material authoring contracts and operations.	stable-candidate
`n:authoring:uv`	`n:authoring`	Own uv authoring contracts and operations.	stable-candidate
`n:authoring:curve`	`n:authoring`	Own curve authoring contracts and operations.	stable-candidate
`n:authoring:sculpt`	`n:authoring`	Own sculpt authoring contracts and operations.	stable-candidate
`n:authoring:brush`	`n:authoring`	Own brush authoring contracts and operations.	stable-candidate
`n:authoring:workspace`	`n:authoring`	Own workspace authoring contracts and operations.	stable-candidate
`n:authoring:project`	`n:authoring`	Own editable project contracts and operations.	stable-candidate
`n:authoring:mesh`	`n:authoring`	Own editable mesh contracts and operations.	stable-candidate
`n:authoring:editing`	`n:authoring`	Own editable editing contracts and operations.	stable-candidate
`n:build`	-	Own isolated build-time source analysis, compilation, toolchains, targets, artifacts, receipts, and proof without entering application runtime composition.	stable-candidate
`n:build:source`	`n:build`	Own read-only project source, immutable dependency identities, content caches, fingerprints, and module graphs.	stable-candidate
`n:build:analysis`	`n:build`	Own real syntax, type, effect, and dependency analysis for build inputs.	stable-candidate

Domain	Parent	Responsibility	Status
`n:build:ir`	`n:build`	Own serializable Kit IR, Execution IR, validation, and source lineage maps.	stable-candidate
`n:build:classification`	`n:build`	Own whole-Kit portability, capability resolution, and fallback selection.	stable-candidate
`n:build:orchestration`	`n:build`	Own normalized requests, target sets, deterministic plans, approvals, execution, and receipts.	stable-candidate
`n:build:compile`	`n:build`	Own deterministic Rust lowering, JavaScript fallback descriptors, runtime ABI, and native link plans.	stable-candidate
`n:build:toolchain`	`n:build`	Own immutable toolchain sources, discovery, approved provisioning, isolated stages, and process execution.	stable-candidate
`n:build:target`	`n:build`	Own target registration and target-specific build providers.	stable-candidate
`n:build:artifact`	`n:build`	Own content-addressed artifact caches, manifests, integrity, and external output.	stable-candidate
`n:build:proof`	`n:build`	Own project immutability, runtime parity, and target validation evidence.	stable-candidate
`n:build:target:web-live`	`n:build:target`	Own verified live ESM loading and content-hash browser caching.	stable-candidate
`n:build:target:web-static`	`n:build:target`	Own self-contained static Web artifact materialization.	stable-candidate
`n:build:target:openxr`	`n:build:target`	Own shared native OpenXR session, input, frame, view, swapchain, and submission contracts.	stable-candidate
`n:build:target:android-xr`	`n:build:target`	Own Android ARM64 OpenXR host generation, Gradle packaging, and APK validation.	stable-candidate
`n:build:target:pcvr`	`n:build:target`	Own Windows x64 OpenXR host generation, executable packaging, and validation.	stable-candidate
`n:build:target:host-native`	`n:build:target`	Own local native and embedded Node contract proof on the build host.	stable-candidate
`n:camel`	-	Own bundled Camel execution and browser/native binding to the installed public Nexus API graph.	experimental
`n:camera`	-	Own camera targets, smoothing, framing, views, projections and portable camera execution records.	stable-candidate
`n:camera:third-person`	`n:camera`	Own renderer-neutral third-person camera follow descriptors.	stable-candidate

Domain	Parent	Responsibility	Status
`n:composition`	-	Own deterministic Domain and Kit discovery, dependency planning, plan identity, and exactly-once apply receipts.	stable-candidate
`n:compute`	-	Own portable compute graphs, resource requirements, dispatch intent, executor hosting, and execution-family contracts.	stable-candidate
`n:compute:graph`	`n:compute`	Own portable compute topology, dependencies, and deterministic plans.	stable-candidate
`n:compute:graph:node`	`n:compute:graph`	Own portable node identity, ports, operations, and requirements.	stable-candidate
`n:compute:graph:dependency`	`n:compute:graph`	Own data, control, and barrier dependencies.	stable-candidate
`n:compute:graph:plan`	`n:compute:graph`	Own deterministic validation, ordering, partitioning, and batching.	stable-candidate
`n:compute:resource`	`n:compute`	Own portable compute resource requirements, access intent, and receipts.	stable-candidate
`n:compute:resource:buffer`	`n:compute:resource`	Own portable compute buffer size, usage, and access semantics.	stable-candidate
`n:compute:resource:image`	`n:compute:resource`	Own portable compute image format, view, and access semantics.	stable-candidate
`n:compute:resource:binding`	`n:compute:resource`	Own portable compute binding slots, layouts, and sets.	stable-candidate
`n:compute:dispatch`	`n:compute`	Own provider-neutral execution intent, workgroups, submissions, and receipts.	stable-candidate
`n:compute:dispatch:workgroup`	`n:compute:dispatch`	Own portable workgroup shape and count semantics.	stable-candidate
`n:compute:host`	`n:compute`	Own compute executor compatibility, deterministic selection, and lifecycle.	stable-candidate
`n:compute:host:capability`	`n:compute:host`	Own compute-specific features, limits, requirements, and profiles.	stable-candidate
`n:compute:host:selection`	`n:compute:host`	Own deterministic compatibility, preferences, and executor selection.	stable-candidate
`n:compute:host:lifecycle`	`n:compute:host`	Own acquisition, readiness, recovery, and release contracts for compute executors.	stable-candidate
`n:compute:host:execution`	`n:compute:host`	Own execution-family classification and realization of portable compute work.	stable-candidate
`n:compute:host:execution:gpu`	`n:compute:host:execution`	Own GPU-class compute execution semantics over Host GPU resources.	stable-candidate

Domain	Parent	Responsibility	Status
`n:compute:host:execution:gpu:vulkan`	`n:compute:host:execution:gpu`	Own portable Vulkan compute contracts over a Host GPU environment without pretending a browser runtime exists.	stable-candidate
`n:compute:host:execution:gpu:opengl`	`n:compute:host:execution:gpu`	Own portable OpenGL compute contracts over a Host GPU environment without pretending a browser runtime exists.	stable-candidate
`n:compute:host:execution:cpu`	`n:compute:host:execution`	Own processor-class compute execution and deterministic CPU fallback.	stable-candidate
`n:compute:host:execution:cpu:javascript`	`n:compute:host:execution:cpu`	Own JavaScript and Worker execution of portable Compute graphs.	stable-candidate
`n:compute:host:execution:cpu:wasm`	`n:compute:host:execution:cpu`	Own WebAssembly compute execution, memory, SIMD, and threading contracts.	stable-candidate
`n:compute:host:execution:cpu:native`	`n:compute:host:execution:cpu`	Own native CPU extension contracts for threads, vector execution, and synchronization.	stable-candidate
`n:compute:model`	`n:compute`	Own model descriptors, registries, inference requests/results, and provider contracts.	stable-candidate
`n:compute:model:inference`	`n:compute:model`	Own provider-neutral inference requests, results, and compute requirements.	stable-candidate
`n:compute:model:inference:provider`	`n:compute:model:inference`	Own inference provider capability and contract semantics without owning runtimes.	stable-candidate
`n:diagnostics`	-	Own renderer-neutral telemetry, health, determinism, performance, replay, and debug evidence descriptors.	stable-candidate
`n:host`	-	Own host capability descriptors, fallback contracts, and shared physical execution-environment ownership while keeping backend handles provider-private.	stable-candidate
`n:host:gpu`	`n:host`	Own the shared GPU environment, portable physical-resource identity, device lifecycle, cross-consumer readiness, and recovery used by Compute and Render.	stable-candidate
`n:host:gpu:capability`	`n:host:gpu`	Own backend-neutral GPU features, limits, profiles, and compatibility requirements.	stable-candidate
`n:host:gpu:device`	`n:host:gpu`	Own portable logical GPU device identity and the device lifecycle shared by GPU consumers.	stable-candidate
`n:host:gpu:device:adapter`	`n:host:gpu:device`	Own portable adapter discovery and selection semantics.	stable-candidate
`n:host:gpu:device:logical-device`	`n:host:gpu:device`	Own logical-device acquisition, identity, generation, and state.	stable-candidate

Domain	Parent	Responsibility	Status
`n:host:gpu:device:queue`	`n:host:gpu:device`	Own portable shared-queue submission ordering and completion receipts.	stable-candidate
`n:host:gpu:device:lifecycle`	`n:host:gpu:device`	Own readiness, loss, release, and generation transitions for the shared logical GPU device.	stable-candidate
`n:host:gpu:resource`	`n:host:gpu`	Own portable shared GPU-resource identity, usage, revision, residency, references, and lifetime.	stable-candidate
`n:host:gpu:synchronization`	`n:host:gpu`	Own engine-level cross-consumer ownership, readiness, transitions, and completion state for shared GPU resources.	stable-candidate
`n:host:gpu:recovery`	`n:host:gpu`	Own shared device-loss records, resource invalidation, and restoration coordination.	stable-candidate
`n:interaction`	-	Own targets, affordances, activation progress, semantic requirements, prompts, and completion results.	stable-candidate
`n:interaction:input`	`n:interaction`	Own semantic input actions, axes, contexts, bindings, dead zones, and adapter contracts.	stable-candidate
`n:interaction:assistance-target`	`n:interaction`	Own assistance target urgency, attachment, completion, loss, and deterministic selection.	stable-candidate
`n:interaction:environmental-affordance`	`n:interaction`	Own portable affordance proximity and activation state.	stable-candidate
`n:interaction:request`	`n:interaction`	Own portable request queue and fulfillment boundaries.	stable-candidate
`n:interaction:request:queue`	`n:interaction:request`	Own queued request patience, fulfillment, expiry, and effect descriptors.	stable-candidate
`n:interaction:request:fulfillment`	`n:interaction:request`	Own spatial request destination, deadline, completion, expiry, and reward state.	stable-candidate
`n:interaction:transfer-zone`	`n:interaction`	Own portable transfer-zone acceptance, dwell, capacity, occupancy, and completion state.	stable-candidate
`n:lighting`	-	Own portable light definitions, reflection policy, negotiation and observation records.	stable-candidate
`n:material`	-	Own visual material descriptions, parameters, instances, portable execution bindings and validation.	stable-candidate
`n:mcp`	-	Own opt-in transport-neutral MCP contracts, provider registration, authorization, and protocol dispatch.	stable-candidate
`n:network`	-	Own portable transport, session, authority, synchronization, and replication contracts.	stable-candidate

Domain	Parent	Responsibility	Status
`n:network:transport`	`n:network`	Own provider-neutral transport capabilities and portable message channels.	stable-candidate
`n:network:multiplayer`	`n:network`	Own the portable multiplayer protocol boundary and deterministic inbound queue contract.	stable-candidate
`n:network:multiplayer:session`	`n:network:multiplayer`	Own match identity, peer readiness, and connection phases.	stable-candidate
`n:network:multiplayer:authority`	`n:network:multiplayer`	Own host and client roles plus state ownership declarations.	stable-candidate
`n:network:multiplayer:tick-sync`	`n:network:multiplayer`	Own deterministic RTT, clock-offset, drift, and remote-tick mapping records.	stable-candidate
`n:network:multiplayer:replication`	`n:network:multiplayer`	Own sequence numbers, acknowledgements, input frames, and snapshot envelopes.	stable-candidate
`n:object`	-	Own renderer-neutral object identity, intrinsic geometry meaning, fidelity, vegetation identity, and placement.	stable-candidate
`n:object:shape`	`n:object`	Own source and derived geometric shapes, provider jobs, qualification, and fallback.	stable-candidate
`n:object:fidelity`	`n:object`	Own valid object forms, fidelity packages, readiness, and contextual adaptation.	stable-candidate
`n:object:vegetation`	`n:object`	Own rooted plant species, instances, lifecycle, and deterministic variation.	stable-candidate
`n:object:vegetation:tree`	`n:object:vegetation`	Own deterministic tree structure, canopy, growth, and fidelity descriptors.	stable-candidate
`n:object:vegetation:foliage`	`n:object:vegetation`	Own deterministic foliage structure and descriptors.	stable-candidate
`n:object:vegetation:ecology`	`n:object:vegetation`	Own deterministic vegetation suitability scoring and species selection.	stable-candidate
`n:object:placement`	`n:object`	Own deterministic placement transforms, grounding, alignment, fit, and validation receipts.	stable-candidate
`n:physics`	-	Own the canonical backend-neutral Physics boundary and compose its atomic capability subdomains.	stable-candidate
`n:physics:contracts`	`n:physics`	Own portable Physics provider, state, command, event, and query boundary schemas.	stable-candidate
`n:physics:lifecycle`	`n:physics`	Own deterministic installation, startup, stepping, shutdown, reset, and snapshot orchestration contracts.	stable-candidate

Domain	Parent	Responsibility	Status
`n:physics:body`	`n:physics`	Own portable provider-neutral body identity, state, mass properties, sleep state, lifecycle, and exact-once registry mutations.	stable-candidate
`n:physics:shape`	`n:physics`	Own portable provider-neutral collision-shape identity, geometry descriptors, validation, and exact-once registration.	stable-candidate
`n:physics:material`	`n:physics`	Own portable physical material identity, coefficients, surface classification, and deterministic pair-combine policy.	stable-candidate
`n:physics:collider`	`n:physics`	Own portable collider identity, attachment, filtering, sensor semantics, lifecycle, and exact-once records.	stable-candidate
`n:physics:detection`	`n:physics`	Own provider-neutral broad-phase and narrow-phase collision classification without contact or solver behavior.	stable-candidate
`n:physics:constraints`	`n:physics`	Own portable constraint descriptors, exact records, lifecycle status, revisions, and break policy semantics.	stable-candidate
`n:physics:world`	`n:physics`	Own portable solver-facing Physics world records, physical fields, Physics time scales, and physical simulation regions.	stable-candidate
`n:policy`	-	Own product-neutral permission, guard, sandbox, and runtime safety decisions.	stable-candidate
`n:presentation`	-	Own renderer-neutral presentation descriptors and output policy contracts.	stable-candidate
`n:presentation:output`	`n:presentation`	Own surface, safe-area, viewport, aspect, bar, and render-resolution policy.	stable-candidate
`n:presentation:animation`	`n:presentation`	Own animation, pose, blend, rig, and timeline descriptors.	stable-candidate
`n:presentation:audio`	`n:presentation`	Own audio cue, music, ambience, mix, volume, and spatial audio descriptors.	stable-candidate
`n:presentation:ui`	`n:presentation`	Own renderer-neutral HUD, menu, prompt, notification, focus, selection, accessibility, and scale descriptors.	stable-candidate
`n:presentation:speech`	`n:presentation`	Own provider-neutral speech requests, voices, utterance lifecycle, and synthesis result contracts.	stable-candidate
`n:presentation:capture`	`n:presentation`	Own observation requests, view sets, framing, capture jobs, progress, and result contracts.	stable-candidate

Domain	Parent	Responsibility	Status
`n:presentation:sky`	`n:presentation`	Own generic sky, atmosphere, cloud, horizon, and celestial descriptors.	stable-candidate
`n:render`	-	Own the canonical backend-neutral render-execution boundary and compose its atomic capability subdomains.	stable-candidate
`n:render:contracts`	`n:render`	Own portable Render provider, resource, frame, resolved-pass, shader-interface, and event boundary schemas.	stable-candidate
`n:render:lifecycle`	`n:render`	Own provider-neutral Render composition installation, startup, shutdown, reset, snapshot, and recovery state.	stable-candidate
`n:render:device`	`n:render`	Own portable Render device contracts, capability negotiation, semantic accounting, lifecycle, loss, and diagnostics.	stable-candidate
`n:render:surface`	`n:render`	Own portable output-surface descriptors, logical regions, format choices, and deterministic transition intents.	stable-candidate
`n:render:resource`	`n:render`	Own portable Render execution-resource identity, references, semantic residency, accounting, operation receipts, and lifecycle state.	stable-candidate
`n:render:buffer`	`n:render`	Own portable logical Buffer descriptors, explicit layouts, semantic typed views, and bounded provider update receipts.	stable-candidate
`n:render:texture`	`n:render`	Own portable logical Texture descriptors, typed views, formats, mip plans, streaming records, and proven subresource residency.	stable-candidate
`n:render:shader`	`n:render`	Own provider-neutral Shader source lineage, module and program composition, variants, compile state, reflection observations, and semantic cache links.	stable-candidate
`n:render:pipeline`	`n:render`	Own portable rendering pipeline state and validation.	stable-candidate
`n:render:batch`	`n:render`	Own portable rendering batch state and validation.	stable-candidate
`n:render:execution`	`n:render`	Own realization-family semantics for portable Render work.	stable-candidate
`n:render:execution:gpu`	`n:render:execution`	Own semantics common to GPU-class rendering over Host GPU resources.	stable-candidate

Domain	Parent	Responsibility	Status
`n:runtime`	-	Own deterministic engine lifecycle, ticks, state mutation contracts, and runtime service installation.	stable-candidate
`n:runtime:realtime`	`n:runtime`	Own deterministic frame context and realtime phase execution.	stable-candidate
`n:runtime:data`	`n:runtime`	Own schemas, snapshots, selectors, migrations, deterministic random streams, and portable data envelopes.	stable-candidate
`n:runtime:transaction`	`n:runtime`	Own portable repeat-safe operation and transaction receipts.	stable-candidate
`n:runtime:persistence`	`n:runtime`	Own save/load targets, save slots, recovery records, and adapter contracts.	stable-candidate
`n:runtime:sequence`	`n:runtime`	Own deterministic sequence nodes, ordered execution, and frame-driven sequence state.	stable-candidate
`n:runtime:startup`	`n:runtime`	Own launch truth, preparation facts, continuation choice, structured failure, and readiness receipts.	stable-candidate
`n:runtime:sequence:schedule`	`n:runtime:sequence`	Own deterministic elapsed-time schedules and occurrence records.	stable-candidate
`n:runtime:data:observation`	`n:runtime:data`	Own bounded portable committed observation history and provenance.	stable-candidate
`n:simulation`	-	Own deterministic simulation objectives, resources, hazards, pressure, checkpoints, timers, and resolution contracts.	stable-candidate
`n:simulation:physics`	`n:simulation`	Own backend-neutral physical bodies, colliders, contacts, constraints, queries, and provider contracts.	stable-candidate
`n:simulation:physics:articulated`	`n:simulation:physics`	Own articulated body topology, joint dynamics inputs, and backend-neutral articulation state.	stable-candidate
`n:simulation:motion`	`n:simulation`	Own intent-to-motion descriptors, movement modes, trajectories, velocity state, movement policies, and deterministic pose solving.	stable-candidate
`n:simulation:motion:articulated`	`n:simulation:motion`	Own target poses, joint limits, articulation motion plans, and drive requests.	stable-candidate
`n:simulation:motion:locomotion`	`n:simulation:motion`	Own deterministic action-to-motion intent and locomotion frame calculation.	stable-candidate
`n:simulation:motion:vehicle`	`n:simulation:motion`	Own deterministic vehicle movement, boost, bounds, and impact frames.	stable-candidate
`n:simulation:physics:world-contact`	`n:simulation:physics`	Own portable world-contact resolution and correction records.	stable-candidate

Domain	Parent	Responsibility	Status
`n:simulation:recovery`	`n:simulation`	Own portable subject recovery records and deterministic recovery state.	stable-candidate
`n:simulation:recovery:soft-respawn`	`n:simulation:recovery`	Own exact-once portable subject recovery records.	stable-candidate
`n:simulation:economy`	`n:simulation`	Own portable economic account and cargo state primitives.	stable-candidate
`n:simulation:economy:accounts`	`n:simulation:economy`	Own finite account balances and economy transaction records.	stable-candidate
`n:simulation:economy:cargo`	`n:simulation:economy`	Own portable cargo inventory, carrying, condition, deposits, and quota state.	stable-candidate
`n:simulation:operations`	`n:simulation`	Own portable facility, occupant, and transport operation primitives.	stable-candidate
`n:simulation:operations:facility`	`n:simulation:operations`	Own deterministic facility capacity, condition, status, cycles, and operation receipts.	stable-candidate
`n:simulation:operations:occupant-flow`	`n:simulation:operations`	Own deterministic occupant spawning, patience, service, and abandonment state.	stable-candidate
`n:simulation:operations:transport-route`	`n:simulation:operations`	Own deterministic transport stops, carriers, capacity, travel, and arrival receipts.	stable-candidate
`n:simulation:hazard-field`	`n:simulation`	Own deterministic bounded hazard state, spawning, motion, and collision queries.	stable-candidate
`n:simulation:pursuit-pressure`	`n:simulation`	Own coherent pursuit distance, warning bands, caught state, recovery, and transition history.	stable-candidate
`n:simulation:progression`	`n:simulation`	Own portable progression capability and lifecycle ownership boundaries.	stable-candidate
`n:simulation:progression:lifecycle`	`n:simulation:progression`	Own prerequisite-gated lifecycle timing, completion, and portable effect descriptors.	stable-candidate
`n:spatial`	-	Own renderer-neutral transforms, coordinate spaces, bounds, zones, distance queries, and deterministic spatial math.	stable-candidate
`n:spatial:scale`	`n:spatial`	Own subject scale, scale anchors, proximity bands, and deterministic scale queries.	stable-candidate
`n:world`	-	Own world identity, cells, partitions, surfaces, deterministic assembly, and world state receipts.	stable-candidate
`n:world:scene`	`n:world`	Own host-neutral scene identity, lifecycle, transition, binding descriptors, and scene snapshots.	stable-candidate

Domain	Parent	Responsibility	Status
`n:world:weather`	`n:world`	Own weather conditions, tendencies, regions, layers, sampling, and deterministic evolution.	stable-candidate
`n:world:foundation`	`n:world`	Own deterministic foundation contributions, composition, sampling, and cell resolution.	stable-candidate
`n:world:feature`	`n:world`	Own semantic world feature definitions, registries, lifecycle, queries, and composition.	stable-candidate
`n:world:feature:landform`	`n:world:feature`	Own semantic elevation and landform feature descriptors.	stable-candidate
`n:world:feature:hydrology`	`n:world:feature`	Own semantic watershed, water path, water body, and wetland feature descriptors.	stable-candidate
`n:world:feature:ecology`	`n:world:feature`	Own semantic biome, habitat, vegetation-region, and ecotone feature descriptors.	stable-candidate
`n:world:feature:settlement`	`n:world:feature`	Own semantic settlement, route, structure, and infrastructure feature descriptors.	stable-candidate
`n:world:feature:atmosphere`	`n:world:feature`	Own semantic cloud, fog, wind, thermal, precipitation, and visibility feature descriptors.	stable-candidate
`n:world:navigation`	`n:world`	Own renderer-neutral navigation graphs, path queries, route fields, and landmark guidance.	stable-candidate
`n:world:navigation:navmesh`	`n:world:navigation`	Own deterministic 2D navigation cells, portals, and 3D waypoint graphs derived from walkability.	stable-candidate
`n:world:navigation:pathfinding`	`n:world:navigation`	Own deterministic path requests, A* resolution, results, and graph adapters.	stable-candidate
`n:world:navigation:route-field`	`n:world:navigation`	Own generic route marker and corridor descriptors plus deterministic proximity queries.	stable-candidate
`n:world:navigation:landmark-guidance`	`n:world:navigation`	Own reusable landmark discovery, reach, completion, priority, and proximity state.	stable-candidate
`n:world:generation`	`n:world`	Own deterministic generic region, connector, point, graph, and walkability generation.	stable-candidate
`n:world:terrain`	`n:world`	Own deterministic terrain layer evaluation, sampling, cell preparation, streaming state, and portable cell evidence.	stable-candidate
`n:world:water-surface`	`n:world`	Own renderer-neutral water zones, currents, drag, depth, wave phase, hazards, and spatial queries.	stable-candidate

Atomic Kits

Kit	Domain	Public subpath	Responsibility
`actor-registry-kit`	`n:actor`	`nexusengine/domains/actor/registry`	Own neutral actor identities and embodiment references.
`creature-registry-kit`	`n:actor:creature`	`nexusengine/domains/actor/creature`	Register and resolve neutral creature embodiment definitions.
`character-registry-kit`	`n:actor:character`	`nexusengine/domains/actor/character`	Register and resolve active embodied character descriptors.
`player-authority-kit`	`n:actor:player`	`nexusengine/domains/actor/player`	Track player identity, possession, control authority, and spawn generations.
`agent-cycle-kit`	`n:agent`	`nexusengine/domains/agent/cycle`	Record observations, action proposals, decision cycles, and execution receipts.
`asset-registry-kit`	`n:asset`	`nexusengine/domains/asset/registry`	Resolve asset manifests and bundles through content-addressed provider jobs.
`authoring-create-service-kit`	`n:authoring:create`	`nexusengine/domains/authoring/create/create-service`	Route creation requests to existing typed Project operations without owning duplicate content.
`authoring-import-registry-kit`	`n:authoring:importing`	`nexusengine/domains/authoring/importing/import-registry`	Own trusted canonical importer discovery and provider selection.
`authoring-import-validation-kit`	`n:authoring:importing`	`nexusengine/domains/authoring/importing/import-validation`	Validate a complete staged canonical import against the existing Project writer.
`authoring-import-commit-kit`	`n:authoring:importing`	`nexusengine/domains/authoring/importing/import-commit`	Commit validated imported documents atomically through Project with source guards.
`authoring-import-service-kit`	`n:authoring:importing`	`nexusengine/domains/authoring/importing/import-service`	Decode external bytes into editable documents and orchestrate staged import.
`authoring-project-package-kit`	`n:authoring:persistence`	`nexusengine/domains/authoring/persistence/project-package`	Encode and verify content-addressed Project snapshots, historical documents and image tiles.
`authoring-storage-provider-registry-kit`	`n:authoring:persistence`	`nexusengine/domains/authoring/persistence/storage-provider-registry`	Own canonical memory, filesystem and IndexedDB storage provider registration.
`authoring-save-kit`	`n:authoring:persistence`	`nexusengine/domains/authoring/persistence/save`	Persist a captured Project package with generation conflict protection.
`authoring-load-kit`	`n:authoring:persistence`	`nexusengine/domains/authoring/persistence/load`	Verify a stored package before atomically restoring the sole Project source authority.
`authoring-persistence-service-kit`	`n:authoring:persistence`	`nexusengine/domains/authoring/persistence/persistence-service`	Expose provider-neutral save and load without an Editor dependency.
`authoring-export-registry-kit`	`n:authoring:publishing:export`	`nexusengine/domains/authoring/publishing/export/export-registry`	Own canonical GLB, FBX and USDZ exporter selection and availability.

Kit	Domain	Public subpath	Responsibility
`authoring-export-capability-kit`	`n:authoring:publishing:export`	`nexusengine/domains/authoring/publishing/export/export-capability`	Inspect the exact delivery packet against a declared format profile.
`authoring-export-validation-kit`	`n:authoring:publishing:export`	`nexusengine/domains/authoring/publishing/export/export-validation`	Parse encoded artifacts and report native validation separately from external proof.
`authoring-export-receipt-kit`	`n:authoring:publishing:export`	`nexusengine/domains/authoring/publishing/export/export-receipt`	Own bounded idempotent export operation receipts; never serialize executable providers.
`authoring-export-service-kit`	`n:authoring:publishing:export`	`nexusengine/domains/authoring/publishing/export/export-service`	Orchestrate encode, native validation, source-guarded publication and export receipts.
`authoring-document-validation-kit`	`n:authoring:validation`	`nexusengine/domains/authoring/validation/document-validation`	Validate an existing document through canonical Project schemas and reference invariants.
`authoring-project-validation-kit`	`n:authoring:validation`	`nexusengine/domains/authoring/validation/project-validation`	Expose read-only validation of the complete current Project.
`authoring-delivery-validation-kit`	`n:authoring:validation`	`nexusengine/domains/authoring/validation/delivery-validation`	Expose read-only publishing preparation diagnostics.
`authoring-format-validation-kit`	`n:authoring:validation`	`nexusengine/domains/authoring/validation/format-validation`	Expose export compatibility and artifact validation through canonical providers.
`authoring-validation-report-kit`	`n:authoring:validation`	`nexusengine/domains/authoring/validation/validation-report`	Provide the public Authoring validation facade and normalized reports.
`authoring-modifier-service-kit`	`n:authoring:modifier`	`nexusengine/domains/authoring/modifier`	Own modifier source operations.
`authoring-publishing-service-kit`	`n:authoring:publishing`	`nexusengine/domains/authoring/publishing`	Own publishing source operations.
`authoring-sequence-service-kit`	`n:authoring:sequence`	`nexusengine/domains/authoring/sequence`	Own sequence source operations.
`authoring-domain-composition-service-kit`	`n:authoring:domain-composition`	`nexusengine/domains/authoring/domain-composition`	Own domain-composition source operations.
`authoring-assembly-service-kit`	`n:authoring:assembly`	`nexusengine/domains/authoring/assembly`	Own assembly source operations.
`authoring-animation-service-kit`	`n:authoring:animation`	`nexusengine/domains/authoring/animation`	Own animation source operations.
`authoring-skin-service-kit`	`n:authoring:skin`	`nexusengine/domains/authoring/skin`	Own skin source operations.
`authoring-rig-service-kit`	`n:authoring:rig`	`nexusengine/domains/authoring/rig`	Own rig source operations.
`authoring-paint-service-kit`	`n:authoring:paint`	`nexusengine/domains/authoring/paint`	Own paint source operations.
`authoring-material-service-kit`	`n:authoring:material`	`nexusengine/domains/authoring/material`	Own material source operations.
`authoring-uv-service-kit`	`n:authoring:uv`	`nexusengine/domains/authoring/uv`	Own uv source operations.

Kit	Domain	Public subpath	Responsibility
`authoring-curve-service-kit`	`n:authoring:curve`	`nexusengine/domains/authoring/curve`	Own curve source operations.
`authoring-sculpt-service-kit`	`n:authoring:sculpt`	`nexusengine/domains/authoring/sculpt`	Own sculpt source operations.
`authoring-brush-service-kit`	`n:authoring:brush`	`nexusengine/domains/authoring/brush`	Own brush source operations.
`authoring-workspace-service-kit`	`n:authoring:workspace`	`nexusengine/domains/authoring/workspace`	Own workspace source operations.
`authoring-domain-contract-kit`	`n:authoring`	`nexusengine/domains/authoring/contract`	Own contract Authoring contracts and operations.
`authoring-project-document-kit`	`n:authoring:project`	`nexusengine/domains/authoring/project`	Own project Authoring contracts and operations.
`authoring-mesh-document-kit`	`n:authoring:mesh`	`nexusengine/domains/authoring/mesh`	Own mesh Authoring contracts and operations.
`authoring-editing-session-kit`	`n:authoring:editing`	`nexusengine/domains/authoring/editing`	Own editing Authoring contracts and operations.
`host-native-target-kit`	`n:build:target:host-native`	`nexusengine/domains/build/target/host-native/host-native-target`	Compile Rust and embedded Node contract bindings for local host proof; no redistributable package claim.
`project-source-kit`	`n:build:source`	`nexusengine/domains/build/source/project-source`	Read a deterministic project inventory without following links or mutating source.
`source-fingerprint-kit`	`n:build:source`	`nexusengine/domains/build/source/source-fingerprint`	Create the canonical SHA-256 project fingerprint.
`dependency-source-kit`	`n:build:source`	`nexusengine/domains/build/source/dependency-source`	Resolve exact dependency source identities and recursive lockfile closure.
`source-cache-kit`	`n:build:source`	`nexusengine/domains/build/source/source-cache`	Store and verify immutable source bytes by SHA-256.
`module-graph-kit`	`n:build:source`	`nexusengine/domains/build/source/module-graph`	Build a deterministic AST-derived module graph.
`javascript-ast-kit`	`n:build:analysis`	`nexusengine/domains/build/analysis/javascript-ast`	Parse JavaScript and TypeScript with a real compiler AST.
`type-analysis-kit`	`n:build:analysis`	`nexusengine/domains/build/analysis/type-analysis`	Run typed compiler diagnostics without emitting or mutating source.
`effect-analysis-kit`	`n:build:analysis`	`nexusengine/domains/build/analysis/effect-analysis`	Classify ambient capabilities and unsupported dynamic effects from AST nodes.
`dependency-analysis-kit`	`n:build:analysis`	`nexusengine/domains/build/analysis/dependency-analysis`	Prove relative and external dependency closure.
`kit-ir-kit`	`n:build:ir`	`nexusengine/domains/build/ir/kit-ir`	Create serializable high-level Kit IR with source lineage.
`execution-ir-kit`	`n:build:ir`	`nexusengine/domains/build/ir/execution-ir`	Create deterministic dependency-ordered Execution IR.
`ir-validation-kit`	`n:build:ir`	`nexusengine/domains/build/ir/ir-validation`	Reject invalid, cyclic, unsupported, or incomplete IR.

Kit	Domain	Public subpath	Responsibility
`source-map-kit`	`n:build:ir`	`nexusengine/domains/build/ir/source-map`	Map generated execution operations to source AST identities.
`portability-classifier-kit`	`n:build:classification`	`nexusengine/domains/build/classification/portability-classifier`	Classify each whole module and composition as native, native-adapter, JavaScript, or unsupported.
`capability-resolution-kit`	`n:build:classification`	`nexusengine/domains/build/classification/capability-resolution`	Resolve target capabilities and exact reviewed substitutions.
`fallback-selection-kit`	`n:build:classification`	`nexusengine/domains/build/classification/fallback-selection`	Select fail-closed whole-Kit fallback per target and profile.
`build-request-kit`	`n:build:orchestration`	`nexusengine/domains/build/orchestration/build-request`	Normalize one project, profile, options, and target set.
`target-set-kit`	`n:build:orchestration`	`nexusengine/domains/build/orchestration/target-set`	Normalize repeated target flags into one sorted unique set.
`build-plan-kit`	`n:build:orchestration`	`nexusengine/domains/build/orchestration/build-plan`	Create the immutable deterministic multi-target plan hash.
`build-approval-kit`	`n:build:orchestration`	`nexusengine/domains/build/orchestration/build-approval`	Require approval for the exact unchanged plan hash.
`build-execution-kit`	`n:build:orchestration`	`nexusengine/domains/build/orchestration/build-execution`	Execute shared stages once and isolated target stages with project immutability proof.
`build-receipt-kit`	`n:build:orchestration`	`nexusengine/domains/build/orchestration/build-receipt`	Persist aggregate and per-target exactly-once Build receipts.
`rust-lowering-kit`	`n:build:compile`	`nexusengine/domains/build/compile/rust-lowering`	Lower supported Execution IR into deterministic Rust source.
`javascript-fallback-kit`	`n:build:compile`	`nexusengine/domains/build/compile/javascript-fallback`	Execute capability-restricted whole-Kit QuickJS-NG fallback.
`runtime-abi-kit`	`n:build:compile`	`nexusengine/domains/build/compile/runtime-abi`	Define the stable native runtime handle and batch-operation ABI.
`native-runtime-link-kit`	`n:build:compile`	`nexusengine/domains/build/compile/native-runtime-link`	Create exact generated-runtime and native-library link plans.
`toolchain-source-kit`	`n:build:toolchain`	`nexusengine/domains/build/toolchain/toolchain-source`	Own immutable official toolchain and native dependency source records.
`toolchain-discovery-kit`	`n:build:toolchain`	`nexusengine/domains/build/toolchain/toolchain-discovery`	Discover installed toolchains without shell evaluation.
`toolchain-provision-kit`	`n:build:toolchain`	`nexusengine/domains/build/toolchain/toolchain-provision`	Provision approved exact official sources on demand after integrity and license checks.
`isolated-stage-kit`	`n:build:toolchain`	`nexusengine/domains/build/toolchain/isolated-stage`	Create content-addressed build stages outside projects.
`process-execution-kit`	`n:build:toolchain`	`nexusengine/domains/build/toolchain/process-execution`	Run argument-array commands inside an allowed Build stage.
`target-registry-kit`	`n:build:target`	`nexusengine/domains/build/target/target-registry`	Register explicit target providers and reject collisions.

Kit	Domain	Public subpath	Responsibility
`web-live-target-kit`	`n:build:target:web-live`	`nexusengine/domains/build/target/web-live/web-live-target`	Emit verified live ESM source, loader, service worker, and cache policy.
`web-static-target-kit`	`n:build:target:web-static`	`nexusengine/domains/build/target/web-static/web-static-target`	Emit a self-contained static Web directory.
`openxr-runtime-kit`	`n:build:target:openxr`	`nexusengine/domains/build/target/openxr/openxr-runtime`	Own OpenXR loader, session, spaces, frame timing, and lifecycle contracts.
`openxr-input-kit`	`n:build:target:openxr`	`nexusengine/domains/build/target/openxr/openxr-input`	Own OpenXR action sets, bindings, haptics, and input snapshots.
`openxr-render-kit`	`n:build:target:openxr`	`nexusengine/domains/build/target/openxr/openxr-render`	Own OpenXR views, swapchains, blend modes, and per-eye submission descriptors.
`android-xr-target-kit`	`n:build:target:android-xr`	`nexusengine/domains/build/target/android-xr/android-xr-target`	Own Android ARM64 lifecycle, SDK/NDK, Gradle, APK, and OpenXR binding stages.
`pcvr-target-kit`	`n:build:target:pcvr`	`nexusengine/domains/build/target/pcvr/pcvr-target`	Own Windows x64 host, OpenXR loader, executable package, and runtime validation stages.
`artifact-cache-kit`	`n:build:artifact`	`nexusengine/domains/build/artifact/artifact-cache`	Reuse successful immutable target artifacts by plan identity.
`artifact-manifest-kit`	`n:build:artifact`	`nexusengine/domains/build/artifact/artifact-manifest`	Create canonical per-target artifact manifests.
`artifact-integrity-kit`	`n:build:artifact`	`nexusengine/domains/build/artifact/artifact-integrity`	Verify every artifact file against SHA-256.
`artifact-output-kit`	`n:build:artifact`	`nexusengine/domains/build/artifact/artifact-output`	Publish immutable artifacts only outside source projects.
`project-immutability-kit`	`n:build:proof`	`nexusengine/domains/build/proof/project-immutability`	Compare before and after project fingerprints byte-for-byte.
`cross-runtime-parity-kit`	`n:build:proof`	`nexusengine/domains/build/proof/cross-runtime-parity`	Compare canonical replay outputs across target runtimes.
`target-validation-kit`	`n:build:proof`	`nexusengine/domains/build/proof/target-validation`	Require target-specific executable artifact validation.
`web-module-linker-kit`	`n:build:compile`	`nexusengine/domains/build/compile/web-module-linker`	Materialize a verified, content-addressed browser module closure from immutable project sources.
`camel-runtime-kit`	`n:camel`	`nexusengine/domains/camel/runtime`	Execute a pinned bundled Camel interpreter against shared Nexus instances.
`camera-descriptor-kit`	`n:camera`	`nexusengine/domains/camera/control`	Manage camera targets, modes, smoothing, and renderer-neutral camera policy.
`camera-framing-kit`	`n:camera`	`nexusengine/domains/camera/framing`	Calculate perspective and orthographic subject framing for a viewport.
`camera-control-math-kit`	`n:camera`	`nexusengine/domains/camera/control-math`	Calculate camera-relative yaw, orbit limits, and root handoff state.

Kit	Domain	Public subpath	Responsibility
`third-person-camera-kit`	`n:camera:third-person`	`nexusengine/domains/camera/third-person`	Produce deterministic renderer-neutral third-person camera descriptors from public Character and Motion bindings.
`camera-world-occlusion-adapter-kit`	`n:camera:third-person`	`nexusengine/domains/camera/adapters/camera-world-occlusion`	Constrain Third-Person Camera descriptors with public terrain and physics query results without owning camera or world state.
`camera-binding-kit`	`n:camera`	`nexusengine/domains/camera/execution/camera-binding`	Own portable camera binding semantics and deterministic state.
`camera-jitter-kit`	`n:camera`	`nexusengine/domains/camera/execution/camera-jitter`	Own portable camera jitter semantics and deterministic state.
`camera-projection-kit`	`n:camera`	`nexusengine/domains/camera/execution/camera-projection`	Own portable camera projection semantics and deterministic state.
`camera-reprojection-kit`	`n:camera`	`nexusengine/domains/camera/execution/camera-reprojection`	Own portable camera reprojection semantics and deterministic state.
`camera-view-kit`	`n:camera`	`nexusengine/domains/camera/execution/camera-view`	Own portable camera view semantics and deterministic state.
`camera-viewport-kit`	`n:camera`	`nexusengine/domains/camera/execution/camera-viewport`	Own portable camera viewport semantics and deterministic state.
`multiview-camera-kit`	`n:camera`	`nexusengine/domains/camera/execution/multiview-camera`	Own portable multiview camera semantics and deterministic state.
`stereo-camera-kit`	`n:camera`	`nexusengine/domains/camera/execution/stereo-camera`	Own portable stereo camera semantics and deterministic state.
`composition-registry-kit`	`n:composition`	`nexusengine/domains/composition/registry`	Maintain normalized composition metadata and produce deterministic plans and receipts.
`compute-graph-kit`	`n:compute`	`nexusengine/domains/compute/graph`	Validate compute descriptors, create deterministic dependency-ordered dispatch plans, and execute them through an injected provider or Compute Host.
`model-registry-kit`	`n:compute:model`	`nexusengine/domains/compute/model`	Register model descriptors and normalize provider-neutral inference requests and results.
`diagnostics-kit`	`n:diagnostics`	`nexusengine/domains/diagnostics/runtime`	Collect serializable telemetry, runtime health, determinism, and performance evidence.
`debug-descriptor-kit`	`n:diagnostics`	`nexusengine/domains/diagnostics/debug`	Record renderer-neutral rays, markers, scalars, and capture packets for diagnostics.
`debug-draw-descriptor-kit`	`n:diagnostics`	`nexusengine/domains/diagnostics/debug-draw`	Create stateless renderer-neutral debug draw descriptors.
`host-capability-kit`	`n:host`	`nexusengine/domains/host/capabilities`	Describe available host capabilities and select declarative fallback modes.
`interaction-kit`	`n:interaction`	`nexusengine/domains/interaction/runtime`	Manage interaction targets, affordances, activation, and results.
`input-contract-kit`	`n:interaction:input`	`nexusengine/domains/interaction/input`	Normalize semantic input actions, axes, contexts, and bindings.

Kit	Domain	Public subpath	Responsibility
`assistance-target-kit`	`n:interaction:assistance-target`	`nexusengine/domains/interaction/assistance-target`	Own assistance target urgency, attachment, terminal completion, loss, and deterministic selection.
`environmental-affordance-kit`	`n:interaction:environmental-affordance`	`nexusengine/domains/interaction/environmental-affordance`	Own read-only affordance proximity queries and exact-once activation progress.
`request-queue-kit`	`n:interaction:request:queue`	`nexusengine/domains/interaction/request/queue`	Own deterministic queued requests, patience, fulfillment, expiry, and portable effect descriptors.
`request-fulfillment-kit`	`n:interaction:request:fulfillment`	`nexusengine/domains/interaction/request/fulfillment`	Own spatial request destinations, deadlines, completion, expiry, and reward totals.
`transfer-zone-kit`	`n:interaction:transfer-zone`	`nexusengine/domains/interaction/transfer-zone`	Own accepted types, dwell, capacity, occupancy, and exact-once transfer completions.
`occupant-request-adapter-kit`	`n:interaction:request:queue`	`nexusengine/domains/interaction/adapters/occupant-request`	Translate Occupant Flow need records into exact-once Request Queue entries.
`transport-request-adapter-kit`	`n:interaction:request:queue`	`nexusengine/domains/interaction/adapters/transport-request`	Translate Transport Route arrivals into exact-once Request Queue fulfillment commands.
`request-economy-adapter-kit`	`n:interaction:request:queue`	`nexusengine/domains/interaction/adapters/request-economy`	Translate fulfilled or expired Request Queue outcomes into exact-once Economy transactions.
`lighting-descriptor-kit`	`n:lighting`	`nexusengine/domains/lighting/descriptions`	Own portable lighting descriptions.
`reflection-descriptor-kit`	`n:lighting`	`nexusengine/domains/lighting/reflection`	Describe renderer-neutral reflection probes, plans, and update policy.
`material-contract-kit`	`n:material`	`nexusengine/domains/material/contract`	Define the canonical backend-neutral Render Material execution boundary.
`material-binding-kit`	`n:material`	`nexusengine/domains/material/binding`	Map portable Material slots to one exact Shader program interface.
`material-parameter-kit`	`n:material`	`nexusengine/domains/material/parameter`	Own typed portable values for exact Material parameter slots.
`texture-binding-kit`	`n:material`	`nexusengine/domains/material/texture-binding`	Bind exact resident Texture views and subresources to Material slots.
`sampler-binding-kit`	`n:material`	`nexusengine/domains/material/sampler-binding`	Own portable sampler state for exact Material sampler slots.
`material-instance-kit`	`n:material`	`nexusengine/domains/material/instance`	Compose one complete portable Material execution instance from exact bindings.
`material-variant-kit`	`n:material`	`nexusengine/domains/material/variant`	Resolve an exact Shader variant and complete Material binding override set.
`material-validation-kit`	`n:material`	`nexusengine/domains/material/validation`	Prove one Material target against exact completed Shader compile and reflection records.

Kit	Domain	Public subpath	Responsibility
`material-cache-kit`	`n:material`	`nexusengine/domains/material/cache`	Link current Material validation to an exact resident material Render Resource.
`material-descriptor-kit`	`n:material`	`nexusengine/domains/material/descriptions`	Own portable material descriptions.
`mcp-registry-kit`	`n:mcp`	`nexusengine/domains/mcp/registry`	Register and dispatch schema-valid MCP providers through an explicit authorization boundary.
`network-contract-kit`	`n:network`	`nexusengine/domains/network/contracts`	Describe portable network messages and synchronization without implementing transport.
`network-transport-contract-kit`	`n:network:transport`	`nexusengine/domains/network/transport`	Validate provider-neutral transport capabilities and portable channel messages.
`multiplayer-contract-kit`	`n:network:multiplayer`	`nexusengine/domains/network/multiplayer`	Own the protocol version and deterministic inbound queue contract.
`multiplayer-session-kit`	`n:network:multiplayer:session`	`nexusengine/domains/network/multiplayer/session`	Manage portable session phases, peer readiness, and disconnect state.
`multiplayer-authority-kit`	`n:network:multiplayer:authority`	`nexusengine/domains/network/multiplayer/authority`	Enforce one authoritative host and portable state ownership declarations.
`multiplayer-tick-sync-kit`	`n:network:multiplayer:tick-sync`	`nexusengine/domains/network/multiplayer/tick-sync`	Map explicit remote timing samples onto deterministic local ticks.
`multiplayer-replication-kit`	`n:network:multiplayer:replication`	`nexusengine/domains/network/multiplayer/replication`	Create sequence-numbered input and snapshot envelopes and reject stale packets.
`object-registry-kit`	`n:object`	`nexusengine/domains/object/registry`	Own object identity and renderer-neutral lifecycle records.
`object-shape-kit`	`n:object:shape`	`nexusengine/domains/object/shape`	Derive and qualify renderer-neutral geometric shape candidates.
`object-fidelity-kit`	`n:object:fidelity`	`nexusengine/domains/object/fidelity`	Package and select valid object fidelity forms.
`object-vegetation-kit`	`n:object:vegetation`	`nexusengine/domains/object/vegetation`	Own deterministic plant species, instances, and lifecycle state.
`object-tree-kit`	`n:object:vegetation:tree`	`nexusengine/domains/object/vegetation/tree`	Produce deterministic tree structure, canopy, growth, and fidelity descriptors.
`object-foliage-kit`	`n:object:vegetation:foliage`	`nexusengine/domains/object/vegetation/foliage`	Produce deterministic foliage structures and descriptors.
`object-vegetation-ecology-kit`	`n:object:vegetation:ecology`	`nexusengine/domains/object/vegetation/ecology`	Score vegetation suitability and select species deterministically.
`object-placement-kit`	`n:object:placement`	`nexusengine/domains/object/placement`	Create, validate, and replay deterministic object placement receipts.
`object-meshoptimizer-shape-provider-kit`	`n:object:shape`	`nexusengine/domains/object/shape/meshoptimizer-provider`	Resolve object shape jobs through an explicitly registered meshoptimizer-compatible provider.

Kit	Domain	Public subpath	Responsibility
`object-shape-fidelity-adapter-kit`	`n:object:fidelity`	`nexusengine/domains/object/adapters/shape-fidelity`	Translate qualified shape records into Object Fidelity form requests.
`object-vegetation-bridge-kit`	`n:object:vegetation`	`nexusengine/domains/object/adapters/vegetation-object`	Project vegetation identities into canonical Object descriptors without owning either state.
`physics-domain-contract-kit`	`n:physics`	`nexusengine/domains/physics/contract`	Expose the canonical backend-neutral Physics ownership and contract boundary.
`physics-provider-contract-kit`	`n:physics:contracts`	`nexusengine/domains/physics/provider-contract`	Describe and validate backend Physics providers without owning a concrete solver.
`physics-state-schema-kit`	`n:physics:contracts`	`nexusengine/domains/physics/state-schema`	Validate and normalize portable Physics snapshots for deterministic replay.
`physics-command-schema-kit`	`n:physics:contracts`	`nexusengine/domains/physics/command-schema`	Define deterministic, exact-once Physics command envelopes.
`physics-event-schema-kit`	`n:physics:contracts`	`nexusengine/domains/physics/event-schema`	Define ordered, portable Physics event envelopes.
`physics-query-schema-kit`	`n:physics:contracts`	`nexusengine/domains/physics/query-schema`	Define read-only portable Physics query request and result envelopes.
`physics-installation-kit`	`n:physics:lifecycle`	`nexusengine/domains/physics/lifecycle/installation`	Own the aggregate phase and provider identity for one installed Physics composition.
`physics-startup-kit`	`n:physics:lifecycle`	`nexusengine/domains/physics/lifecycle/startup`	Own deterministic startup requests and provider-readiness receipts.
`physics-step-kit`	`n:physics:lifecycle`	`nexusengine/domains/physics/lifecycle/step`	Own deterministic Physics step requests, completion ordering, and provider-neutral frame receipts.
`physics-shutdown-kit`	`n:physics:lifecycle`	`nexusengine/domains/physics/lifecycle/shutdown`	Own deterministic provider shutdown requests and completion receipts.
`physics-reset-kit`	`n:physics:lifecycle`	`nexusengine/domains/physics/lifecycle/reset`	Reset composed Physics lifecycle state atomically through public capability APIs.
`physics-snapshot-kit`	`n:physics:lifecycle`	`nexusengine/domains/physics/lifecycle/snapshot`	Capture and atomically restore portable snapshots of composed Physics lifecycle state.
`body-identity-kit`	`n:physics:body`	`nexusengine/domains/physics/body/identity`	Normalize stable portable Physics body identity, tags, and metadata.
`body-type-kit`	`n:physics:body`	`nexusengine/domains/physics/body/type`	Normalize static, dynamic, and kinematic Physics body modes.
`body-pose-kit`	`n:physics:body`	`nexusengine/domains/physics/body/pose`	Normalize body position and canonical quaternion orientation.
`body-velocity-kit`	`n:physics:body`	`nexusengine/domains/physics/body/velocity`	Normalize finite linear and angular Physics body velocity descriptors.
`body-force-kit`	`n:physics:body`	`nexusengine/domains/physics/body/force`	Normalize portable force, torque, and impulse accumulator descriptors.

Kit	Domain	Public subpath	Responsibility
`body-mass-kit`	`n:physics:body`	`nexusengine/domains/physics/body/mass`	Normalize body mass, inverse mass, and center-of-mass descriptors.
`body-inertia-kit`	`n:physics:body`	`nexusengine/domains/physics/body/inertia`	Normalize principal inertia, inverse inertia, and local inertia orientation.
`body-damping-kit`	`n:physics:body`	`nexusengine/domains/physics/body/damping`	Normalize finite nonnegative linear and angular damping descriptors.
`body-sleep-kit`	`n:physics:body`	`nexusengine/domains/physics/body/sleep`	Normalize body sleep state and explicit exact-once sleep commands.
`body-wake-kit`	`n:physics:body`	`nexusengine/domains/physics/body/wake`	Normalize explicit exact-once Physics body wake commands.
`body-lifecycle-kit`	`n:physics:body`	`nexusengine/domains/physics/body/lifecycle`	Normalize active and disabled body lifecycle state and exact transition commands.
`body-state-kit`	`n:physics:body`	`nexusengine/domains/physics/body/state`	Compose atomic portable body descriptors into one coherent provider-neutral body state.
`body-registry-kit`	`n:physics:body`	`nexusengine/domains/physics/body/registry`	Own portable Physics body records and exact-once lifecycle transitions without solver execution.
`shape-identity-kit`	`n:physics:shape`	`nexusengine/domains/physics/shape/identity`	Normalize stable portable Physics shape identity, type, and metadata.
`shape-validation-kit`	`n:physics:shape`	`nexusengine/domains/physics/shape/validation`	Validate any canonical portable Physics shape descriptor without mutation.
`sphere-shape-kit`	`n:physics:shape`	`nexusengine/domains/physics/shape/sphere`	Normalize portable sphere collision geometry.
`box-shape-kit`	`n:physics:shape`	`nexusengine/domains/physics/shape/box`	Normalize portable box collision geometry.
`capsule-shape-kit`	`n:physics:shape`	`nexusengine/domains/physics/shape/capsule`	Normalize portable capsule collision geometry.
`cylinder-shape-kit`	`n:physics:shape`	`nexusengine/domains/physics/shape/cylinder`	Normalize portable cylinder collision geometry.
`cone-shape-kit`	`n:physics:shape`	`nexusengine/domains/physics/shape/cone`	Normalize portable cone collision geometry.
`plane-shape-kit`	`n:physics:shape`	`nexusengine/domains/physics/shape/plane`	Normalize portable infinite-plane collision geometry.
`convex-shape-kit`	`n:physics:shape`	`nexusengine/domains/physics/shape/convex`	Normalize portable convex-hull collision geometry.
`triangle-mesh-shape-kit`	`n:physics:shape`	`nexusengine/domains/physics/shape/triangle-mesh`	Normalize indexed portable triangle-mesh collision geometry.
`heightfield-shape-kit`	`n:physics:shape`	`nexusengine/domains/physics/shape/heightfield`	Normalize sampled portable heightfield collision geometry.
`compound-shape-kit`	`n:physics:shape`	`nexusengine/domains/physics/shape/compound`	Normalize portable compound collision-shape references and local poses.

Kit	Domain	Public subpath	Responsibility
`scaled-shape-kit`	`n:physics:shape`	`nexusengine/domains/physics/shape/scaled`	Normalize portable positive nonuniform scaling of a referenced collision shape.
`shape-registry-kit`	`n:physics:shape`	`nexusengine/domains/physics/shape/registry`	Own exact-once deterministic registration and lookup of portable Physics shapes.
`friction-material-kit`	`n:physics:material`	`nexusengine/domains/physics/material/friction`	Normalize portable isotropic and anisotropic physical friction descriptors.
`restitution-material-kit`	`n:physics:material`	`nexusengine/domains/physics/material/restitution`	Normalize physical restitution coefficient and activation-threshold descriptors.
`density-material-kit`	`n:physics:material`	`nexusengine/domains/physics/material/density`	Normalize positive SI physical mass-density descriptors.
`surface-material-kit`	`n:physics:material`	`nexusengine/domains/physics/material/surface`	Normalize renderer-neutral physical surface classification and tags.
`material-combine-policy-kit`	`n:physics:material`	`nexusengine/domains/physics/material/combine-policy`	Resolve physical material pairs with deterministic symmetric coefficient-combine policy.
`physics-material-kit`	`n:physics:material`	`nexusengine/domains/physics/material/registry`	Own immutable portable physical material records and exact-once registry mutations.
`collider-identity-kit`	`n:physics:collider`	`nexusengine/domains/physics/collider/identity`	Normalize stable portable Physics collider identity, tags, and metadata.
`collider-attachment-kit`	`n:physics:collider`	`nexusengine/domains/physics/collider/attachment`	Normalize a collider attachment to public Body and Shape registry identities.
`collider-pose-kit`	`n:physics:collider`	`nexusengine/domains/physics/collider/pose`	Normalize provider-neutral collider-local position and orientation descriptors.
`collider-material-kit`	`n:physics:collider`	`nexusengine/domains/physics/collider/material`	Normalize a collider reference to one public Physics material identity.
`collision-layer-kit`	`n:physics:collider`	`nexusengine/domains/physics/collider/layer`	Normalize one bounded provider-neutral collision layer.
`collision-mask-kit`	`n:physics:collider`	`nexusengine/domains/physics/collider/mask`	Normalize a deterministic bounded collision-layer set and its portable bit value.
`collision-group-kit`	`n:physics:collider`	`nexusengine/domains/physics/collider/group`	Normalize a named collision layer-and-mask policy descriptor.
`collider-filter-kit`	`n:physics:collider`	`nexusengine/domains/physics/collider/filter`	Normalize provider-neutral collider layer, mask, group, and exclusion descriptors.
`sensor-collider-kit`	`n:physics:collider`	`nexusengine/domains/physics/collider/sensor`	Normalize non-solving sensor semantics independently from collision detection and event dispatch.

Kit	Domain	Public subpath	Responsibility
`trigger-collider-kit`	`n:physics:collider`	`nexusengine/domains/physics/collider/trigger`	Normalize event-selection semantics for a sensor-backed trigger collider.
`collider-lifecycle-kit`	`n:physics:collider`	`nexusengine/domains/physics/collider/lifecycle`	Normalize provider-neutral enabled and disabled collider lifecycle state and commands.
`collider-registry-kit`	`n:physics:collider`	`nexusengine/domains/physics/collider/registry`	Own portable collider records, revisions, reference validation, and exact-once mutations.
`collision-detection-result-kit`	`n:physics:detection`	`nexusengine/domains/physics/detection/result`	Normalize finite portable collision results and stable result ordering.
`broad-phase-pair-kit`	`n:physics:detection`	`nexusengine/domains/physics/detection/broad-phase-pair`	Normalize, filter, deduplicate, and stably order broad-phase pairs.
`spatial-partition-kit`	`n:physics:detection`	`nexusengine/domains/physics/detection/spatial-partition`	Own exact-once portable broad-phase proxy records and deterministic bounds queries.
`dynamic-tree-kit`	`n:physics:detection`	`nexusengine/domains/physics/detection/dynamic-tree`	Build and query deterministic immutable AABB trees from portable proxies.
`sweep-and-prune-kit`	`n:physics:detection`	`nexusengine/domains/physics/detection/sweep-and-prune`	Generate deterministic broad-phase pairs by sorted-axis interval sweeping.
`shape-intersection-kit`	`n:physics:detection`	`nexusengine/domains/physics/detection/shape-intersection`	Resolve exact analytic primitive and convex-plane shape intersections.
`gjk-detection-kit`	`n:physics:detection`	`nexusengine/domains/physics/detection/gjk`	Determine convex support-shape separation or intersection with deterministic GJK simplex evolution.
`epa-penetration-kit`	`n:physics:detection`	`nexusengine/domains/physics/detection/epa`	Expand an intersecting GJK simplex into deterministic convex penetration witnesses.
`continuous-collision-kit`	`n:physics:detection`	`nexusengine/domains/physics/detection/continuous-collision`	Compute exact linear sphere-sphere time of impact and reject unsupported sweep pairs.
`narrow-phase-kit`	`n:physics:detection`	`nexusengine/domains/physics/detection/narrow-phase`	Dispatch supported analytic and convex algorithms into one portable collision result.
`broad-phase-kit`	`n:physics:detection`	`nexusengine/domains/physics/detection/broad-phase`	Own canonical Detection discovery and deterministic broad-phase strategy selection.
`ball-socket-constraint-kit`	`n:physics:constraints`	`nexusengine/domains/physics/constraints/ball-socket`	Normalize portable ball-socket constraint descriptors without provider or solver execution.
`cone-twist-constraint-kit`	`n:physics:constraints`	`nexusengine/domains/physics/constraints/cone-twist`	Normalize portable cone-twist constraint descriptors without provider or solver execution.
`distance-constraint-kit`	`n:physics:constraints`	`nexusengine/domains/physics/constraints/distance`	Normalize portable bounded-distance constraint descriptors without provider or solver execution.

Kit	Domain	Public subpath	Responsibility
`drive-constraint-kit`	`n:physics:constraints`	`nexusengine/domains/physics/constraints/drive`	Normalize portable positional and velocity drive constraint descriptors.
`fixed-constraint-kit`	`n:physics:constraints`	`nexusengine/domains/physics/constraints/fixed`	Normalize portable fixed constraint descriptors without provider or solver execution.
`hinge-constraint-kit`	`n:physics:constraints`	`nexusengine/domains/physics/constraints/hinge`	Normalize portable local-axis hinge constraint descriptors.
`limit-constraint-kit`	`n:physics:constraints`	`nexusengine/domains/physics/constraints/limit`	Normalize portable linear and angular limit constraint descriptors.
`motor-constraint-kit`	`n:physics:constraints`	`nexusengine/domains/physics/constraints/motor`	Normalize portable bounded motor constraint descriptors.
`slider-constraint-kit`	`n:physics:constraints`	`nexusengine/domains/physics/constraints/slider`	Normalize portable local-axis slider constraint descriptors.
`spring-constraint-kit`	`n:physics:constraints`	`nexusengine/domains/physics/constraints/spring`	Normalize portable linear and angular spring constraint descriptors.
`constraint-break-kit`	`n:physics:constraints`	`nexusengine/domains/physics/constraints/break`	Normalize and purely evaluate portable constraint break thresholds.
`constraint-registry-kit`	`n:physics:constraints`	`nexusengine/domains/physics/constraints/registry`	Own deterministic portable constraint records, terminal break state, and exact-once mutations.
`physics-world-settings-kit`	`n:physics:world`	`nexusengine/domains/physics/world/settings`	Normalize portable Physics coordinate, unit, bounds, and out-of-bounds settings.
`gravity-field-kit`	`n:physics:world`	`nexusengine/domains/physics/world/gravity-field`	Own portable deterministic uniform and point-gravity field records and sampling.
`force-field-kit`	`n:physics:world`	`nexusengine/domains/physics/world/force-field`	Own portable deterministic non-gravity force and acceleration field records and sampling.
`wind-field-kit`	`n:physics:world`	`nexusengine/domains/physics/world/wind-field`	Own portable deterministic physical flow-velocity field records and sampling.
`time-scale-kit`	`n:physics:world`	`nexusengine/domains/physics/world/time-scale`	Own portable deterministic Physics-only time-scale records and delta resolution.
`simulation-region-kit`	`n:physics:world`	`nexusengine/domains/physics/world/simulation-region`	Own portable physical simulation activation regions and deterministic point resolution.
`physics-world-kit`	`n:physics:world`	`nexusengine/domains/physics/world/registry`	Own immutable Physics world records and compose public field, scale, and region capabilities into read-only samples.
`policy-kit`	`n:policy`	`nexusengine/domains/policy/guard`	Evaluate declarative runtime safety and permission rules.
`presentation-registry-kit`	`n:presentation`	`nexusengine/domains/presentation/registry`	Register renderer-neutral presentation capabilities and descriptors.

Kit	Domain	Public subpath	Responsibility
`presentation-output-kit`	`n:presentation:output`	`nexusengine/domains/presentation/output`	Calculate renderer-neutral surfaces, safe areas, viewports, aspect policy, and render resolution.
`animation-descriptor-kit`	`n:presentation:animation`	`nexusengine/domains/presentation/animation`	Manage animation clips, poses, blends, transitions, and timeline descriptors.
`rig-transform-kit`	`n:presentation:animation`	`nexusengine/domains/presentation/animation/rig`	Create neutral joint, limb, and rig transform descriptors.
`audio-descriptor-kit`	`n:presentation:audio`	`nexusengine/domains/presentation/audio`	Manage renderer-neutral audio cues, mix groups, volume policy, and spatial audio descriptors.
`ui-descriptor-kit`	`n:presentation:ui`	`nexusengine/domains/presentation/ui`	Manage renderer-neutral UI, focus, selection, prompt, notification, and accessibility descriptors.
`ui-scale-kit`	`n:presentation:ui`	`nexusengine/domains/presentation/ui/scale`	Calculate deterministic reference-resolution and UI scale policy.
`speech-contract-kit`	`n:presentation:speech`	`nexusengine/domains/presentation/speech`	Manage provider-neutral speech requests, voices, utterance lifecycle, and synthesis results.
`capture-contract-kit`	`n:presentation:capture`	`nexusengine/domains/presentation/capture`	Manage observation requests, view sets, framing, capture jobs, progress, and result contracts.
`sky-descriptor-kit`	`n:presentation:sky`	`nexusengine/domains/presentation/sky`	Create generic sky, horizon, atmosphere, cloud, and celestial descriptors.
`render-domain-contract-kit`	`n:render`	`nexusengine/domains/render/contract`	Expose the canonical backend-neutral Render ownership and execution-contract boundary.
`render-provider-contract-kit`	`n:render:contracts`	`nexusengine/domains/render/provider-contract`	Validate concrete Render providers without executing or retaining provider code or handles.
`render-resource-schema-kit`	`n:render:contracts`	`nexusengine/domains/render/resource-schema`	Validate and normalize portable Render resource records without exposing backend handles.
`render-frame-schema-kit`	`n:render:contracts`	`nexusengine/domains/render/frame-schema`	Validate and normalize portable Render frame execution records.
`render-pass-schema-kit`	`n:render:contracts`	`nexusengine/domains/render/pass-schema`	Validate and normalize resolved provider-facing Render pass records without owning Presentation graph planning.
`shader-schema-kit`	`n:render:contracts`	`nexusengine/domains/render/shader-schema`	Validate and normalize portable shader-interface records shared by Render providers.
`render-event-schema-kit`	`n:render:contracts`	`nexusengine/domains/render/event-schema`	Validate and normalize ordered portable Render lifecycle and execution events.
`render-installation-kit`	`n:render:lifecycle`	`nexusengine/domains/render/lifecycle/installation`	Own the aggregate phase, provider generation and controlled handover status for one Render composition.

Kit	Domain	Public subpath	Responsibility
`render-startup-kit`	`n:render:lifecycle`	`nexusengine/domains/render/lifecycle/startup`	Own deterministic startup requests and provider-readiness receipts.
`render-shutdown-kit`	`n:render:lifecycle`	`nexusengine/domains/render/lifecycle/shutdown`	Own deterministic provider shutdown requests and completion receipts.
`render-recovery-kit`	`n:render:lifecycle`	`nexusengine/domains/render/lifecycle/recovery`	Coordinate deterministic recovery from a failed Render provider lifecycle without owning provider repair execution.
`render-reset-kit`	`n:render:lifecycle`	`nexusengine/domains/render/lifecycle/reset`	Reset composed Render lifecycle state atomically through public capability APIs.
`render-snapshot-kit`	`n:render:lifecycle`	`nexusengine/domains/render/lifecycle/snapshot`	Capture and atomically restore portable snapshots of composed Render lifecycle state.
`render-device-contract-kit`	`n:render:device`	`nexusengine/domains/render/device/contract`	Define the portable identity and ownership boundary for one Render device.
`device-feature-kit`	`n:render:device`	`nexusengine/domains/render/device/feature`	Own canonical Render device feature declarations and deterministic requirement negotiation.
`device-limit-kit`	`n:render:device`	`nexusengine/domains/render/device/limit`	Own portable Render device limit profiles and deterministic requirement checks.
`device-capability-kit`	`n:render:device`	`nexusengine/domains/render/device/capability`	Compose a portable Render device identity, feature set, and limit profile into one capability record.
`device-memory-kit`	`n:render:device`	`nexusengine/domains/render/device/memory`	Own portable memory budgets, semantic reservations, and exact-once accounting receipts.
`device-queue-kit`	`n:render:device`	`nexusengine/domains/render/device/queue`	Own logical Render queue descriptors and exact-once submission and completion receipts.
`device-lifecycle-kit`	`n:render:device`	`nexusengine/domains/render/device/lifecycle`	Own portable acquisition, readiness, loss, failure, recovery, and release state for one selected Render device.
`device-loss-kit`	`n:render:device`	`nexusengine/domains/render/device/loss`	Own exact-once Render device loss incidents and externally proven resolution records.
`device-diagnostics-kit`	`n:render:device`	`nexusengine/domains/render/device/diagnostics`	Project deterministic read-only diagnostics from public Render device capabilities.
`render-surface-kit`	`n:render:surface`	`nexusengine/domains/render/surface/render-surface`	Own portable base Render surface descriptors and exact-once lifecycle records.
`surface-format-kit`	`n:render:surface`	`nexusengine/domains/render/surface/format`	Own portable color, depth, alpha, sample, and HDR surface format selections.

Kit	Domain	Public subpath	Responsibility
`window-surface-kit`	`n:render:surface`	`nexusengine/domains/render/surface/window`	Own portable window-surface descriptors without host window handles or platform transitions.
`offscreen-surface-kit`	`n:render:surface`	`nexusengine/domains/render/surface/offscreen`	Own portable offscreen layer, sample, and usage policy while base Surface owns dimensions.
`swapchain-surface-kit`	`n:render:surface`	`nexusengine/domains/render/surface/swapchain`	Own portable swapchain requests without creating GPU swapchains or provider handles.
`viewport-kit`	`n:render:surface`	`nexusengine/domains/render/surface/viewport`	Own bounded portable viewport regions and depth ranges.
`scissor-kit`	`n:render:surface`	`nexusengine/domains/render/surface/scissor`	Own bounded portable scissor regions without issuing provider commands.
`resize-kit`	`n:render:surface`	`nexusengine/domains/render/surface/resize`	Own portable resize intents without mutating host or provider surfaces.
`fullscreen-kit`	`n:render:surface`	`nexusengine/domains/render/surface/fullscreen`	Own portable fullscreen enter and exit intents without platform execution.
`render-resource-contract-kit`	`n:render:resource`	`nexusengine/domains/render/resource/contract`	Define portable Render execution-resource identity, lifecycle, operation, and provider receipt contracts.
`resource-identity-kit`	`n:render:resource`	`nexusengine/domains/render/resource/identity`	Own deterministic Render execution-resource identities, revisions, and dependency lineage.
`resource-state-kit`	`n:render:resource`	`nexusengine/domains/render/resource/state`	Define portable Render resource phases and legal lifecycle transitions.
`resource-reference-kit`	`n:render:resource`	`nexusengine/domains/render/resource/reference`	Own exact, portable references to Render execution-resource identities.
`resource-integrity-kit`	`n:render:resource`	`nexusengine/domains/render/resource/integrity`	Record portable integrity comparisons for exact Render resource identities.
`resource-cache-kit`	`n:render:resource`	`nexusengine/domains/render/resource/cache`	Index reusable provider resources by exact portable content identity and deterministic access order.
`resource-budget-kit`	`n:render:resource`	`nexusengine/domains/render/resource/budget`	Map exact Render resource identities to existing Device Memory reservations without duplicating capacity authority.
`resource-upload-kit`	`n:render:resource`	`nexusengine/domains/render/resource/upload`	Record exact Render resource upload requests, provider receipts, and failures against completed Device Queue submissions.
`resource-release-kit`	`n:render:resource`	`nexusengine/domains/render/resource/release`	Record exact Render resource release requests, provider receipts, and failures after reference safety checks.

Kit	Domain	Public subpath	Responsibility
`resource-lifecycle-kit`	`n:render:resource`	`nexusengine/domains/render/resource/lifecycle`	Own the portable lifecycle state of exact Render resource identities using explicit upload and release receipts.
`buffer-resource-kit`	`n:render:buffer`	`nexusengine/domains/render/buffer/resource`	Own portable logical Buffer records, exact content revisions, and bounded provider update receipts.
`buffer-layout-kit`	`n:render:buffer`	`nexusengine/domains/render/buffer/layout`	Own explicit portable Buffer field formats, member offsets, alignments, and stride.
`vertex-buffer-kit`	`n:render:buffer`	`nexusengine/domains/render/buffer/vertex`	Own logical Vertex Buffer views with exact resource, layout, count, and range validation.
`index-buffer-kit`	`n:render:buffer`	`nexusengine/domains/render/buffer/index`	Own logical Index Buffer views with exact resource, format, count, alignment, and range validation.
`uniform-buffer-kit`	`n:render:buffer`	`nexusengine/domains/render/buffer/uniform`	Own logical Uniform Buffer ranges with exact layout, size, and explicit dynamic-alignment validation.
`storage-buffer-kit`	`n:render:buffer`	`nexusengine/domains/render/buffer/storage`	Own logical Storage Buffer ranges with explicit access, layout, element-count, and range validation.
`instance-buffer-kit`	`n:render:buffer`	`nexusengine/domains/render/buffer/instance`	Own logical Instance Buffer views with exact resource, layout, count, and range validation.
`indirect-buffer-kit`	`n:render:buffer`	`nexusengine/domains/render/buffer/indirect`	Own logical Indirect Buffer command ranges with exact type, count, stride, alignment, and range validation.
`texture-format-kit`	`n:render:texture`	`nexusengine/domains/render/texture/format`	Own portable Texture format aspects, block layout, and provider-neutral capability declarations.
`texture-resource-kit`	`n:render:texture`	`nexusengine/domains/render/texture/resource`	Own exact logical Texture records derived from canonical Render Resource identities and portable formats.
`texture-2d-kit`	`n:render:texture`	`nexusengine/domains/render/texture/2d`	Own logical 2D Texture views with exact identity and mip-range validation.
`texture-cube-kit`	`n:render:texture`	`nexusengine/domains/render/texture/cube`	Own logical Cube Texture views with exact identity, six-face, and mip-range validation.
`texture-array-kit`	`n:render:texture`	`nexusengine/domains/render/texture/array`	Own logical 2D Texture Array views with exact identity, layer-range, and mip-range validation.
`render-target-texture-kit`	`n:render:texture`	`nexusengine/domains/render/texture/render-target`	Own logical color attachment Texture views with exact format and subresource qualification.

Kit	Domain	Public subpath	Responsibility
`depth-texture-kit`	`n:render:texture`	`nexusengine/domains/render/texture/depth`	Own logical depth-stencil Texture views with exact aspect, format, and subresource qualification.
`shadow-texture-kit`	`n:render:texture`	`nexusengine/domains/render/texture/shadow`	Own provider-neutral shadow-readable depth views without owning authored shadow policy or execution.
`mipmap-kit`	`n:render:texture`	`nexusengine/domains/render/texture/mipmap`	Own explicit portable Texture mip-chain plans with exact contiguous levels and source identities.
`texture-stream-kit`	`n:render:texture`	`nexusengine/domains/render/texture/stream`	Own exact Texture subresource stream requests and portable provider completion or failure receipts.
`texture-residency-kit`	`n:render:texture`	`nexusengine/domains/render/texture/residency`	Own desired and proven resident Texture subresources derived from completed stream receipts.
`shader-contract-kit`	`n:render:shader`	`nexusengine/domains/render/shader/contract`	Define the canonical provider-neutral Shader execution boundary and stage topology.
`shader-language-kit`	`n:render:shader`	`nexusengine/domains/render/shader/language`	Own portable Shader language capabilities, source kinds, stages, and feature requirements.
`shader-source-kit`	`n:render:shader`	`nexusengine/domains/render/shader/source`	Own immutable text or binary Shader source revisions and exact integrity.
`shader-include-kit`	`n:render:shader`	`nexusengine/domains/render/shader/include`	Own immutable Shader include records and a deterministic acyclic dependency graph.
`shader-module-kit`	`n:render:shader`	`nexusengine/domains/render/shader/module`	Own one portable Shader stage module, entry point, and exact source closure.
`shader-program-kit`	`n:render:shader`	`nexusengine/domains/render/shader/program`	Own linked portable Shader program topology and its canonical interface.
`shader-variant-kit`	`n:render:shader`	`nexusengine/domains/render/shader/variant`	Own exact Shader define and specialization selections with deterministic identity.
`shader-permutation-kit`	`n:render:shader`	`nexusengine/domains/render/shader/permutation`	Own bounded deterministic Shader permutation axes and read-only expansion.
`shader-error-kit`	`n:render:shader`	`nexusengine/domains/render/shader/error`	Normalize portable Shader diagnostics without owning compiler execution or repair.
`shader-compile-kit`	`n:render:shader`	`nexusengine/domains/render/shader/compile`	Own exact-once logical Shader compile requests and provider completion or failure receipts.
`shader-reflection-kit`	`n:render:shader`	`nexusengine/domains/render/shader/reflection`	Validate normalized provider reflection against the completed compile and program interface.

Kit	Domain	Public subpath	Responsibility
`shader-cache-kit`	`n:render:shader`	`nexusengine/domains/render/shader/cache`	Link completed Shader compiles to resident shader-program Render Resources and select deterministic eviction candidates.
`render-layer-graph-kit`	`n:render:pipeline`	`nexusengine/domains/render/pipeline`	Validate renderer-neutral render layers, dependencies, and pass ordering.
`render-batch-kit`	`n:render:batch`	`nexusengine/domains/render/batch`	Own portable instance batch state, capacity, cell updates and flush records.
`runtime-lifecycle-kit`	`n:runtime`	`nexusengine/domains/runtime/lifecycle`	Own deterministic runtime lifecycle and Kit installation receipts.
`realtime-runtime-kit`	`n:runtime:realtime`	`nexusengine/domains/runtime/realtime`	Create deterministic realtime frame context and phase execution.
`runtime-data-kit`	`n:runtime:data`	`nexusengine/domains/runtime/data`	Provide deterministic schemas, snapshots, selectors, migrations, and data envelopes.
`transaction-ledger-kit`	`n:runtime:transaction`	`nexusengine/domains/runtime/transaction`	Record repeat-safe operation keys and immutable transaction receipts.
`persistence-contract-kit`	`n:runtime:persistence`	`nexusengine/domains/runtime/persistence`	Describe save/load targets, slots, recovery records, and persistence adapter contracts.
`runtime-sequence-kit`	`n:runtime:sequence`	`nexusengine/domains/runtime/sequence`	Install deterministic sequence node definitions and execution state.
`runtime-startup-kit`	`n:runtime:startup`	`nexusengine/domains/runtime/startup`	Coordinate deterministic startup preparation and readiness receipts.
`schedule-kit`	`n:runtime:sequence:schedule`	`nexusengine/domains/runtime/sequence/schedule`	Advance deterministic repeatable and one-shot elapsed-time schedules without losing residual time.
`observation-history-kit`	`n:runtime:data:observation`	`nexusengine/domains/runtime/data/observation`	Retain bounded committed portable observations with deterministic identity and replay checks.
`simulation-state-kit`	`n:simulation`	`nexusengine/domains/simulation/runtime`	Manage deterministic simulation objectives, resources, hazards, timers, and resolution receipts.
`simulation-physics-contract-kit`	`n:simulation:physics`	`nexusengine/domains/simulation/physics`	Describe physical bodies, colliders, contacts, constraints, queries, and provider boundaries.
`articulated-physics-kit`	`n:simulation:physics:articulated`	`nexusengine/domains/simulation/physics/articulated`	Manage backend-neutral articulated body topology and joint dynamics state.
`motion-contract-kit`	`n:simulation:motion`	`nexusengine/domains/simulation/motion`	Manage intent-to-motion descriptors, trajectories, velocity state, and movement policies.
`two-bone-ik-kit`	`n:simulation:motion`	`nexusengine/domains/simulation/motion/two-bone-ik`	Solve deterministic two-bone inverse-kinematics poses.

Kit	Domain	Public subpath	Responsibility
`articulated-motion-kit`	`n:simulation:motion:articulated`	`nexusengine/domains/simulation/motion/articulated`	Create target poses, joint limits, articulation plans, and drive requests.
`articulated-motion-drive-adapter-kit`	`n:simulation:physics:articulated`	`nexusengine/domains/simulation/adapters/articulated-drive`	Translate articulated motion plans into backend-neutral physics drive requests.
`action-locomotion-kit`	`n:simulation:motion:locomotion`	`nexusengine/domains/simulation/motion/locomotion`	Convert action commands into deterministic renderer-neutral Motion intents and locomotion frames.
`vehicle-dynamics-kit`	`n:simulation:motion:vehicle`	`nexusengine/domains/simulation/motion/vehicle`	Advance deterministic vehicle motion, boost, bounds, and impact state without owning surface policy.
`world-contact-kit`	`n:simulation:physics:world-contact`	`nexusengine/domains/simulation/physics/world-contact`	Resolve portable world contact, slope, impact, stability, and correction records without implementing a physics backend.
`soft-respawn-kit`	`n:simulation:recovery:soft-respawn`	`nexusengine/domains/simulation/recovery/soft-respawn`	Produce exact-once coherent subject recovery records at configured portable points.
`economy-account-kit`	`n:simulation:economy:accounts`	`nexusengine/domains/simulation/economy/accounts`	Own finite account balances and exact-once portable economy transaction records.
`cargo-manifest-kit`	`n:simulation:economy:cargo`	`nexusengine/domains/simulation/economy/cargo`	Own portable cargo inventory, capacity, condition, pickup, deposit, and quota state.
`facility-operations-kit`	`n:simulation:operations:facility`	`nexusengine/domains/simulation/operations/facility`	Own deterministic facility capacity, condition, status, cycle, and portable output receipts.
`occupant-flow-kit`	`n:simulation:operations:occupant-flow`	`nexusengine/domains/simulation/operations/occupant-flow`	Own deterministic occupant spawning, patience, service, and abandonment state.
`transport-route-kit`	`n:simulation:operations:transport-route`	`nexusengine/domains/simulation/operations/transport-route`	Own deterministic stops, carriers, capacity, calls, travel progress, and arrival receipts.
`hazard-field-kit`	`n:simulation:hazard-field`	`nexusengine/domains/simulation/hazard-field`	Own deterministic bounded hazards, verified spawn identities, motion, and read-only collision queries.
`pursuit-pressure-kit`	`n:simulation:pursuit-pressure`	`nexusengine/domains/simulation/pursuit-pressure`	Own coherent pursuit distance, warning bands, caught state, recovery, and transition history.
`lifecycle-progression-kit`	`n:simulation:progression:lifecycle`	`nexusengine/domains/simulation/progression/lifecycle`	Own prerequisite-gated lifecycle start, timing, completion, and portable effect descriptors.
`locomotion-contact-response-adapter-kit`	`n:simulation:motion:locomotion`	`nexusengine/domains/simulation/adapters/locomotion-contact-response`	Translate Locomotion frames and World Contact results into corrected Motion frames without owning either state.
`vehicle-water-response-adapter-kit`	`n:simulation:motion:vehicle`	`nexusengine/domains/simulation/adapters/vehicle-water-response`	Translate Vehicle state and Water Surface queries into portable drag, current, and buoyancy responses.

Kit	Domain	Public subpath	Responsibility
`lifecycle-economy-adapter-kit`	`n:simulation:progression:lifecycle`	`nexusengine/domains/simulation/adapters/lifecycle-economy`	Translate accepted Lifecycle costs and Economy effects into exact-once Economy transactions.
`lifecycle-facility-adapter-kit`	`n:simulation:progression:lifecycle`	`nexusengine/domains/simulation/adapters/lifecycle-facility`	Translate accepted Lifecycle facility effects into exact-once Facility Operations commands.
`facility-economy-adapter-kit`	`n:simulation:operations:facility`	`nexusengine/domains/simulation/adapters/facility-economy`	Translate Facility output and upkeep receipts into exact-once Economy transactions.
`spatial-contract-kit`	`n:spatial`	`nexusengine/domains/spatial/contracts`	Describe transforms, bounds, zones, spaces, and spatial query requests.
`spatial-angle-math-kit`	`n:spatial`	`nexusengine/domains/spatial/angle-math`	Normalize, compare, and interpolate angular values.
`spatial-vector-math-kit`	`n:spatial`	`nexusengine/domains/spatial/vector-math`	Create and operate on renderer-neutral vector values.
`spatial-transform-math-kit`	`n:spatial`	`nexusengine/domains/spatial/transform-math`	Calculate deterministic transforms, bases, interpolation, and planar projections.
`spatial-quaternion-math-kit`	`n:spatial`	`nexusengine/domains/spatial/quaternion-math`	Create, compose, normalize, rotate, and interpolate quaternions.
`spatial-scale-kit`	`n:spatial:scale`	`nexusengine/domains/spatial/scale`	Manage deterministic subject scale, scale anchors, proximity bands, and scale transitions.
`world-state-kit`	`n:world`	`nexusengine/domains/world/runtime`	Manage world identity, cells, partitions, surfaces, and deterministic assembly state.
`scene-lifecycle-kit`	`n:world:scene`	`nexusengine/domains/world/scene`	Manage host-neutral scene identity, lifecycle, transitions, bindings, and snapshots.
`weather-state-kit`	`n:world:weather`	`nexusengine/domains/world/weather`	Manage deterministic weather conditions, tendencies, regions, and sampling.
`layered-weather-kit`	`n:world:weather`	`nexusengine/domains/world/weather/layers`	Compose and evolve deterministic altitude-aware weather layers.
`world-foundation-kit`	`n:world:foundation`	`nexusengine/domains/world/foundation`	Compose deterministic world foundation definitions, contributions, sampling, and cell resolution.
`world-feature-kit`	`n:world:feature`	`nexusengine/domains/world/feature`	Manage semantic world feature definitions, registry, lifecycle, queries, and composition.
`semantic-world-feature-kit`	`n:world:feature`	`nexusengine/domains/world/feature/semantic`	Create one semantic world feature Domain Service Kit from a bounded feature specification.
`landform-feature-kit`	`n:world:feature:landform`	`nexusengine/domains/world/feature/landform`	Create semantic elevation and landform feature descriptors and lifecycle state.
`hydrology-feature-kit`	`n:world:feature:hydrology`	`nexusengine/domains/world/feature/hydrology`	Create semantic watershed, water path, water body, and wetland feature descriptors.

Kit	Domain	Public subpath	Responsibility
`ecology-feature-kit`	`n:world:feature:ecology`	`nexusengine/domains/world/feature/ecology`	Create semantic biome, habitat, vegetation-region, and ecotone feature descriptors.
`settlement-feature-kit`	`n:world:feature:settlement`	`nexusengine/domains/world/feature/settlement`	Create semantic settlement, route, structure, and infrastructure feature descriptors.
`atmosphere-feature-kit`	`n:world:feature:atmosphere`	`nexusengine/domains/world/feature/atmosphere`	Create semantic cloud, fog, wind, thermal, precipitation, and visibility feature descriptors.
`navmesh-kit`	`n:world:navigation:navmesh`	`nexusengine/domains/world/navigation/navmesh`	Build deterministic 2D navigation meshes and portable 3D waypoint graphs from walkability.
`pathfinding-kit`	`n:world:navigation:pathfinding`	`nexusengine/domains/world/navigation/pathfinding`	Resolve deterministic A* path requests over portable grid and navigation graph adapters.
`route-field-kit`	`n:world:navigation:route-field`	`nexusengine/domains/world/navigation/route-field`	Manage reusable route marker and corridor descriptors plus pure proximity queries.
`landmark-guidance-kit`	`n:world:navigation:landmark-guidance`	`nexusengine/domains/world/navigation/landmark-guidance`	Manage deterministic landmark discovery, reach, completion, priority, and proximity state.
`procedural-generation-kit`	`n:world:generation`	`nexusengine/domains/world/generation`	Generate deterministic generic regions, connectors, points, graphs, and walkability from complete normalized configuration.
`terrain-kit`	`n:world:terrain`	`nexusengine/domains/world/terrain`	Evaluate deterministic terrain layers and manage portable sampled terrain cells without rendering ownership.
`water-surface-kit`	`n:world:water-surface`	`nexusengine/domains/world/water-surface`	Manage renderer-neutral water zones, currents, drag, wave phase, hazards, and pure queries.

GENERATED REFERENCE

Kit Ownership Ledger

Generated from Domain manifest v2 and the production source inventory. Null compliance fields are intentionally unproven; they are never inferred as true.

Registry SHA-256: `147e7302e0563cdf246cc33c197ba32fec606f783817d345244544e3ce486c76`

- Source modules: 1731
- Manifest-proven public atoms: 385
- Manifest-owned internal modules: 1311
- Root contract modules: 25
- Optional provider modules: 10
- Unreviewed modules: 0
- Violations: 0

Path	Owner	Review	Destination
`kits/catalog.js`	`n:render`	optional-provider-kit	NexusEngine optional Graphics Language Kit
`kits/render/graphics-language-kit/contracts.js`	`n:render`	optional-provider-kit	NexusEngine optional Graphics Language Kit
`kits/render/graphics-language-kit/hosts.js`	`n:render`	optional-provider-kit	NexusEngine optional Graphics Language Kit
`kits/render/graphics-language-kit/index.js`	`n:render`	optional-provider-kit	NexusEngine optional Graphics Language Kit
`kits/render/graphics-language-kit/manifest.js`	`n:render`	optional-provider-kit	NexusEngine optional Graphics Language Kit
`kits/render/graphics-language-kit/material.js`	`n:render`	optional-provider-kit	NexusEngine optional Graphics Language Kit
`kits/render/graphics-language-kit/provider.js`	`n:render`	optional-provider-kit	NexusEngine optional Graphics Language Kit
`kits/render/graphics-language-kit/runtime/artifacts/nexusgl.mjs`	`n:render`	optional-provider-kit	NexusEngine optional Graphics Language Kit
`kits/render/graphics-language-kit/runtime/loader.js`	`n:render`	optional-provider-kit	NexusEngine optional Graphics Language Kit
`kits/render/graphics-language-kit/session.js`	`n:render`	optional-provider-kit	NexusEngine optional Graphics Language Kit
`src/core-domains/actor/character/index.js`	`n:actor`	manifest-owned-internal	NexusEngine Core
`src/core-domains/actor/character/kits/character-kit/contracts.js`	`n:actor`	manifest-owned-internal	NexusEngine Core
`src/core-domains/actor/character/kits/character-kit/index.js`	`n:actor`	manifest-proven-public-atom	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/actor/creature/index.js`	`n:actor`	manifest-owned-internal	NexusEngine Core
`src/core-domains/actor/creature/kits/creature-kit/contracts.js`	`n:actor`	manifest-owned-internal	NexusEngine Core
`src/core-domains/actor/creature/kits/creature-kit/index.js`	`n:actor`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/actor/domain.manifest.js`	`n:actor`	manifest-owned-internal	NexusEngine Core
`src/core-domains/actor/index.js`	`n:actor`	manifest-owned-internal	NexusEngine Core
`src/core-domains/actor/kits/actor-registry-kit/index.js`	`n:actor`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/actor/player/index.js`	`n:actor`	manifest-owned-internal	NexusEngine Core
`src/core-domains/actor/player/kits/player-kit/contracts.js`	`n:actor`	manifest-owned-internal	NexusEngine Core
`src/core-domains/actor/player/kits/player-kit/index.js`	`n:actor`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/agent/domain.manifest.js`	`n:agent`	manifest-owned-internal	NexusEngine Core
`src/core-domains/agent/index.js`	`n:agent`	manifest-owned-internal	NexusEngine Core
`src/core-domains/agent/kits/agent-kit/action-proposals.js`	`n:agent`	manifest-owned-internal	NexusEngine Core
`src/core-domains/agent/kits/agent-kit/adapters.js`	`n:agent`	manifest-owned-internal	NexusEngine Core
`src/core-domains/agent/kits/agent-kit/agent-state.js`	`n:agent`	manifest-owned-internal	NexusEngine Core
`src/core-domains/agent/kits/agent-kit/decision-cycle.js`	`n:agent`	manifest-owned-internal	NexusEngine Core
`src/core-domains/agent/kits/agent-kit/execution-ledger.js`	`n:agent`	manifest-owned-internal	NexusEngine Core
`src/core-domains/agent/kits/agent-kit/index.js`	`n:agent`	manifest-proven-public-atom	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/agent/kits/agent-kit/observations.js`	`n:agent`	manifest-owned-internal	NexusEngine Core
`src/core-domains/asset/domain.manifest.js`	`n:asset`	manifest-owned-internal	NexusEngine Core
`src/core-domains/asset/index.js`	`n:asset`	manifest-owned-internal	NexusEngine Core
`src/core-domains/asset/kits/asset-kit/cache-provider.js`	`n:asset`	manifest-owned-internal	NexusEngine Core
`src/core-domains/asset/kits/asset-kit/descriptors.js`	`n:asset`	manifest-owned-internal	NexusEngine Core
`src/core-domains/asset/kits/asset-kit/index.js`	`n:asset`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/asset/kits/asset-kit/provider.js`	`n:asset`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/animation/index.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/animation/kits/authoring-animation-service-kit/index.js`	`n:authoring`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/authoring/animation/kits/authoring-animation-service-kit/kit.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/animation/services.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/animation/subdomain.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/assembly/index.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/assembly/kits/authoring-assembly-service-kit/index.js`	`n:authoring`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/authoring/assembly/kits/authoring-assembly-service-kit/kit.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/assembly/scatter.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/authoring/assembly/services.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/assembly/subdomain.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/brush/index.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/brush/kits/authoring-brush-service-kit/index.js`	`n:authoring`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/authoring/brush/kits/authoring-brush-service-kit/kit.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/brush/services.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/brush/subdomain.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/contracts/io-kit.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/contracts/io.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/contracts/ledger.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/contracts/provider-registry.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/contracts/service-lifecycle.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/contracts/services.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/contracts/tool-schema.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/contracts/transforms.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/contracts/value.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/authoring/create/index.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/create/kits/authoring-create-service-kit/index.js`	`n:authoring`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/authoring/create/kits/authoring-create-service-kit/kit.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/create/services.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/create/subdomain.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/curve/index.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/curve/kits/authoring-curve-service-kit/index.js`	`n:authoring`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/authoring/curve/kits/authoring-curve-service-kit/kit.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/curve/services.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/curve/subdomain.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/domain-composition/index.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/domain-composition/kits/authoring-domain-composition-service-kit/index.js`	`n:authoring`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/authoring/domain-composition/kits/authoring-domain-composition-service-kit/kit.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/domain-composition/services.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/authoring/domain-composition/subdomain.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/domain.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/editing/index.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/editing/kits/authoring-editing-session-kit/index.js`	`n:authoring`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/authoring/editing/kits/authoring-editing-session-kit/kit.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/editing/selection.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/editing/services.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/editing/subdomain.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/importing/index.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/importing/kits/authoring-import-commit-kit/index.js`	`n:authoring`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/authoring/importing/kits/authoring-import-commit-kit/kit.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/importing/kits/authoring-import-registry-kit/index.js`	`n:authoring`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/authoring/importing/kits/authoring-import-registry-kit/kit.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/importing/kits/authoring-import-service-kit/index.js`	`n:authoring`	manifest-proven-public-atom	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/authoring/importing/kits/authoring-import-service-kit/kit.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/importing/kits/authoring-import-validation-kit/index.js`	`n:authoring`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/authoring/importing/kits/authoring-import-validation-kit/kit.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/importing/providers/fbx/decode.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/importing/providers/fbx/index.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/importing/providers/fbx/texture-helpers.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/importing/providers/glb/decode.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/importing/providers/glb/index.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/importing/providers/obj/decode.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/importing/providers/obj/index.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/importing/providers/usdz/decode.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/importing/providers/usdz/index.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/importing/scene-documents.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/importing/services.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/importing/subdomain.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/index.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/authoring/io-kits.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/io.manifests.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/kits/authoring-domain-contract-kit/index.js`	`n:authoring`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/authoring/kits/authoring-domain-contract-kit/kit.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/material/index.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/material/kits/authoring-material-service-kit/index.js`	`n:authoring`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/authoring/material/kits/authoring-material-service-kit/kit.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/material/services.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/material/subdomain.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/mesh/bevel.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/mesh/evaluate.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/mesh/geometry.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/mesh/index.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/mesh/kits/authoring-mesh-document-kit/index.js`	`n:authoring`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/authoring/mesh/kits/authoring-mesh-document-kit/kit.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/mesh/primitives.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/authoring/mesh/services.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/mesh/subdomain.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/mesh/surface-query.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/mesh/topology.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/modifier/boolean.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/modifier/decimation.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/modifier/evaluate.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/modifier/index.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/modifier/kits/authoring-modifier-service-kit/index.js`	`n:authoring`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/authoring/modifier/kits/authoring-modifier-service-kit/kit.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/modifier/services.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/modifier/subdivision.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/modifier/subdomain.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/paint/image.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/paint/index.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/paint/kits/authoring-paint-service-kit/index.js`	`n:authoring`	manifest-proven-public-atom	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/authoring/paint/kits/authoring-paint-service-kit/kit.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/paint/services.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/paint/subdomain.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/persistence/index.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/persistence/kits/authoring-load-kit/index.js`	`n:authoring`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/authoring/persistence/kits/authoring-load-kit/kit.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/persistence/kits/authoring-persistence-service-kit/index.js`	`n:authoring`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/authoring/persistence/kits/authoring-persistence-service-kit/kit.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/persistence/kits/authoring-project-package-kit/index.js`	`n:authoring`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/authoring/persistence/kits/authoring-project-package-kit/kit.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/persistence/kits/authoring-save-kit/index.js`	`n:authoring`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/authoring/persistence/kits/authoring-save-kit/kit.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/persistence/kits/authoring-storage-provider-registry-kit/index.js`	`n:authoring`	manifest-proven-public-atom	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/authoring/persistence/kits/authoring-storage-provider-registry-kit/kit.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/persistence/project-package.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/persistence/providers/filesystem/index.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/persistence/providers/indexeddb/index.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/persistence/providers/memory/index.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/persistence/services.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/persistence/subdomain.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/project/index.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/project/kits/authoring-project-document-kit/index.js`	`n:authoring`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/authoring/project/kits/authoring-project-document-kit/kit.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/project/store.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/project/subdomain.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/publishing/export/codecs/fbx-binary.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/publishing/export/codecs/png.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/publishing/export/codecs/usda.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/authoring/publishing/export/codecs/zip.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/publishing/export/contracts/packet.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/publishing/export/index.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/publishing/export/kits/authoring-export-capability-kit/index.js`	`n:authoring`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/authoring/publishing/export/kits/authoring-export-capability-kit/kit.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/publishing/export/kits/authoring-export-receipt-kit/index.js`	`n:authoring`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/authoring/publishing/export/kits/authoring-export-receipt-kit/kit.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/publishing/export/kits/authoring-export-registry-kit/index.js`	`n:authoring`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/authoring/publishing/export/kits/authoring-export-registry-kit/kit.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/publishing/export/kits/authoring-export-service-kit/index.js`	`n:authoring`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/authoring/publishing/export/kits/authoring-export-service-kit/kit.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/publishing/export/kits/authoring-export-validation-kit/index.js`	`n:authoring`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/authoring/publishing/export/kits/authoring-export-validation-kit/kit.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/authoring/publishing/export/providers/fbx/encode.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/publishing/export/providers/fbx/index.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/publishing/export/providers/glb/encode.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/publishing/export/providers/glb/index.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/publishing/export/providers/usdz/encode.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/publishing/export/providers/usdz/index.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/publishing/export/publishing/artifact-output.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/publishing/export/services.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/publishing/export/subdomain.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/publishing/index.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/publishing/kits/authoring-publishing-service-kit/index.js`	`n:authoring`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/authoring/publishing/kits/authoring-publishing-service-kit/kit.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/publishing/services.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/publishing/subdomain.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/authoring/rig/index.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/rig/kits/authoring-rig-service-kit/index.js`	`n:authoring`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/authoring/rig/kits/authoring-rig-service-kit/kit.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/rig/services.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/rig/subdomain.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/sculpt/index.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/sculpt/kits/authoring-sculpt-service-kit/index.js`	`n:authoring`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/authoring/sculpt/kits/authoring-sculpt-service-kit/kit.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/sculpt/services.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/sculpt/subdomain.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/sequence/index.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/sequence/kits/authoring-sequence-service-kit/index.js`	`n:authoring`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/authoring/sequence/kits/authoring-sequence-service-kit/kit.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/sequence/services.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/sequence/subdomain.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/authoring/skin/index.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/skin/kits/authoring-skin-service-kit/index.js`	`n:authoring`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/authoring/skin/kits/authoring-skin-service-kit/kit.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/skin/services.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/skin/subdomain.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/uv/diagnostics.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/uv/index.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/uv/kits/authoring-uv-service-kit/index.js`	`n:authoring`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/authoring/uv/kits/authoring-uv-service-kit/kit.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/uv/services.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/uv/subdomain.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/validation/index.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/validation/kits/authoring-delivery-validation-kit/index.js`	`n:authoring`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/authoring/validation/kits/authoring-delivery-validation-kit/kit.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/validation/kits/authoring-document-validation-kit/index.js`	`n:authoring`	manifest-proven-public-atom	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/authoring/validation/kits/authoring-document-validation-kit/kit.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/validation/kits/authoring-format-validation-kit/index.js`	`n:authoring`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/authoring/validation/kits/authoring-format-validation-kit/kit.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/validation/kits/authoring-project-validation-kit/index.js`	`n:authoring`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/authoring/validation/kits/authoring-project-validation-kit/kit.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/validation/kits/authoring-validation-report-kit/index.js`	`n:authoring`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/authoring/validation/kits/authoring-validation-report-kit/kit.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/validation/services.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/validation/subdomain.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/workspace/index.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/workspace/kits/authoring-workspace-service-kit/index.js`	`n:authoring`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/authoring/workspace/kits/authoring-workspace-service-kit/kit.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/workspace/services.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core
`src/core-domains/authoring/workspace/subdomain.manifest.js`	`n:authoring`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/build/adapters/mcp/build-mcp-provider.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/analysis/index.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/analysis/kits/dependency-analysis-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/analysis/kits/dependency-analysis-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/analysis/kits/dependency-analysis-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/analysis/kits/dependency-analysis-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/analysis/kits/dependency-analysis-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/analysis/kits/effect-analysis-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/analysis/kits/effect-analysis-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/analysis/kits/effect-analysis-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/analysis/kits/effect-analysis-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/analysis/kits/effect-analysis-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/analysis/kits/javascript-ast-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/analysis/kits/javascript-ast-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/build/analysis/kits/javascript-ast-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/analysis/kits/javascript-ast-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/analysis/kits/javascript-ast-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/analysis/kits/type-analysis-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/analysis/kits/type-analysis-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/analysis/kits/type-analysis-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/analysis/kits/type-analysis-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/analysis/kits/type-analysis-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/analysis/subdomain.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/artifact/index.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/artifact/kits/artifact-cache-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/artifact/kits/artifact-cache-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/artifact/kits/artifact-cache-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/artifact/kits/artifact-cache-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/artifact/kits/artifact-cache-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/build/artifact/kits/artifact-integrity-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/artifact/kits/artifact-integrity-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/artifact/kits/artifact-integrity-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/artifact/kits/artifact-integrity-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/artifact/kits/artifact-integrity-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/artifact/kits/artifact-manifest-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/artifact/kits/artifact-manifest-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/artifact/kits/artifact-manifest-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/artifact/kits/artifact-manifest-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/artifact/kits/artifact-manifest-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/artifact/kits/artifact-output-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/artifact/kits/artifact-output-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/artifact/kits/artifact-output-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/build/artifact/kits/artifact-output-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/artifact/kits/artifact-output-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/artifact/subdomain.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/atomic-kit.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/classification/index.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/classification/kits/capability-resolution-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/classification/kits/capability-resolution-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/classification/kits/capability-resolution-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/classification/kits/capability-resolution-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/classification/kits/capability-resolution-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/classification/kits/fallback-selection-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/classification/kits/fallback-selection-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/classification/kits/fallback-selection-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/classification/kits/fallback-selection-kit/resolution.js`	`n:build`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/build/classification/kits/fallback-selection-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/classification/kits/fallback-selection-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/classification/kits/portability-classifier-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/classification/kits/portability-classifier-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/classification/kits/portability-classifier-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/classification/kits/portability-classifier-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/classification/kits/portability-classifier-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/classification/subdomain.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/compile/index.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/compile/kits/javascript-fallback-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/compile/kits/javascript-fallback-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/compile/kits/javascript-fallback-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/compile/kits/javascript-fallback-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/compile/kits/javascript-fallback-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/build/compile/kits/native-runtime-link-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/compile/kits/native-runtime-link-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/compile/kits/native-runtime-link-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/compile/kits/native-runtime-link-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/compile/kits/native-runtime-link-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/compile/kits/runtime-abi-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/compile/kits/runtime-abi-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/compile/kits/runtime-abi-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/compile/kits/runtime-abi-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/compile/kits/runtime-abi-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/compile/kits/rust-lowering-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/compile/kits/rust-lowering-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/compile/kits/rust-lowering-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/compile/kits/rust-lowering-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/compile/kits/rust-lowering-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/build/compile/kits/web-module-linker-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/compile/kits/web-module-linker-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/compile/kits/web-module-linker-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/compile/kits/web-module-linker-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/compile/kits/web-module-linker-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/compile/subdomain.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/domain.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/index.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/ir/index.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/ir/kits/execution-ir-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/ir/kits/execution-ir-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/ir/kits/execution-ir-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/ir/kits/execution-ir-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/ir/kits/execution-ir-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/ir/kits/ir-validation-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/ir/kits/ir-validation-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/build/ir/kits/ir-validation-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/ir/kits/ir-validation-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/ir/kits/ir-validation-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/ir/kits/kit-ir-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/ir/kits/kit-ir-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/ir/kits/kit-ir-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/ir/kits/kit-ir-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/ir/kits/kit-ir-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/ir/kits/source-map-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/ir/kits/source-map-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/ir/kits/source-map-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/ir/kits/source-map-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/ir/kits/source-map-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/ir/subdomain.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/kit-manifests.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/manifest-input.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/orchestration/index.js`	`n:build`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/build/orchestration/kits/build-approval-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/orchestration/kits/build-approval-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/orchestration/kits/build-approval-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/orchestration/kits/build-approval-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/orchestration/kits/build-approval-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/orchestration/kits/build-execution-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/orchestration/kits/build-execution-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/orchestration/kits/build-execution-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/orchestration/kits/build-execution-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/orchestration/kits/build-execution-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/orchestration/kits/build-plan-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/orchestration/kits/build-plan-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/orchestration/kits/build-plan-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/build/orchestration/kits/build-plan-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/orchestration/kits/build-plan-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/orchestration/kits/build-receipt-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/orchestration/kits/build-receipt-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/orchestration/kits/build-receipt-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/orchestration/kits/build-receipt-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/orchestration/kits/build-receipt-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/orchestration/kits/build-request-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/orchestration/kits/build-request-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/orchestration/kits/build-request-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/orchestration/kits/build-request-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/orchestration/kits/build-request-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/orchestration/kits/target-set-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/orchestration/kits/target-set-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/build/orchestration/kits/target-set-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/orchestration/kits/target-set-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/orchestration/kits/target-set-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/orchestration/subdomain.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/proof/index.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/proof/kits/cross-runtime-parity-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/proof/kits/cross-runtime-parity-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/proof/kits/cross-runtime-parity-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/proof/kits/cross-runtime-parity-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/proof/kits/cross-runtime-parity-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/proof/kits/project-immutability-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/proof/kits/project-immutability-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/proof/kits/project-immutability-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/proof/kits/project-immutability-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/build/proof/kits/project-immutability-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/proof/kits/target-validation-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/proof/kits/target-validation-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/proof/kits/target-validation-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/proof/kits/target-validation-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/proof/kits/target-validation-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/proof/subdomain.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/source/index.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/source/kits/dependency-source-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/source/kits/dependency-source-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/source/kits/dependency-source-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/source/kits/dependency-source-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/source/kits/dependency-source-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/source/kits/module-graph-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/source/kits/module-graph-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/build/source/kits/module-graph-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/source/kits/module-graph-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/source/kits/module-graph-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/source/kits/project-source-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/source/kits/project-source-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/source/kits/project-source-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/source/kits/project-source-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/source/kits/project-source-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/source/kits/source-cache-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/source/kits/source-cache-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/source/kits/source-cache-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/source/kits/source-cache-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/source/kits/source-cache-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/source/kits/source-fingerprint-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/source/kits/source-fingerprint-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/source/kits/source-fingerprint-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/build/source/kits/source-fingerprint-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/source/kits/source-fingerprint-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/source/subdomain.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/subdomain-manifests.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/android-xr/index.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/android-xr/kits/android-xr-target-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/android-xr/kits/android-xr-target-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/target/android-xr/kits/android-xr-target-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/android-xr/kits/android-xr-target-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/android-xr/kits/android-xr-target-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/android-xr/subdomain.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/host-native/index.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/host-native/kits/host-native-target-kit/bindings.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/host-native/kits/host-native-target-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/build/target/host-native/kits/host-native-target-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/host-native/kits/host-native-target-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/host-native/subdomain.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/index.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/kits/target-registry-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/kits/target-registry-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/target/kits/target-registry-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/kits/target-registry-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/kits/target-registry-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/native-target-helpers.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/openxr/index.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/openxr/kits/openxr-input-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/openxr/kits/openxr-input-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/target/openxr/kits/openxr-input-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/openxr/kits/openxr-input-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/build/target/openxr/kits/openxr-input-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/openxr/kits/openxr-render-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/openxr/kits/openxr-render-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/target/openxr/kits/openxr-render-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/openxr/kits/openxr-render-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/openxr/kits/openxr-render-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/openxr/kits/openxr-runtime-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/openxr/kits/openxr-runtime-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/target/openxr/kits/openxr-runtime-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/openxr/kits/openxr-runtime-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/openxr/kits/openxr-runtime-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/openxr/subdomain.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/pcvr/index.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/pcvr/kits/pcvr-target-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/build/target/pcvr/kits/pcvr-target-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/target/pcvr/kits/pcvr-target-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/pcvr/kits/pcvr-target-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/pcvr/kits/pcvr-target-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/pcvr/subdomain.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/subdomain.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/web-live/index.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/web-live/kits/web-live-target-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/web-live/kits/web-live-target-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/target/web-live/kits/web-live-target-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/web-live/kits/web-live-target-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/web-live/kits/web-live-target-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/web-live/subdomain.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/web-static/index.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/web-static/kits/web-static-target-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/build/target/web-static/kits/web-static-target-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/target/web-static/kits/web-static-target-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/web-static/kits/web-static-target-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/web-static/kits/web-static-target-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/web-static/subdomain.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/target/web-target-helpers.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/toolchain/index.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/toolchain/kits/isolated-stage-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/toolchain/kits/isolated-stage-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/toolchain/kits/isolated-stage-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/toolchain/kits/isolated-stage-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/toolchain/kits/isolated-stage-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/toolchain/kits/process-execution-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/toolchain/kits/process-execution-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/build/toolchain/kits/process-execution-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/toolchain/kits/process-execution-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/toolchain/kits/process-execution-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/toolchain/kits/toolchain-discovery-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/toolchain/kits/toolchain-discovery-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/toolchain/kits/toolchain-discovery-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/toolchain/kits/toolchain-discovery-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/toolchain/kits/toolchain-discovery-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/toolchain/kits/toolchain-provision-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/toolchain/kits/toolchain-provision-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/toolchain/kits/toolchain-provision-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/toolchain/kits/toolchain-provision-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/toolchain/kits/toolchain-provision-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/build/toolchain/kits/toolchain-source-kit/contracts.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/toolchain/kits/toolchain-source-kit/index.js`	`n:build`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/build/toolchain/kits/toolchain-source-kit/kit.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/toolchain/kits/toolchain-source-kit/services.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/toolchain/kits/toolchain-source-kit/state.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/build/toolchain/subdomain.manifest.js`	`n:build`	manifest-owned-internal	NexusEngine Core
`src/core-domains/camel/domain.manifest.js`	`n:camel`	manifest-owned-internal	NexusEngine Core
`src/core-domains/camel/execution/runtime.js`	`n:camel`	manifest-owned-internal	NexusEngine Core
`src/core-domains/camel/index.js`	`n:camel`	manifest-owned-internal	NexusEngine Core
`src/core-domains/camel/kits/runtime-kit/index.js`	`n:camel`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/camel/nexus-api/binding.js`	`n:camel`	manifest-owned-internal	NexusEngine Core
`src/core-domains/camel/runtime/asyncify.js`	`n:camel`	manifest-owned-internal	NexusEngine Core
`src/core-domains/camel/runtime/browser-worker.js`	`n:camel`	manifest-owned-internal	NexusEngine Core
`src/core-domains/camel/runtime/browser.js`	`n:camel`	manifest-owned-internal	NexusEngine Core
`src/core-domains/camel/runtime/native.js`	`n:camel`	manifest-owned-internal	NexusEngine Core
`src/core-domains/camel/runtime/wire.js`	`n:camel`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/camera/adapters/camera-world-occlusion-adapter-kit/contracts.js`	`n:camera`	manifest-owned-internal	NexusEngine Core
`src/core-domains/camera/adapters/camera-world-occlusion-adapter-kit/index.js`	`n:camera`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/camera/adapters/camera-world-occlusion-adapter-kit/kit.manifest.js`	`n:camera`	manifest-owned-internal	NexusEngine Core
`src/core-domains/camera/adapters/camera-world-occlusion-adapter-kit/services.js`	`n:camera`	manifest-owned-internal	NexusEngine Core
`src/core-domains/camera/adapters/camera-world-occlusion-adapter-kit/state.js`	`n:camera`	manifest-owned-internal	NexusEngine Core
`src/core-domains/camera/contracts.js`	`n:camera`	manifest-owned-internal	NexusEngine Core
`src/core-domains/camera/domain.manifest.js`	`n:camera`	manifest-owned-internal	NexusEngine Core
`src/core-domains/camera/execution/camera-contracts.js`	`n:camera`	manifest-owned-internal	NexusEngine Core
`src/core-domains/camera/execution/camera-kit.js`	`n:camera`	manifest-owned-internal	NexusEngine Core
`src/core-domains/camera/execution/camera-manifests.js`	`n:camera`	manifest-owned-internal	NexusEngine Core
`src/core-domains/camera/execution/index.js`	`n:camera`	manifest-owned-internal	NexusEngine Core
`src/core-domains/camera/execution/kits/camera-binding-kit/contracts.js`	`n:camera`	manifest-owned-internal	NexusEngine Core
`src/core-domains/camera/execution/kits/camera-binding-kit/index.js`	`n:camera`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/camera/execution/kits/camera-binding-kit/kit.manifest.js`	`n:camera`	manifest-owned-internal	NexusEngine Core
`src/core-domains/camera/execution/kits/camera-jitter-kit/contracts.js`	`n:camera`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/camera/execution/kits/camera-jitter-kit/index.js`	`n:camera`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/camera/execution/kits/camera-jitter-kit/kit.manifest.js`	`n:camera`	manifest-owned-internal	NexusEngine Core
`src/core-domains/camera/execution/kits/camera-projection-kit/contracts.js`	`n:camera`	manifest-owned-internal	NexusEngine Core
`src/core-domains/camera/execution/kits/camera-projection-kit/index.js`	`n:camera`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/camera/execution/kits/camera-projection-kit/kit.manifest.js`	`n:camera`	manifest-owned-internal	NexusEngine Core
`src/core-domains/camera/execution/kits/camera-reprojection-kit/contracts.js`	`n:camera`	manifest-owned-internal	NexusEngine Core
`src/core-domains/camera/execution/kits/camera-reprojection-kit/index.js`	`n:camera`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/camera/execution/kits/camera-reprojection-kit/kit.manifest.js`	`n:camera`	manifest-owned-internal	NexusEngine Core
`src/core-domains/camera/execution/kits/camera-view-kit/contracts.js`	`n:camera`	manifest-owned-internal	NexusEngine Core
`src/core-domains/camera/execution/kits/camera-view-kit/index.js`	`n:camera`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/camera/execution/kits/camera-view-kit/kit.manifest.js`	`n:camera`	manifest-owned-internal	NexusEngine Core
`src/core-domains/camera/execution/kits/camera-viewport-kit/contracts.js`	`n:camera`	manifest-owned-internal	NexusEngine Core
`src/core-domains/camera/execution/kits/camera-viewport-kit/index.js`	`n:camera`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/camera/execution/kits/camera-viewport-kit/kit.manifest.js`	`n:camera`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/camera/execution/kits/multiview-camera-kit/contracts.js`	`n:camera`	manifest-owned-internal	NexusEngine Core
`src/core-domains/camera/execution/kits/multiview-camera-kit/index.js`	`n:camera`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/camera/execution/kits/multiview-camera-kit/kit.manifest.js`	`n:camera`	manifest-owned-internal	NexusEngine Core
`src/core-domains/camera/execution/kits/stereo-camera-kit/contracts.js`	`n:camera`	manifest-owned-internal	NexusEngine Core
`src/core-domains/camera/execution/kits/stereo-camera-kit/index.js`	`n:camera`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/camera/execution/kits/stereo-camera-kit/kit.manifest.js`	`n:camera`	manifest-owned-internal	NexusEngine Core
`src/core-domains/camera/index.js`	`n:camera`	manifest-owned-internal	NexusEngine Core
`src/core-domains/camera/kits/camera-control-kit.js`	`n:camera`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/camera/kits/camera-kit/index.js`	`n:camera`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/camera/kits/camera-kit/smoothing.js`	`n:camera`	manifest-owned-internal	NexusEngine Core
`src/core-domains/camera/kits/framing-kit/index.js`	`n:camera`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/camera/kits/framing-kit/orthographic-fit.js`	`n:camera`	manifest-owned-internal	NexusEngine Core
`src/core-domains/camera/kits/framing-kit/perspective-fit.js`	`n:camera`	manifest-owned-internal	NexusEngine Core
`src/core-domains/camera/third-person/index.js`	`n:camera`	manifest-owned-internal	NexusEngine Core
`src/core-domains/camera/third-person/kits/third-person-camera-kit/contracts.js`	`n:camera`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/camera/third-person/kits/third-person-camera-kit/index.js`	`n:camera`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/camera/third-person/kits/third-person-camera-kit/kit.manifest.js`	`n:camera`	manifest-owned-internal	NexusEngine Core
`src/core-domains/camera/third-person/kits/third-person-camera-kit/services.js`	`n:camera`	manifest-owned-internal	NexusEngine Core
`src/core-domains/camera/third-person/kits/third-person-camera-kit/state.js`	`n:camera`	manifest-owned-internal	NexusEngine Core
`src/core-domains/camera/third-person/subdomain.manifest.js`	`n:camera`	manifest-owned-internal	NexusEngine Core
`src/core-domains/catalog.js`	`n:composition`	manifest-infrastructure	NexusEngine Core
`src/core-domains/composition/adapters/mcp/composition-mcp-provider.js`	`n:composition`	manifest-owned-internal	NexusEngine Core
`src/core-domains/composition/adapters/mcp/generated-guide-resources.js`	`n:composition`	manifest-owned-internal	NexusEngine Core
`src/core-domains/composition/domain.manifest.js`	`n:composition`	manifest-owned-internal	NexusEngine Core
`src/core-domains/composition/index.js`	`n:composition`	manifest-owned-internal	NexusEngine Core
`src/core-domains/composition/kits/composition-registry-kit/composition-tree.js`	`n:composition`	manifest-owned-internal	NexusEngine Core
`src/core-domains/composition/kits/composition-registry-kit/index.js`	`n:composition`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/composition/kits/composition-registry-kit/registry.js`	`n:composition`	manifest-owned-internal	NexusEngine Core
`src/core-domains/composition/kits/composition-registry-kit/services.js`	`n:composition`	manifest-owned-internal	NexusEngine Core
`src/core-domains/composition/recipes/restored-behavior-recipes.js`	`n:composition`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/composition/services/composition-apply-controller.js`	`n:composition`	manifest-owned-internal	NexusEngine Core
`src/core-domains/compute/dispatch/index.js`	`n:compute`	manifest-owned-internal	NexusEngine Core
`src/core-domains/compute/domain.manifest.js`	`n:compute`	manifest-owned-internal	NexusEngine Core
`src/core-domains/compute/graph/index.js`	`n:compute`	manifest-owned-internal	NexusEngine Core
`src/core-domains/compute/host/execution/cpu/javascript/index.js`	`n:compute`	manifest-owned-internal	NexusEngine Core
`src/core-domains/compute/host/execution/cpu/native/index.js`	`n:compute`	manifest-owned-internal	NexusEngine Core
`src/core-domains/compute/host/execution/cpu/wasm/index.js`	`n:compute`	manifest-owned-internal	NexusEngine Core
`src/core-domains/compute/host/execution/gpu/opengl/index.js`	`n:compute`	manifest-owned-internal	NexusEngine Core
`src/core-domains/compute/host/execution/gpu/vulkan/index.js`	`n:compute`	manifest-owned-internal	NexusEngine Core
`src/core-domains/compute/host/index.js`	`n:compute`	manifest-owned-internal	NexusEngine Core
`src/core-domains/compute/index.js`	`n:compute`	manifest-owned-internal	NexusEngine Core
`src/core-domains/compute/kits/compute-kit/descriptors.js`	`n:compute`	manifest-owned-internal	NexusEngine Core
`src/core-domains/compute/kits/compute-kit/index.js`	`n:compute`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/compute/kits/compute-kit/provider.js`	`n:compute`	manifest-owned-internal	NexusEngine Core
`src/core-domains/compute/model/kits/model-kit/adapters.js`	`n:compute`	manifest-owned-internal	NexusEngine Core
`src/core-domains/compute/model/kits/model-kit/index.js`	`n:compute`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/compute/model/kits/model-kit/inference-request.js`	`n:compute`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/compute/model/kits/model-kit/inference-result.js`	`n:compute`	manifest-owned-internal	NexusEngine Core
`src/core-domains/compute/model/kits/model-kit/model-descriptors.js`	`n:compute`	manifest-owned-internal	NexusEngine Core
`src/core-domains/compute/model/kits/model-kit/model-registry.js`	`n:compute`	manifest-owned-internal	NexusEngine Core
`src/core-domains/compute/portable.js`	`n:compute`	manifest-owned-internal	NexusEngine Core
`src/core-domains/compute/resource/index.js`	`n:compute`	manifest-owned-internal	NexusEngine Core
`src/core-domains/diagnostics/domain.manifest.js`	`n:diagnostics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/diagnostics/index.js`	`n:diagnostics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/diagnostics/kits/debug-descriptor-kit/index.js`	`n:diagnostics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/diagnostics/kits/debug-draw-kit.js`	`n:diagnostics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/diagnostics/kits/diagnostics-kit/index.js`	`n:diagnostics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/domain-kit.js`	`n:composition`	manifest-infrastructure	NexusEngine Core
`src/core-domains/domain-manifest.js`	`n:composition`	manifest-infrastructure	NexusEngine Core
`src/core-domains/host/domain.manifest.js`	`n:host`	manifest-owned-internal	NexusEngine Core
`src/core-domains/host/gpu/index.js`	`n:host`	manifest-owned-internal	NexusEngine Core
`src/core-domains/host/index.js`	`n:host`	manifest-owned-internal	NexusEngine Core
`src/core-domains/host/kits/host-capability-kit/index.js`	`n:host`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/index.js`	`n:composition`	manifest-infrastructure	NexusEngine Core
`src/core-domains/interaction/adapters/occupant-request-adapter-kit/contracts.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/adapters/occupant-request-adapter-kit/index.js`	`n:interaction`	manifest-proven-public-atom	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/interaction/adapters/occupant-request-adapter-kit/kit.manifest.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/adapters/occupant-request-adapter-kit/services.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/adapters/occupant-request-adapter-kit/state.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/adapters/request-economy-adapter-kit/contracts.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/adapters/request-economy-adapter-kit/index.js`	`n:interaction`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/interaction/adapters/request-economy-adapter-kit/kit.manifest.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/adapters/request-economy-adapter-kit/services.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/adapters/request-economy-adapter-kit/state.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/adapters/transport-request-adapter-kit/contracts.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/adapters/transport-request-adapter-kit/index.js`	`n:interaction`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/interaction/adapters/transport-request-adapter-kit/kit.manifest.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/adapters/transport-request-adapter-kit/services.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/adapters/transport-request-adapter-kit/state.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/interaction/assistance-target/index.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/assistance-target/kits/assistance-target-kit/contracts.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/assistance-target/kits/assistance-target-kit/index.js`	`n:interaction`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/interaction/assistance-target/kits/assistance-target-kit/kit.manifest.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/assistance-target/kits/assistance-target-kit/services.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/assistance-target/kits/assistance-target-kit/state.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/assistance-target/subdomain.manifest.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/domain.manifest.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/environmental-affordance/index.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/environmental-affordance/kits/environmental-affordance-kit/contracts.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/environmental-affordance/kits/environmental-affordance-kit/index.js`	`n:interaction`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/interaction/environmental-affordance/kits/environmental-affordance-kit/kit.manifest.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/environmental-affordance/kits/environmental-affordance-kit/services.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/interaction/environmental-affordance/kits/environmental-affordance-kit/state.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/environmental-affordance/subdomain.manifest.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/index.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/input/kits/input-kit/actions.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/input/kits/input-kit/adapters.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/input/kits/input-kit/bindings.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/input/kits/input-kit/contexts.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/input/kits/input-kit/index.js`	`n:interaction`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/interaction/input/kits/input-kit/intent.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/kits/interaction-kit/activation.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/kits/interaction-kit/affordances.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/kits/interaction-kit/index.js`	`n:interaction`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/interaction/kits/interaction-kit/interactors.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/kits/interaction-kit/prompts.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/kits/interaction-kit/results.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/kits/interaction-kit/targets.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/interaction/request/fulfillment/index.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/request/fulfillment/kits/request-fulfillment-kit/contracts.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/request/fulfillment/kits/request-fulfillment-kit/index.js`	`n:interaction`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/interaction/request/fulfillment/kits/request-fulfillment-kit/kit.manifest.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/request/fulfillment/kits/request-fulfillment-kit/services.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/request/fulfillment/kits/request-fulfillment-kit/state.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/request/fulfillment/subdomain.manifest.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/request/index.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/request/queue/index.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/request/queue/kits/request-queue-kit/contracts.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/request/queue/kits/request-queue-kit/index.js`	`n:interaction`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/interaction/request/queue/kits/request-queue-kit/kit.manifest.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/request/queue/kits/request-queue-kit/services.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/request/queue/kits/request-queue-kit/state.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/interaction/request/queue/subdomain.manifest.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/request/subdomain.manifest.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/restored-behavior-manifests.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/transfer-zone/index.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/transfer-zone/kits/transfer-zone-kit/contracts.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/transfer-zone/kits/transfer-zone-kit/index.js`	`n:interaction`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/interaction/transfer-zone/kits/transfer-zone-kit/kit.manifest.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/transfer-zone/kits/transfer-zone-kit/services.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/transfer-zone/kits/transfer-zone-kit/state.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/interaction/transfer-zone/subdomain.manifest.js`	`n:interaction`	manifest-owned-internal	NexusEngine Core
`src/core-domains/lighting/descriptors.js`	`n:lighting`	manifest-owned-internal	NexusEngine Core
`src/core-domains/lighting/domain.manifest.js`	`n:lighting`	manifest-owned-internal	NexusEngine Core
`src/core-domains/lighting/index.js`	`n:lighting`	manifest-owned-internal	NexusEngine Core
`src/core-domains/lighting/kits/lighting-descriptor-kit/index.js`	`n:lighting`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/lighting/reflection/adapters.js`	`n:lighting`	manifest-owned-internal	NexusEngine Core
`src/core-domains/lighting/reflection/contract.js`	`n:lighting`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/lighting/reflection/index.js`	`n:lighting`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/manifest-domain-service-kit.js`	`n:composition`	manifest-infrastructure	NexusEngine Core
`src/core-domains/manifest-input.js`	`n:composition`	manifest-infrastructure	NexusEngine Core
`src/core-domains/manifest-kit-contract.js`	`n:composition`	manifest-infrastructure	NexusEngine Core
`src/core-domains/material/appearance.js`	`n:material`	manifest-owned-internal	NexusEngine Core
`src/core-domains/material/domain.manifest.js`	`n:material`	manifest-owned-internal	NexusEngine Core
`src/core-domains/material/index.js`	`n:material`	manifest-owned-internal	NexusEngine Core
`src/core-domains/material/kits/material-binding-kit/contracts.js`	`n:material`	manifest-owned-internal	NexusEngine Core
`src/core-domains/material/kits/material-binding-kit/index.js`	`n:material`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/material/kits/material-binding-kit/kit.manifest.js`	`n:material`	manifest-owned-internal	NexusEngine Core
`src/core-domains/material/kits/material-cache-kit/contracts.js`	`n:material`	manifest-owned-internal	NexusEngine Core
`src/core-domains/material/kits/material-cache-kit/index.js`	`n:material`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/material/kits/material-cache-kit/kit.manifest.js`	`n:material`	manifest-owned-internal	NexusEngine Core
`src/core-domains/material/kits/material-contract-kit/contracts.js`	`n:material`	manifest-owned-internal	NexusEngine Core
`src/core-domains/material/kits/material-contract-kit/index.js`	`n:material`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/material/kits/material-contract-kit/kit.manifest.js`	`n:material`	manifest-owned-internal	NexusEngine Core
`src/core-domains/material/kits/material-descriptor-kit/index.js`	`n:material`	manifest-proven-public-atom	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/material/kits/material-instance-kit/contracts.js`	`n:material`	manifest-owned-internal	NexusEngine Core
`src/core-domains/material/kits/material-instance-kit/index.js`	`n:material`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/material/kits/material-instance-kit/kit.manifest.js`	`n:material`	manifest-owned-internal	NexusEngine Core
`src/core-domains/material/kits/material-parameter-kit/contracts.js`	`n:material`	manifest-owned-internal	NexusEngine Core
`src/core-domains/material/kits/material-parameter-kit/index.js`	`n:material`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/material/kits/material-parameter-kit/kit.manifest.js`	`n:material`	manifest-owned-internal	NexusEngine Core
`src/core-domains/material/kits/material-validation-kit/contracts.js`	`n:material`	manifest-owned-internal	NexusEngine Core
`src/core-domains/material/kits/material-validation-kit/index.js`	`n:material`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/material/kits/material-validation-kit/kit.manifest.js`	`n:material`	manifest-owned-internal	NexusEngine Core
`src/core-domains/material/kits/material-variant-kit/contracts.js`	`n:material`	manifest-owned-internal	NexusEngine Core
`src/core-domains/material/kits/material-variant-kit/index.js`	`n:material`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/material/kits/material-variant-kit/kit.manifest.js`	`n:material`	manifest-owned-internal	NexusEngine Core
`src/core-domains/material/kits/sampler-binding-kit/contracts.js`	`n:material`	manifest-owned-internal	NexusEngine Core
`src/core-domains/material/kits/sampler-binding-kit/index.js`	`n:material`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/material/kits/sampler-binding-kit/kit.manifest.js`	`n:material`	manifest-owned-internal	NexusEngine Core
`src/core-domains/material/kits/texture-binding-kit/contracts.js`	`n:material`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/material/kits/texture-binding-kit/index.js`	`n:material`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/material/kits/texture-binding-kit/kit.manifest.js`	`n:material`	manifest-owned-internal	NexusEngine Core
`src/core-domains/material/material-contracts.js`	`n:material`	manifest-owned-internal	NexusEngine Core
`src/core-domains/material/material-manifests.js`	`n:material`	manifest-owned-internal	NexusEngine Core
`src/core-domains/material/material-registry-kit.js`	`n:material`	manifest-owned-internal	NexusEngine Core
`src/core-domains/material/procedural.js`	`n:material`	manifest-owned-internal	NexusEngine Core
`src/core-domains/mcp/adapters/node-mcp-sdk-adapter/index.js`	`n:mcp`	manifest-owned-internal	NexusEngine Core
`src/core-domains/mcp/contracts/contract-utilities.js`	`n:mcp`	manifest-owned-internal	NexusEngine Core
`src/core-domains/mcp/contracts/prompt-contract.js`	`n:mcp`	manifest-owned-internal	NexusEngine Core
`src/core-domains/mcp/contracts/provider-contract.js`	`n:mcp`	manifest-owned-internal	NexusEngine Core
`src/core-domains/mcp/contracts/resource-contract.js`	`n:mcp`	manifest-owned-internal	NexusEngine Core
`src/core-domains/mcp/contracts/result-contract.js`	`n:mcp`	manifest-owned-internal	NexusEngine Core
`src/core-domains/mcp/contracts/tool-contract.js`	`n:mcp`	manifest-owned-internal	NexusEngine Core
`src/core-domains/mcp/domain.manifest.js`	`n:mcp`	manifest-owned-internal	NexusEngine Core
`src/core-domains/mcp/index.js`	`n:mcp`	manifest-owned-internal	NexusEngine Core
`src/core-domains/mcp/kits/mcp-registry-kit/authorization.js`	`n:mcp`	manifest-owned-internal	NexusEngine Core
`src/core-domains/mcp/kits/mcp-registry-kit/index.js`	`n:mcp`	manifest-proven-public-atom	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/mcp/kits/mcp-registry-kit/registry.js`	`n:mcp`	manifest-owned-internal	NexusEngine Core
`src/core-domains/mcp/state/registry-snapshot.js`	`n:mcp`	manifest-owned-internal	NexusEngine Core
`src/core-domains/network/domain.manifest.js`	`n:network`	manifest-owned-internal	NexusEngine Core
`src/core-domains/network/index.js`	`n:network`	manifest-owned-internal	NexusEngine Core
`src/core-domains/network/kits/network-kit/index.js`	`n:network`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/network/multiplayer/authority/contracts.js`	`n:network`	manifest-owned-internal	NexusEngine Core
`src/core-domains/network/multiplayer/authority/index.js`	`n:network`	manifest-owned-internal	NexusEngine Core
`src/core-domains/network/multiplayer/authority/kits/multiplayer-authority-kit/index.js`	`n:network`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/network/multiplayer/authority/subdomain.manifest.js`	`n:network`	manifest-owned-internal	NexusEngine Core
`src/core-domains/network/multiplayer/contracts.js`	`n:network`	manifest-owned-internal	NexusEngine Core
`src/core-domains/network/multiplayer/index.js`	`n:network`	manifest-owned-internal	NexusEngine Core
`src/core-domains/network/multiplayer/kits/multiplayer-contract-kit/index.js`	`n:network`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/network/multiplayer/replication/contracts.js`	`n:network`	manifest-owned-internal	NexusEngine Core
`src/core-domains/network/multiplayer/replication/index.js`	`n:network`	manifest-owned-internal	NexusEngine Core
`src/core-domains/network/multiplayer/replication/kits/multiplayer-replication-kit/index.js`	`n:network`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/network/multiplayer/replication/subdomain.manifest.js`	`n:network`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/network/multiplayer/session/contracts.js`	`n:network`	manifest-owned-internal	NexusEngine Core
`src/core-domains/network/multiplayer/session/index.js`	`n:network`	manifest-owned-internal	NexusEngine Core
`src/core-domains/network/multiplayer/session/kits/multiplayer-session-kit/index.js`	`n:network`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/network/multiplayer/session/subdomain.manifest.js`	`n:network`	manifest-owned-internal	NexusEngine Core
`src/core-domains/network/multiplayer/subdomain.manifest.js`	`n:network`	manifest-owned-internal	NexusEngine Core
`src/core-domains/network/multiplayer/tick-sync/contracts.js`	`n:network`	manifest-owned-internal	NexusEngine Core
`src/core-domains/network/multiplayer/tick-sync/index.js`	`n:network`	manifest-owned-internal	NexusEngine Core
`src/core-domains/network/multiplayer/tick-sync/kits/multiplayer-tick-sync-kit/index.js`	`n:network`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/network/multiplayer/tick-sync/subdomain.manifest.js`	`n:network`	manifest-owned-internal	NexusEngine Core
`src/core-domains/network/portable.js`	`n:network`	manifest-owned-internal	NexusEngine Core
`src/core-domains/network/transport/contracts.js`	`n:network`	manifest-owned-internal	NexusEngine Core
`src/core-domains/network/transport/index.js`	`n:network`	manifest-owned-internal	NexusEngine Core
`src/core-domains/network/transport/kits/network-transport-contract-kit/index.js`	`n:network`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/network/transport/subdomain.manifest.js`	`n:network`	manifest-owned-internal	NexusEngine Core
`src/core-domains/object/adapters/object-shape-fidelity-adapter-kit/index.js`	`n:object`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/object/contracts/object-descriptor.js`	`n:object`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/object/domain.manifest.js`	`n:object`	manifest-owned-internal	NexusEngine Core
`src/core-domains/object/fidelity/index.js`	`n:object`	manifest-owned-internal	NexusEngine Core
`src/core-domains/object/fidelity/kits/object-fidelity-kit/descriptors.js`	`n:object`	manifest-owned-internal	NexusEngine Core
`src/core-domains/object/fidelity/kits/object-fidelity-kit/index.js`	`n:object`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/object/index.js`	`n:object`	manifest-owned-internal	NexusEngine Core
`src/core-domains/object/kits/object-registry-kit/index.js`	`n:object`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/object/placement/contracts/placement-descriptor.js`	`n:object`	manifest-owned-internal	NexusEngine Core
`src/core-domains/object/placement/index.js`	`n:object`	manifest-owned-internal	NexusEngine Core
`src/core-domains/object/placement/kits/object-placement-kit/index.js`	`n:object`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/object/placement/kits/object-placement-kit/placement-math.js`	`n:object`	manifest-owned-internal	NexusEngine Core
`src/core-domains/object/placement/kits/object-placement-kit/placement-operations.js`	`n:object`	manifest-owned-internal	NexusEngine Core
`src/core-domains/object/placement/kits/object-placement-kit/placement-validation.js`	`n:object`	manifest-owned-internal	NexusEngine Core
`src/core-domains/object/shape/index.js`	`n:object`	manifest-owned-internal	NexusEngine Core
`src/core-domains/object/shape/kits/object-shape-kit/descriptors.js`	`n:object`	manifest-owned-internal	NexusEngine Core
`src/core-domains/object/shape/kits/object-shape-kit/index.js`	`n:object`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/object/shape/kits/object-shape-kit/metrics.js`	`n:object`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/object/shape/kits/object-shape-kit/profiles.js`	`n:object`	manifest-owned-internal	NexusEngine Core
`src/core-domains/object/shape/kits/object-shape-kit/provider.js`	`n:object`	manifest-owned-internal	NexusEngine Core
`src/core-domains/object/shape/kits/object-shape-kit/qualification.js`	`n:object`	manifest-owned-internal	NexusEngine Core
`src/core-domains/object/shape/providers/meshoptimizer-shape-provider-kit/index.js`	`n:object`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/object/shape/providers/meshoptimizer-shape-provider-kit/meshoptimizer-provider.js`	`n:object`	manifest-owned-internal	NexusEngine Core
`src/core-domains/object/shape/providers/meshoptimizer-shape-provider-kit/reference-provider.js`	`n:object`	manifest-owned-internal	NexusEngine Core
`src/core-domains/object/state/object-registry-state.js`	`n:object`	manifest-owned-internal	NexusEngine Core
`src/core-domains/object/vegetation/adapters/vegetation-object-bridge-kit/index.js`	`n:object`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/object/vegetation/ecology-domain/index.js`	`n:object`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/object/vegetation/foliage-domain/index.js`	`n:object`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/object/vegetation/index.js`	`n:object`	manifest-owned-internal	NexusEngine Core
`src/core-domains/object/vegetation/kits/object-vegetation-kit/contracts.js`	`n:object`	manifest-owned-internal	NexusEngine Core
`src/core-domains/object/vegetation/kits/object-vegetation-kit/index.js`	`n:object`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/object/vegetation/tree-domain/index.js`	`n:object`	manifest-proven-public-atom	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/physics/body/atomic-body-kit.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/body/body-contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/body/body-manifests.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/body/index.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/body/kits/body-damping-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/body/kits/body-damping-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/body/kits/body-damping-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/body/kits/body-force-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/body/kits/body-force-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/body/kits/body-force-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/body/kits/body-identity-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/body/kits/body-identity-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/body/kits/body-identity-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/body/kits/body-inertia-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/body/kits/body-inertia-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/body/kits/body-inertia-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/physics/body/kits/body-lifecycle-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/body/kits/body-lifecycle-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/body/kits/body-lifecycle-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/body/kits/body-mass-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/body/kits/body-mass-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/body/kits/body-mass-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/body/kits/body-pose-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/body/kits/body-pose-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/body/kits/body-pose-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/body/kits/body-registry-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/body/kits/body-registry-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/body/kits/body-registry-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/body/kits/body-registry-kit/step-commit.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/body/kits/body-sleep-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/body/kits/body-sleep-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/body/kits/body-sleep-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/physics/body/kits/body-state-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/body/kits/body-state-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/body/kits/body-state-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/body/kits/body-type-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/body/kits/body-type-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/body/kits/body-type-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/body/kits/body-velocity-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/body/kits/body-velocity-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/body/kits/body-velocity-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/body/kits/body-wake-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/body/kits/body-wake-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/body/kits/body-wake-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/body/subdomain.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/collider/atomic-collider-kit.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/collider/collider-contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/collider/collider-manifests.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/collider/index.js`	`n:physics`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/physics/collider/kits/collider-attachment-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/collider/kits/collider-attachment-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/collider/kits/collider-attachment-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/collider/kits/collider-filter-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/collider/kits/collider-filter-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/collider/kits/collider-filter-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/collider/kits/collider-identity-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/collider/kits/collider-identity-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/collider/kits/collider-identity-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/collider/kits/collider-lifecycle-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/collider/kits/collider-lifecycle-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/collider/kits/collider-lifecycle-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/collider/kits/collider-material-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/physics/collider/kits/collider-material-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/collider/kits/collider-material-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/collider/kits/collider-pose-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/collider/kits/collider-pose-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/collider/kits/collider-pose-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/collider/kits/collider-registry-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/collider/kits/collider-registry-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/collider/kits/collider-registry-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/collider/kits/collision-group-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/collider/kits/collision-group-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/collider/kits/collision-group-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/collider/kits/collision-layer-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/collider/kits/collision-layer-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/physics/collider/kits/collision-layer-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/collider/kits/collision-mask-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/collider/kits/collision-mask-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/collider/kits/collision-mask-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/collider/kits/sensor-collider-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/collider/kits/sensor-collider-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/collider/kits/sensor-collider-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/collider/kits/trigger-collider-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/collider/kits/trigger-collider-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/collider/kits/trigger-collider-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/collider/subdomain.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/constraints/atomic-constraint-kit.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/constraints/constraints-contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/constraints/constraints-manifests.js`	`n:physics`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/physics/constraints/index.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/constraints/kits/ball-socket-constraint-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/constraints/kits/ball-socket-constraint-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/constraints/kits/ball-socket-constraint-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/constraints/kits/cone-twist-constraint-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/constraints/kits/cone-twist-constraint-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/constraints/kits/cone-twist-constraint-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/constraints/kits/constraint-break-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/constraints/kits/constraint-break-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/constraints/kits/constraint-break-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/constraints/kits/constraint-registry-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/constraints/kits/constraint-registry-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/constraints/kits/constraint-registry-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/physics/constraints/kits/distance-constraint-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/constraints/kits/distance-constraint-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/constraints/kits/distance-constraint-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/constraints/kits/drive-constraint-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/constraints/kits/drive-constraint-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/constraints/kits/drive-constraint-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/constraints/kits/fixed-constraint-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/constraints/kits/fixed-constraint-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/constraints/kits/fixed-constraint-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/constraints/kits/hinge-constraint-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/constraints/kits/hinge-constraint-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/constraints/kits/hinge-constraint-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/constraints/kits/limit-constraint-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/physics/constraints/kits/limit-constraint-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/constraints/kits/limit-constraint-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/constraints/kits/motor-constraint-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/constraints/kits/motor-constraint-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/constraints/kits/motor-constraint-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/constraints/kits/slider-constraint-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/constraints/kits/slider-constraint-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/constraints/kits/slider-constraint-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/constraints/kits/spring-constraint-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/constraints/kits/spring-constraint-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/constraints/kits/spring-constraint-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/constraints/subdomain.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/contracts/contract-manifests.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/contracts/index.js`	`n:physics`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/physics/contracts/kits/physics-command-schema-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/contracts/kits/physics-command-schema-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/contracts/kits/physics-command-schema-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/contracts/kits/physics-domain-contract-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/contracts/kits/physics-domain-contract-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/contracts/kits/physics-domain-contract-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/contracts/kits/physics-event-schema-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/contracts/kits/physics-event-schema-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/contracts/kits/physics-event-schema-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/contracts/kits/physics-provider-contract-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/contracts/kits/physics-provider-contract-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/contracts/kits/physics-provider-contract-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/contracts/kits/physics-query-schema-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/physics/contracts/kits/physics-query-schema-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/contracts/kits/physics-query-schema-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/contracts/kits/physics-state-schema-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/contracts/kits/physics-state-schema-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/contracts/kits/physics-state-schema-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/contracts/portable-value.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/contracts/subdomain.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/detection/detection-algorithms.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/detection/detection-contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/detection/detection-kit.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/detection/detection-manifests.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/detection/detection-math.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/detection/index.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/detection/kits/broad-phase-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/detection/kits/broad-phase-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/physics/detection/kits/broad-phase-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/detection/kits/broad-phase-pair-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/detection/kits/broad-phase-pair-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/detection/kits/broad-phase-pair-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/detection/kits/collision-detection-result-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/detection/kits/collision-detection-result-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/detection/kits/collision-detection-result-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/detection/kits/continuous-collision-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/detection/kits/continuous-collision-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/detection/kits/continuous-collision-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/detection/kits/dynamic-tree-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/detection/kits/dynamic-tree-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/detection/kits/dynamic-tree-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/physics/detection/kits/epa-penetration-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/detection/kits/epa-penetration-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/detection/kits/epa-penetration-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/detection/kits/gjk-detection-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/detection/kits/gjk-detection-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/detection/kits/gjk-detection-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/detection/kits/narrow-phase-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/detection/kits/narrow-phase-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/detection/kits/narrow-phase-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/detection/kits/shape-intersection-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/detection/kits/shape-intersection-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/detection/kits/shape-intersection-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/detection/kits/spatial-partition-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/detection/kits/spatial-partition-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/physics/detection/kits/spatial-partition-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/detection/kits/sweep-and-prune-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/detection/kits/sweep-and-prune-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/detection/kits/sweep-and-prune-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/detection/subdomain.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/domain.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/index.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/lifecycle/index.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/lifecycle/kits/physics-installation-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/lifecycle/kits/physics-installation-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/lifecycle/kits/physics-installation-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/lifecycle/kits/physics-reset-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/lifecycle/kits/physics-reset-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/lifecycle/kits/physics-reset-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/physics/lifecycle/kits/physics-shutdown-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/lifecycle/kits/physics-shutdown-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/lifecycle/kits/physics-shutdown-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/lifecycle/kits/physics-snapshot-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/lifecycle/kits/physics-snapshot-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/lifecycle/kits/physics-snapshot-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/lifecycle/kits/physics-startup-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/lifecycle/kits/physics-startup-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/lifecycle/kits/physics-startup-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/lifecycle/kits/physics-step-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/lifecycle/kits/physics-step-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/lifecycle/kits/physics-step-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/lifecycle/lifecycle-contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/lifecycle/lifecycle-manifests.js`	`n:physics`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/physics/lifecycle/subdomain.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/material/index.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/material/kits/density-material-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/material/kits/density-material-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/material/kits/density-material-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/material/kits/friction-material-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/material/kits/friction-material-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/material/kits/friction-material-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/material/kits/material-combine-policy-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/material/kits/material-combine-policy-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/material/kits/material-combine-policy-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/material/kits/physics-material-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/material/kits/physics-material-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/material/kits/physics-material-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/physics/material/kits/restitution-material-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/material/kits/restitution-material-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/material/kits/restitution-material-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/material/kits/surface-material-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/material/kits/surface-material-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/material/kits/surface-material-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/material/material-contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/material/material-manifests.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/material/subdomain.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/shape/index.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/shape/kits/box-shape/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/shape/kits/box-shape/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/shape/kits/box-shape/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/shape/kits/capsule-shape/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/shape/kits/capsule-shape/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/physics/shape/kits/capsule-shape/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/shape/kits/compound-shape/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/shape/kits/compound-shape/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/shape/kits/compound-shape/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/shape/kits/cone-shape/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/shape/kits/cone-shape/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/shape/kits/cone-shape/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/shape/kits/convex-shape/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/shape/kits/convex-shape/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/shape/kits/convex-shape/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/shape/kits/cylinder-shape/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/shape/kits/cylinder-shape/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/shape/kits/cylinder-shape/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/shape/kits/heightfield-shape/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/shape/kits/heightfield-shape/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/shape/kits/heightfield-shape/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/shape/kits/plane-shape/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/physics/shape/kits/plane-shape/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/shape/kits/plane-shape/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/shape/kits/scaled-shape/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/shape/kits/scaled-shape/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/shape/kits/scaled-shape/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/shape/kits/shape-identity/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/shape/kits/shape-identity/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/shape/kits/shape-identity/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/shape/kits/shape-registry/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/shape/kits/shape-registry/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/shape/kits/shape-registry/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/shape/kits/shape-validation/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/shape/kits/shape-validation/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/shape/kits/shape-validation/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/shape/kits/sphere-shape/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/shape/kits/sphere-shape/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/shape/kits/sphere-shape/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/physics/shape/kits/triangle-mesh-shape/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/shape/kits/triangle-mesh-shape/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/shape/kits/triangle-mesh-shape/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/shape/shape-contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/shape/shape-manifests.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/shape/subdomain.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/world/index.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/world/kits/force-field-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/world/kits/force-field-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/world/kits/force-field-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/world/kits/gravity-field-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/world/kits/gravity-field-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/world/kits/gravity-field-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/world/kits/physics-world-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/world/kits/physics-world-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/world/kits/physics-world-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/physics/world/kits/physics-world-settings-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/world/kits/physics-world-settings-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/world/kits/physics-world-settings-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/world/kits/simulation-region-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/world/kits/simulation-region-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/world/kits/simulation-region-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/world/kits/time-scale-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/world/kits/time-scale-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/world/kits/time-scale-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/world/kits/wind-field-kit/contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/world/kits/wind-field-kit/index.js`	`n:physics`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/physics/world/kits/wind-field-kit/kit.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/world/record-registry.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/world/subdomain.manifest.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/world/world-contracts.js`	`n:physics`	manifest-owned-internal	NexusEngine Core
`src/core-domains/physics/world/world-manifests.js`	`n:physics`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/policy/domain.manifest.js`	`n:policy`	manifest-owned-internal	NexusEngine Core
`src/core-domains/policy/index.js`	`n:policy`	manifest-owned-internal	NexusEngine Core
`src/core-domains/policy/kits/policy-kit/index.js`	`n:policy`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/presentation/animation/kits/animation-kit/index.js`	`n:presentation`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/presentation/animation/kits/rig-transform-kit.js`	`n:presentation`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/presentation/audio/kits/audio-kit/index.js`	`n:presentation`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/presentation/capture/index.js`	`n:presentation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/presentation/capture/kits/capture-kit/descriptors.js`	`n:presentation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/presentation/capture/kits/capture-kit/index.js`	`n:presentation`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/presentation/capture/kits/capture-kit/provider.js`	`n:presentation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/presentation/contracts.js`	`n:presentation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/presentation/domain.manifest.js`	`n:presentation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/presentation/index.js`	`n:presentation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/presentation/kits/presentation-kit/index.js`	`n:presentation`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/presentation/output/kits/output-kit/index.js`	`n:presentation`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/presentation/output/kits/output-kit/math.js`	`n:presentation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/presentation/requests/vfx-descriptors.js`	`n:presentation`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/presentation/sky/kits/sky-kit/index.js`	`n:presentation`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/presentation/speech/index.js`	`n:presentation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/presentation/speech/kits/speech-kit/index.js`	`n:presentation`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/presentation/ui/kits/ui-kit/index.js`	`n:presentation`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/presentation/ui/kits/ui-kit/spatial-panel.js`	`n:presentation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/presentation/ui/kits/ui-scale-kit/index.js`	`n:presentation`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/presentation/ui/kits/ui-scale-kit/math.js`	`n:presentation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/batch/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/batch/instance-batches.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/buffer/buffer-contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/buffer/buffer-manifests.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/buffer/buffer-registry-kit.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/buffer/index.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/buffer/kits/buffer-layout-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/buffer/kits/buffer-layout-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/buffer/kits/buffer-layout-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/render/buffer/kits/buffer-resource-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/buffer/kits/buffer-resource-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/buffer/kits/buffer-resource-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/buffer/kits/index-buffer-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/buffer/kits/index-buffer-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/buffer/kits/index-buffer-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/buffer/kits/indirect-buffer-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/buffer/kits/indirect-buffer-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/buffer/kits/indirect-buffer-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/buffer/kits/instance-buffer-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/buffer/kits/instance-buffer-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/buffer/kits/instance-buffer-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/buffer/kits/storage-buffer-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/buffer/kits/storage-buffer-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/render/buffer/kits/storage-buffer-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/buffer/kits/uniform-buffer-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/buffer/kits/uniform-buffer-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/buffer/kits/uniform-buffer-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/buffer/kits/vertex-buffer-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/buffer/kits/vertex-buffer-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/buffer/kits/vertex-buffer-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/buffer/subdomain.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/contracts/contract-manifests.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/contracts/index.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/contracts/kits/render-domain-contract-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/contracts/kits/render-domain-contract-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/contracts/kits/render-domain-contract-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/contracts/kits/render-event-schema-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/render/contracts/kits/render-event-schema-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/contracts/kits/render-event-schema-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/contracts/kits/render-frame-schema-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/contracts/kits/render-frame-schema-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/contracts/kits/render-frame-schema-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/contracts/kits/render-pass-schema-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/contracts/kits/render-pass-schema-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/contracts/kits/render-pass-schema-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/contracts/kits/render-provider-contract-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/contracts/kits/render-provider-contract-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/contracts/kits/render-provider-contract-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/contracts/kits/render-resource-schema-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/contracts/kits/render-resource-schema-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/render/contracts/kits/render-resource-schema-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/contracts/kits/shader-schema-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/contracts/kits/shader-schema-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/contracts/kits/shader-schema-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/contracts/portable-value.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/contracts/subdomain.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/contracts/visual-contributions.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/device/device-contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/device/device-manifests.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/device/index.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/device/kits/device-capability-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/device/kits/device-capability-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/device/kits/device-capability-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/device/kits/device-diagnostics-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/device/kits/device-diagnostics-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/render/device/kits/device-diagnostics-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/device/kits/device-feature-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/device/kits/device-feature-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/device/kits/device-feature-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/device/kits/device-lifecycle-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/device/kits/device-lifecycle-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/device/kits/device-lifecycle-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/device/kits/device-limit-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/device/kits/device-limit-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/device/kits/device-limit-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/device/kits/device-loss-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/device/kits/device-loss-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/device/kits/device-loss-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/device/kits/device-memory-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/device/kits/device-memory-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/render/device/kits/device-memory-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/device/kits/device-queue-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/device/kits/device-queue-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/device/kits/device-queue-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/device/kits/render-device-contract-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/device/kits/render-device-contract-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/device/kits/render-device-contract-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/device/subdomain.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/domain.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/index.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/lifecycle/index.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/lifecycle/kits/render-installation-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/lifecycle/kits/render-installation-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/lifecycle/kits/render-installation-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/lifecycle/kits/render-recovery-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/render/lifecycle/kits/render-recovery-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/lifecycle/kits/render-recovery-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/lifecycle/kits/render-reset-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/lifecycle/kits/render-reset-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/lifecycle/kits/render-reset-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/lifecycle/kits/render-shutdown-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/lifecycle/kits/render-shutdown-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/lifecycle/kits/render-shutdown-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/lifecycle/kits/render-snapshot-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/lifecycle/kits/render-snapshot-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/lifecycle/kits/render-snapshot-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/lifecycle/kits/render-startup-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/lifecycle/kits/render-startup-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/lifecycle/kits/render-startup-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/render/lifecycle/lifecycle-contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/lifecycle/lifecycle-manifests.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/lifecycle/subdomain.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/pipeline/contract.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/pipeline/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/pipeline/quality-profiles.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/pipeline/render-descriptors.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/resource/index.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/resource/kits/render-resource-contract-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/resource/kits/render-resource-contract-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/resource/kits/render-resource-contract-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/resource/kits/resource-budget-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/resource/kits/resource-budget-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/resource/kits/resource-budget-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/resource/kits/resource-cache-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/render/resource/kits/resource-cache-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/resource/kits/resource-cache-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/resource/kits/resource-identity-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/resource/kits/resource-identity-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/resource/kits/resource-identity-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/resource/kits/resource-integrity-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/resource/kits/resource-integrity-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/resource/kits/resource-integrity-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/resource/kits/resource-lifecycle-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/resource/kits/resource-lifecycle-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/resource/kits/resource-lifecycle-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/resource/kits/resource-reference-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/resource/kits/resource-reference-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/render/resource/kits/resource-reference-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/resource/kits/resource-release-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/resource/kits/resource-release-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/resource/kits/resource-release-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/resource/kits/resource-state-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/resource/kits/resource-state-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/resource/kits/resource-state-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/resource/kits/resource-upload-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/resource/kits/resource-upload-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/resource/kits/resource-upload-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/resource/resource-contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/resource/resource-manifests.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/resource/subdomain.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/shader/index.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/shader/kits/shader-cache-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/render/shader/kits/shader-cache-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/shader/kits/shader-cache-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/shader/kits/shader-compile-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/shader/kits/shader-compile-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/shader/kits/shader-compile-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/shader/kits/shader-contract-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/shader/kits/shader-contract-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/shader/kits/shader-contract-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/shader/kits/shader-error-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/shader/kits/shader-error-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/shader/kits/shader-error-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/shader/kits/shader-include-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/shader/kits/shader-include-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/shader/kits/shader-include-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/render/shader/kits/shader-language-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/shader/kits/shader-language-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/shader/kits/shader-language-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/shader/kits/shader-module-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/shader/kits/shader-module-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/shader/kits/shader-module-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/shader/kits/shader-permutation-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/shader/kits/shader-permutation-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/shader/kits/shader-permutation-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/shader/kits/shader-program-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/shader/kits/shader-program-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/shader/kits/shader-program-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/shader/kits/shader-reflection-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/shader/kits/shader-reflection-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/render/shader/kits/shader-reflection-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/shader/kits/shader-source-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/shader/kits/shader-source-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/shader/kits/shader-source-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/shader/kits/shader-variant-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/shader/kits/shader-variant-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/shader/kits/shader-variant-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/shader/shader-contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/shader/shader-manifests.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/shader/shader-registry-kit.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/shader/subdomain.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/surface/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/surface/index.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/surface/kits/fullscreen-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/surface/kits/fullscreen-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/surface/kits/fullscreen-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/render/surface/kits/offscreen-surface-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/surface/kits/offscreen-surface-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/surface/kits/offscreen-surface-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/surface/kits/render-surface-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/surface/kits/render-surface-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/surface/kits/render-surface-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/surface/kits/resize-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/surface/kits/resize-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/surface/kits/resize-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/surface/kits/scissor-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/surface/kits/scissor-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/surface/kits/scissor-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/surface/kits/surface-format-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/surface/kits/surface-format-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/surface/kits/surface-format-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/render/surface/kits/swapchain-surface-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/surface/kits/swapchain-surface-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/surface/kits/swapchain-surface-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/surface/kits/viewport-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/surface/kits/viewport-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/surface/kits/viewport-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/surface/kits/window-surface-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/surface/kits/window-surface-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/surface/kits/window-surface-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/surface/subdomain.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/surface/surface-contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/surface/surface-kit.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/surface/surface-manifests.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/texture/index.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/texture/kits/depth-texture-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/render/texture/kits/depth-texture-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/texture/kits/depth-texture-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/texture/kits/mipmap-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/texture/kits/mipmap-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/texture/kits/mipmap-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/texture/kits/render-target-texture-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/texture/kits/render-target-texture-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/texture/kits/render-target-texture-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/texture/kits/shadow-texture-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/texture/kits/shadow-texture-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/texture/kits/shadow-texture-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/texture/kits/texture-2d-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/texture/kits/texture-2d-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/texture/kits/texture-2d-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/texture/kits/texture-array-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/render/texture/kits/texture-array-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/texture/kits/texture-array-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/texture/kits/texture-cube-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/texture/kits/texture-cube-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/texture/kits/texture-cube-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/texture/kits/texture-format-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/texture/kits/texture-format-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/texture/kits/texture-format-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/texture/kits/texture-residency-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/texture/kits/texture-residency-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/texture/kits/texture-residency-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/texture/kits/texture-resource-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/texture/kits/texture-resource-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/texture/kits/texture-resource-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/render/texture/kits/texture-stream-kit/contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/texture/kits/texture-stream-kit/index.js`	`n:render`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/render/texture/kits/texture-stream-kit/kit.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/texture/subdomain.manifest.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/texture/texture-contracts.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/texture/texture-manifests.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/render/texture/texture-registry-kit.js`	`n:render`	manifest-owned-internal	NexusEngine Core
`src/core-domains/runtime/data/kits/data-kit/index.js`	`n:runtime`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/runtime/data/kits/data-kit/ledger.js`	`n:runtime`	manifest-owned-internal	NexusEngine Core
`src/core-domains/runtime/data/kits/data-kit/migration.js`	`n:runtime`	manifest-owned-internal	NexusEngine Core
`src/core-domains/runtime/data/kits/data-kit/package-service.js`	`n:runtime`	manifest-owned-internal	NexusEngine Core
`src/core-domains/runtime/data/kits/data-kit/schema.js`	`n:runtime`	manifest-owned-internal	NexusEngine Core
`src/core-domains/runtime/data/kits/data-kit/selectors.js`	`n:runtime`	manifest-owned-internal	NexusEngine Core
`src/core-domains/runtime/data/kits/data-kit/services.js`	`n:runtime`	manifest-owned-internal	NexusEngine Core
`src/core-domains/runtime/data/kits/data-kit/snapshot.js`	`n:runtime`	manifest-owned-internal	NexusEngine Core
`src/core-domains/runtime/data/observation/contracts.js`	`n:runtime`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/runtime/data/observation/kits/observation-history-kit/index.js`	`n:runtime`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/runtime/data/observation/kits/observation-history-kit/kit.manifest.js`	`n:runtime`	manifest-owned-internal	NexusEngine Core
`src/core-domains/runtime/data/observation/kits/observation-history-kit/state.js`	`n:runtime`	manifest-owned-internal	NexusEngine Core
`src/core-domains/runtime/data/observation/subdomain.manifest.js`	`n:runtime`	manifest-owned-internal	NexusEngine Core
`src/core-domains/runtime/domain.manifest.js`	`n:runtime`	manifest-owned-internal	NexusEngine Core
`src/core-domains/runtime/index.js`	`n:runtime`	manifest-owned-internal	NexusEngine Core
`src/core-domains/runtime/kits/runtime-lifecycle-kit/index.js`	`n:runtime`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/runtime/persistence/kits/persistence-kit/index.js`	`n:runtime`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/runtime/realtime/contracts/surfaces.js`	`n:runtime`	manifest-owned-internal	NexusEngine Core
`src/core-domains/runtime/realtime/contracts/tick-context-scheduler.js`	`n:runtime`	manifest-owned-internal	NexusEngine Core
`src/core-domains/runtime/realtime/kits/realtime-kit/index.js`	`n:runtime`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/runtime/sequence/kits/sequence-kit/index.js`	`n:runtime`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/runtime/sequence/runtime/sequence-node-kit.js`	`n:runtime`	manifest-owned-internal	NexusEngine Core
`src/core-domains/runtime/sequence/runtime/sequence-node-library.js`	`n:runtime`	manifest-owned-internal	NexusEngine Core
`src/core-domains/runtime/sequence/runtime/sequence-node-runtime.js`	`n:runtime`	manifest-owned-internal	NexusEngine Core
`src/core-domains/runtime/sequence/runtime/sequence-runtime.js`	`n:runtime`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/runtime/sequence/schedule/index.js`	`n:runtime`	manifest-owned-internal	NexusEngine Core
`src/core-domains/runtime/sequence/schedule/kits/schedule-kit/contracts.js`	`n:runtime`	manifest-owned-internal	NexusEngine Core
`src/core-domains/runtime/sequence/schedule/kits/schedule-kit/index.js`	`n:runtime`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/runtime/sequence/schedule/kits/schedule-kit/kit.manifest.js`	`n:runtime`	manifest-owned-internal	NexusEngine Core
`src/core-domains/runtime/sequence/schedule/kits/schedule-kit/services.js`	`n:runtime`	manifest-owned-internal	NexusEngine Core
`src/core-domains/runtime/sequence/schedule/kits/schedule-kit/state.js`	`n:runtime`	manifest-owned-internal	NexusEngine Core
`src/core-domains/runtime/sequence/schedule/subdomain.manifest.js`	`n:runtime`	manifest-owned-internal	NexusEngine Core
`src/core-domains/runtime/startup/core-assets-startup-bridge.js`	`n:runtime`	manifest-owned-internal	NexusEngine Core
`src/core-domains/runtime/startup/index.js`	`n:runtime`	manifest-owned-internal	NexusEngine Core
`src/core-domains/runtime/startup/kits/startup-kit/index.js`	`n:runtime`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/runtime/transaction/kits/transaction-ledger-kit/index.js`	`n:runtime`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/simulation/adapters/facility-economy-adapter-kit/contracts.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/adapters/facility-economy-adapter-kit/index.js`	`n:simulation`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/simulation/adapters/facility-economy-adapter-kit/kit.manifest.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/simulation/adapters/facility-economy-adapter-kit/services.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/adapters/facility-economy-adapter-kit/state.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/adapters/lifecycle-economy-adapter-kit/contracts.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/adapters/lifecycle-economy-adapter-kit/index.js`	`n:simulation`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/simulation/adapters/lifecycle-economy-adapter-kit/kit.manifest.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/adapters/lifecycle-economy-adapter-kit/services.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/adapters/lifecycle-economy-adapter-kit/state.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/adapters/lifecycle-facility-adapter-kit/contracts.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/adapters/lifecycle-facility-adapter-kit/index.js`	`n:simulation`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/simulation/adapters/lifecycle-facility-adapter-kit/kit.manifest.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/adapters/lifecycle-facility-adapter-kit/services.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/adapters/lifecycle-facility-adapter-kit/state.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/adapters/locomotion-contact-response-adapter-kit/contracts.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/simulation/adapters/locomotion-contact-response-adapter-kit/index.js`	`n:simulation`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/simulation/adapters/locomotion-contact-response-adapter-kit/kit.manifest.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/adapters/locomotion-contact-response-adapter-kit/services.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/adapters/locomotion-contact-response-adapter-kit/state.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/adapters/vehicle-water-response-adapter-kit/contracts.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/adapters/vehicle-water-response-adapter-kit/index.js`	`n:simulation`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/simulation/adapters/vehicle-water-response-adapter-kit/kit.manifest.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/adapters/vehicle-water-response-adapter-kit/services.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/adapters/vehicle-water-response-adapter-kit/state.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/domain.manifest.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/economy/accounts/index.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/economy/accounts/kits/economy-account-kit/contracts.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/economy/accounts/kits/economy-account-kit/index.js`	`n:simulation`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/simulation/economy/accounts/kits/economy-account-kit/kit.manifest.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/simulation/economy/accounts/kits/economy-account-kit/services.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/economy/accounts/kits/economy-account-kit/state.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/economy/accounts/subdomain.manifest.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/economy/cargo/index.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/economy/cargo/kits/cargo-manifest-kit/contracts.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/economy/cargo/kits/cargo-manifest-kit/index.js`	`n:simulation`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/simulation/economy/cargo/kits/cargo-manifest-kit/kit.manifest.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/economy/cargo/kits/cargo-manifest-kit/services.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/economy/cargo/kits/cargo-manifest-kit/state.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/economy/cargo/subdomain.manifest.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/economy/index.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/economy/subdomain.manifest.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/hazard-field/index.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/hazard-field/kits/hazard-field-kit/contracts.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/simulation/hazard-field/kits/hazard-field-kit/index.js`	`n:simulation`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/simulation/hazard-field/kits/hazard-field-kit/kit.manifest.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/hazard-field/kits/hazard-field-kit/services.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/hazard-field/kits/hazard-field-kit/state.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/hazard-field/subdomain.manifest.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/index.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/kits/simulation-kit/checkpoints.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/kits/simulation-kit/hazards.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/kits/simulation-kit/index.js`	`n:simulation`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/simulation/kits/simulation-kit/objectives.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/kits/simulation-kit/pressure.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/kits/simulation-kit/resolution.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/kits/simulation-kit/resources.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/kits/simulation-kit/services.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/kits/simulation-kit/timers.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/motion/articulated-motion-domain/contracts.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/simulation/motion/articulated-motion-domain/index.js`	`n:simulation`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/simulation/motion/index.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/motion/kits/motion-kit/contracts.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/motion/kits/motion-kit/index.js`	`n:simulation`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/simulation/motion/kits/two-bone-ik-kit/index.js`	`n:simulation`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/simulation/motion/locomotion/index.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/motion/locomotion/kits/action-locomotion-kit/contracts.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/motion/locomotion/kits/action-locomotion-kit/index.js`	`n:simulation`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/simulation/motion/locomotion/kits/action-locomotion-kit/kit.manifest.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/motion/locomotion/kits/action-locomotion-kit/services.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/motion/locomotion/kits/action-locomotion-kit/state.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/motion/locomotion/subdomain.manifest.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/motion/vehicle/index.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/motion/vehicle/kits/vehicle-dynamics-kit/contracts.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/simulation/motion/vehicle/kits/vehicle-dynamics-kit/index.js`	`n:simulation`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/simulation/motion/vehicle/kits/vehicle-dynamics-kit/kit.manifest.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/motion/vehicle/kits/vehicle-dynamics-kit/services.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/motion/vehicle/kits/vehicle-dynamics-kit/state.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/motion/vehicle/subdomain.manifest.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/operations/facility/index.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/operations/facility/kits/facility-operations-kit/contracts.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/operations/facility/kits/facility-operations-kit/index.js`	`n:simulation`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/simulation/operations/facility/kits/facility-operations-kit/kit.manifest.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/operations/facility/kits/facility-operations-kit/services.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/operations/facility/kits/facility-operations-kit/state.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/operations/facility/subdomain.manifest.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/operations/index.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/operations/occupant-flow/index.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/simulation/operations/occupant-flow/kits/occupant-flow-kit/contracts.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/operations/occupant-flow/kits/occupant-flow-kit/index.js`	`n:simulation`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/simulation/operations/occupant-flow/kits/occupant-flow-kit/kit.manifest.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/operations/occupant-flow/kits/occupant-flow-kit/services.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/operations/occupant-flow/kits/occupant-flow-kit/state.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/operations/occupant-flow/subdomain.manifest.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/operations/subdomain.manifest.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/operations/transport-route/index.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/operations/transport-route/kits/transport-route-kit/contracts.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/operations/transport-route/kits/transport-route-kit/index.js`	`n:simulation`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/simulation/operations/transport-route/kits/transport-route-kit/kit.manifest.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/operations/transport-route/kits/transport-route-kit/services.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/operations/transport-route/kits/transport-route-kit/state.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/simulation/operations/transport-route/subdomain.manifest.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/physics/adapters/articulated-motion-drive-adapter/index.js`	`n:simulation`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/simulation/physics/articulated-dynamics-domain/contracts.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/physics/articulated-dynamics-domain/index.js`	`n:simulation`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/simulation/physics/index.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/physics/kits/physics-kit/catalog-factory.js`	`n:simulation`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/simulation/physics/kits/physics-kit/index.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/physics/kits/physics-kit/provider.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/physics/world-contact/index.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/physics/world-contact/kits/world-contact-kit/contracts.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/physics/world-contact/kits/world-contact-kit/index.js`	`n:simulation`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/simulation/physics/world-contact/kits/world-contact-kit/kit.manifest.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/physics/world-contact/kits/world-contact-kit/services.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/physics/world-contact/kits/world-contact-kit/state.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/simulation/physics/world-contact/subdomain.manifest.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/progression/index.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/progression/lifecycle/index.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/progression/lifecycle/kits/lifecycle-progression-kit/contracts.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/progression/lifecycle/kits/lifecycle-progression-kit/index.js`	`n:simulation`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/simulation/progression/lifecycle/kits/lifecycle-progression-kit/kit.manifest.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/progression/lifecycle/kits/lifecycle-progression-kit/services.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/progression/lifecycle/kits/lifecycle-progression-kit/state.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/progression/lifecycle/subdomain.manifest.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/progression/subdomain.manifest.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/pursuit-pressure/index.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/pursuit-pressure/kits/pursuit-pressure-kit/contracts.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/pursuit-pressure/kits/pursuit-pressure-kit/index.js`	`n:simulation`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/simulation/pursuit-pressure/kits/pursuit-pressure-kit/kit.manifest.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/simulation/pursuit-pressure/kits/pursuit-pressure-kit/services.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/pursuit-pressure/kits/pursuit-pressure-kit/state.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/pursuit-pressure/subdomain.manifest.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/recovery/index.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/recovery/soft-respawn/index.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/recovery/soft-respawn/kits/soft-respawn-kit/contracts.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/recovery/soft-respawn/kits/soft-respawn-kit/index.js`	`n:simulation`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/simulation/recovery/soft-respawn/kits/soft-respawn-kit/kit.manifest.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/recovery/soft-respawn/kits/soft-respawn-kit/services.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/recovery/soft-respawn/kits/soft-respawn-kit/state.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/recovery/soft-respawn/subdomain.manifest.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/recovery/subdomain.manifest.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/simulation/restored-behavior-manifests.js`	`n:simulation`	manifest-owned-internal	NexusEngine Core
`src/core-domains/spatial/domain.manifest.js`	`n:spatial`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/spatial/index.js`	`n:spatial`	manifest-owned-internal	NexusEngine Core
`src/core-domains/spatial/kits/angle-math-kit.js`	`n:spatial`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/spatial/kits/quaternion-math-kit.js`	`n:spatial`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/spatial/kits/spatial-kit/index.js`	`n:spatial`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/spatial/kits/transform-math-kit.js`	`n:spatial`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/spatial/kits/vector-math-kit.js`	`n:spatial`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/spatial/scale/index.js`	`n:spatial`	manifest-owned-internal	NexusEngine Core
`src/core-domains/spatial/scale/kits/spatial-scale-kit/contracts.js`	`n:spatial`	manifest-owned-internal	NexusEngine Core
`src/core-domains/spatial/scale/kits/spatial-scale-kit/index.js`	`n:spatial`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/spatial/scale/kits/spatial-scale-kit/kit.manifest.js`	`n:spatial`	manifest-owned-internal	NexusEngine Core
`src/core-domains/spatial/scale/kits/spatial-scale-kit/services.js`	`n:spatial`	manifest-owned-internal	NexusEngine Core
`src/core-domains/spatial/scale/kits/spatial-scale-kit/state.js`	`n:spatial`	manifest-owned-internal	NexusEngine Core
`src/core-domains/spatial/scale/subdomain.manifest.js`	`n:spatial`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/adapters/terrain-provider-adapter/index.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/domain.manifest.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/generation/index.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/generation/kits/procedural-generation-kit/contracts.js`	`n:world`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/world/generation/kits/procedural-generation-kit/index.js`	`n:world`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/world/generation/kits/procedural-generation-kit/kit.manifest.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/generation/kits/procedural-generation-kit/services.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/generation/kits/procedural-generation-kit/state.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/generation/subdomain.manifest.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/index.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/kits/world-builder-runtime-kit/index.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/kits/world-cell-kit/index.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/kits/world-effect-provider-kit/index.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/kits/world-partition-kit/index.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/kits/world-surface-kit/index.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/navigation/index.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/navigation/landmark-guidance/index.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/navigation/landmark-guidance/kits/landmark-guidance-kit/contracts.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/navigation/landmark-guidance/kits/landmark-guidance-kit/index.js`	`n:world`	manifest-proven-public-atom	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/world/navigation/landmark-guidance/kits/landmark-guidance-kit/kit.manifest.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/navigation/landmark-guidance/kits/landmark-guidance-kit/services.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/navigation/landmark-guidance/kits/landmark-guidance-kit/state.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/navigation/landmark-guidance/subdomain.manifest.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/navigation/navmesh/index.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/navigation/navmesh/kits/navmesh-kit/contracts.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/navigation/navmesh/kits/navmesh-kit/index.js`	`n:world`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/world/navigation/navmesh/kits/navmesh-kit/kit.manifest.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/navigation/navmesh/kits/navmesh-kit/services.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/navigation/navmesh/kits/navmesh-kit/state.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/navigation/navmesh/subdomain.manifest.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/navigation/pathfinding/index.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/navigation/pathfinding/kits/pathfinding-kit/contracts.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/navigation/pathfinding/kits/pathfinding-kit/index.js`	`n:world`	manifest-proven-public-atom	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/world/navigation/pathfinding/kits/pathfinding-kit/kit.manifest.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/navigation/pathfinding/kits/pathfinding-kit/services.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/navigation/pathfinding/kits/pathfinding-kit/state.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/navigation/pathfinding/subdomain.manifest.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/navigation/route-field/index.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/navigation/route-field/kits/route-field-kit/contracts.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/navigation/route-field/kits/route-field-kit/index.js`	`n:world`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/world/navigation/route-field/kits/route-field-kit/kit.manifest.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/navigation/route-field/kits/route-field-kit/services.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/navigation/route-field/kits/route-field-kit/state.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/navigation/route-field/subdomain.manifest.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/navigation/subdomain.manifest.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/partitions/quadtree-partition-kit/index.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/partitions/uniform-grid-partition-kit/index.js`	`n:world`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/world/portable.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/preparation/world-patch-preparation-controller.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/restored-behavior-manifests.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/scene/kits/scene-kit/constants.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/scene/kits/scene-kit/descriptors.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/scene/kits/scene-kit/host-contract.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/scene/kits/scene-kit/index.js`	`n:world`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/world/scene/kits/scene-kit/ledgers.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/scene/kits/scene-kit/scene-registry.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/scene/kits/scene-kit/scene-transition-ledger.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/scene/kits/scene-kit/state.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/scene/kits/scene-kit/transitions.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/scene/kits/scene-kit/utils.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/snapshot.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/state.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/surfaces/curved-horizon-surface-kit/index.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/surfaces/flat-world-surface-kit/index.js`	`n:world`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/world/terrain/index.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/terrain/kits/terrain-kit/contracts.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/terrain/kits/terrain-kit/index.js`	`n:world`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/world/terrain/kits/terrain-kit/kit.manifest.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/terrain/kits/terrain-kit/services.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/terrain/kits/terrain-kit/state.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/terrain/lod-descriptors.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/terrain/subdomain.manifest.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/validation.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/water-surface/index.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/water-surface/kits/water-surface-kit/contracts.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/water-surface/kits/water-surface-kit/index.js`	`n:world`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/world/water-surface/kits/water-surface-kit/kit.manifest.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/water-surface/kits/water-surface-kit/services.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/water-surface/kits/water-surface-kit/state.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/water-surface/subdomain.manifest.js`	`n:world`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/world/weather/contracts.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/weather/index.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/weather/layered-weather-domain/index.js`	`n:world`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/world/weather/weather-domain.js`	`n:world`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/world/world-domain.js`	`n:world`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/world/world-feature-domain/atmosphere-feature-domain/index.js`	`n:world`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/world/world-feature-domain/contracts.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/world-feature-domain/ecology-feature-domain/index.js`	`n:world`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/world/world-feature-domain/feature-family-domain-kit/index.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/world-feature-domain/hydrology-feature-domain/index.js`	`n:world`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/world/world-feature-domain/index.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/world-feature-domain/kits/feature-composition-kit/index.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/world-feature-domain/kits/feature-definition-kit/index.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/world-feature-domain/kits/feature-lifecycle-kit/index.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/world-feature-domain/kits/feature-query-kit/index.js`	`n:world`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/world/world-feature-domain/kits/feature-registry-kit/index.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/world-feature-domain/kits/semantic-feature-kit/index.js`	`n:world`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/world/world-feature-domain/landform-feature-domain/contracts.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/world-feature-domain/landform-feature-domain/index.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/world-feature-domain/landform-feature-domain/kits/canyon-feature-kit/index.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/world-feature-domain/landform-feature-domain/kits/cliff-feature-kit/index.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/world-feature-domain/landform-feature-domain/kits/landform-feature-kits.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/world-feature-domain/landform-feature-domain/kits/mountain-feature-kit/index.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/world-feature-domain/landform-feature-domain/kits/plateau-feature-kit/index.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/world-feature-domain/landform-feature-domain/landform-feature-domain.js`	`n:world`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/world/world-feature-domain/settlement-feature-domain/index.js`	`n:world`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/world/world-feature-domain/snapshot.js`	`n:world`	manifest-owned-internal	NexusEngine Core

Path	Owner	Review	Destination
`src/core-domains/world/world-feature-domain/state.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/world-feature-domain/validation.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/world-feature-domain/world-feature-domain.js`	`n:world`	manifest-proven-public-atom	NexusEngine Core
`src/core-domains/world/world-foundation-domain/contracts.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/world-foundation-domain/index.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/world-foundation-domain/kits/foundation-cell-resolution-kit/index.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/world-foundation-domain/kits/foundation-composition-kit/index.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/world-foundation-domain/kits/foundation-definition-kit/index.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/world-foundation-domain/kits/foundation-sampling-kit/index.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/world-foundation-domain/snapshot.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/world-foundation-domain/state.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/world-foundation-domain/validation.js`	`n:world`	manifest-owned-internal	NexusEngine Core
`src/core-domains/world/world-foundation-domain/world-foundation-domain.js`	`n:world`	manifest-proven-public-atom	NexusEngine Core
`src/domain-api.js`	`engine-root`	root-contract	NexusEngine minimal root

Path	Owner	Review	Destination
`src/domain-path.js`	`engine-root`	root-contract	NexusEngine minimal root
`src/domain-service-kit.js`	`engine-root`	root-contract	NexusEngine minimal root
`src/ecs.js`	`engine-root`	root-contract	NexusEngine minimal root
`src/engine.js`	`engine-root`	root-contract	NexusEngine minimal root
`src/foundation/completion-ledger.js`	`engine-foundation`	root-contract	NexusEngine foundation
`src/foundation/deterministic-replay.js`	`engine-foundation`	root-contract	NexusEngine foundation
`src/foundation/idempotency-ledger.js`	`engine-foundation`	root-contract	NexusEngine foundation
`src/foundation/index.js`	`engine-foundation`	root-contract	NexusEngine foundation
`src/foundation/progress-timer.js`	`engine-foundation`	root-contract	NexusEngine foundation
`src/foundation/promotion-guard.js`	`engine-foundation`	root-contract	NexusEngine foundation
`src/foundation/seeded-random.js`	`engine-foundation`	root-contract	NexusEngine foundation
`src/foundation/serializable-state.js`	`engine-foundation`	root-contract	NexusEngine foundation
`src/foundation/sha256.js`	`engine-foundation`	root-contract	NexusEngine foundation
`src/foundation/snapshot.js`	`engine-foundation`	root-contract	NexusEngine foundation
`src/index.js`	`engine-root`	root-contract	NexusEngine minimal root
`src/release.js`	`engine-root`	root-contract	NexusEngine minimal root
`src/runtime-kit.js`	`engine-root`	root-contract	NexusEngine minimal root

GENERATED REFERENCE

NexusEngine 0.0.4 Restored Behaviors

This guide is generated from `0.0.4-restored-behaviors.json`. It is a hard migration: no root forwarding exports and no legacy snapshot coercion.

- Historical modules: **26**
- Restored behavior atoms: **27**
- Optional adapters: **9**
- Composition recipes: **6**

Commit Lineage

Code	Commit	Meaning
`I`	`56aed83b618553057b3da4e200a99701287a9ad2`	Initial implementations
`N`	`d05aac6219645ba383f6f4c5c03216dc887f14f6`	Navigation and procedural generation
`K`	`99292c7d1d93ca2d50a3d8a8a6f8601154670498`	Expanded Kit implementations
`T`	`8b57b03904889cdbc71021d3bdb1d4070af5c8d3`	Seamless streamed-terrain normals
`R`	`8b968c201b713c2cb00447e7765f0e2c16dcec8a`	Pre-consolidation Core refactor
`S`	`a9adca5b3620f996f00860358c4864dd4bdfa6d9`	Last complete source snapshot
`X`	`a685444344244384 91be1398e3f3d5aced5bc5ee`	Removal commit

Migration Index

Historical source	Status	New atom(s)	Import(s)
`src/action-movement-kit.js`	implemented-and-proven	`action-locomotion-kit`	`nexusengine/domains/simulation/motion/locomotion`
`src/assistance-target-kit.js`	implemented-and-proven	`assistance-target-kit`	`nexusengine/domains/interaction/assistance-target`
`src/cargo-manifest-kit.js`	implemented-and-proven	`cargo-manifest-kit`	`nexusengine/domains/simulation/economy/cargo`
`src/character-camera-kit.js`	implemented-and-proven	`third-person-camera-kit`	`nexusengine/domains/camera/third-person`
`src/economy-kit.js`	implemented-and-proven	`economy-account-kit`	`nexusengine/domains/simulation/economy/accounts`
`src/environmental-affordance-kit.js`	implemented-and-proven	`environmental-affordance-kit`	`nexusengine/domains/interaction/environmental-affordance`
`src/facility-operations-kit.js`	implemented-and-proven	`facility-operations-kit`	`nexusengine/domains/simulation/operations/facility`
`src/hazard-field-kit.js`	implemented-and-proven	`hazard-field-kit`	`nexusengine/domains/simulation/hazard-field`

Historical source	Status	New atom(s)	Import(s)
`src/landmark-guidance-kit.js`	implemented-and-proven	`landmark-guidance-kit`	`nexusengine/domains/world/navigation/landmark-guidance`
`src/lifecycle-progression-kit.js`	implemented-and-proven	`lifecycle-progression-kit`	`nexusengine/domains/simulation/progression/lifecycle`
`src/navmesh-kit.js`	implemented-and-proven	`navmesh-kit`	`nexusengine/domains/world/navigation/navmesh`
`src/occupant-flow-kit.js`	implemented-and-proven	`occupant-flow-kit`	`nexusengine/domains/simulation/operations/occupant-flow`
`src/pathfinding-kit.js`	implemented-and-proven	`pathfinding-kit`	`nexusengine/domains/world/navigation/pathfinding`
`src/procedural-kit.js`	implemented-and-proven	`procedural-generation-kit`	`nexusengine/domains/world/generation`
`src/pursuit-pressure-kit.js`	implemented-and-proven	`pursuit-pressure-kit`	`nexusengine/domains/simulation/pursuit-pressure`
`src/request-fulfillment-kit.js`	implemented-and-proven	`request-fulfillment-kit`	`nexusengine/domains/interaction/request/fulfillment`
`src/request-queue-kit.js`	implemented-and-proven	`request-queue-kit`	`nexusengine/domains/interaction/request/queue`
`src/route-field-kit.js`	implemented-and-proven	`route-field-kit`	`nexusengine/domains/world/navigation/route-field`
`src/schedule-kit.js`	implemented-and-proven	`schedule-kit`	`nexusengine/domains/runtime/sequence/schedule`
`src/spatial-scale-kit.js`	implemented-and-proven	`spatial-scale-kit`	`nexusengine/domains/spatial/scale`
`src/terrain-kit.js`	implemented-and-proven	`terrain-kit`	`nexusengine/domains/world/terrain`
`src/transfer-zone-kit.js`	implemented-and-proven	`transfer-zone-kit`	`nexusengine/domains/interaction/transfer-zone`
`src/transport-route-kit.js`	implemented-and-proven	`transport-route-kit`	`nexusengine/domains/simulation/operations/transport-route`
`src/vehicle-dynamics-kit.js`	implemented-and-proven	`vehicle-dynamics-kit`	`nexusengine/domains/simulation/motion/vehicle`
`src/water-surface-kit.js`	implemented-and-proven	`water-surface-kit`	`nexusengine/domains/world/water-surface`
`src/world-physics-kit.js`	implemented-and-proven	`world-contact-kit`, `soft-respawn-kit`	`nexusengine/domains/simulation/physics/world-contact` `nexusengine/domains/simulation/recovery/soft-respawn`

1. src/action-movement-kit.js

Status: `implemented-and-proven` | SHA-256 at `S`: `bdf54ee25fb3e8d8cc623a80d16cd3575844e89f2d9210ede0dfc96a2fcf978d` | Lineage: `I` -> `K` -> `S` -> `X`

Historical factories: `createLocomotionKit`, `createCharacterMovementKit`, `createActionMovementKit`. Historical helpers: None. Historical resources: None. Historical events: None. Historical engine APIs: None.

Import Cutover

Before:

```
import { createLocomotionKit, createCharacterMovementKit, createActionMovementKit } from
"nexusengine";
```

After:

```
import { createActionLocomotionKit } from "nexusengine/domains/simulation/motion/locomotion";

engine.n.actionLocomotion.step({ operationId: "movement:1", delta: 1 / 60, input })
```

Required Changes

- Configuration: Merge action, character, and locomotion options into one Action Locomotion config. Keep input mapping, camera, contact, and recovery policy outside the atom.
- Snapshot: Map position, velocity, grounded, movement mode, and tuning fields into the new Action Locomotion snapshot. Do not load the old resource object directly.
- Events: Read transition names from step() receipts; glide events are emitted only on state transitions.
- Optional adapters: `locomotion-contact-response-adapter-kit`.
- Consumer status: `not-started-outside-restoration-wave`.

Corrected Defects

- Three factory aliases represented one implementation.
- Ground snapping could occur before killY recovery evaluation.
- Glide events could repeat without a transition.

Proof: `src/core-domains/simulation/tests/restored-motion-behaviors.mjs`, `src/core-domains/simulation/tests/restored-integration-adapters.mjs`.

2. src/assistance-target-kit.js

Status: `implemented-and-proven` | SHA-256 at `S`: `15f8e2e4972126bc8b254cf323cfbfdb8ef658b63edb853c776d6af314fdb04e` | Lineage: `K` -> `S` -> `X`

Historical factories: `createAssistanceTargetKit`. Historical helpers: None. Historical resources: `AssistanceTargetState`. Historical events: `AssistanceTargetStabilized`, `AssistanceTargetAttached`, `AssistanceTargetCompleted`, `AssistanceTargetLost`. Historical engine APIs: `engine.assistanceTargets`.

Import Cutover

Before:

```
import { createAssistanceTargetKit } from "./src/assistance-target-kit.js";
```

After:

```
import { createAssistanceTargetKit } from "nexusengine/domains/interaction/assistance-target";

engine.n.assistanceTargets.complete({ operationId: "assist:complete:1", targetId })
```

Required Changes

- Configuration: Keep target identity, urgency, position, and terminal policy; use the semantic target dataset shape.
- Snapshot: Map targets and counts into getSnapshot(); legacy resource identities are not accepted.
- Events: Use exact transition receipts and Domain events instead of importing legacy event constants.

- Optional adapters: None.
- Consumer status: `not-started-outside-restoration-wave`.

Corrected Defects

- Lost and completed terminal states could conflict.
- Terminal targets could transition again.
- Result references and nearest ties were unstable.

Proof: `src/core-domains/interaction/tests/restored-interaction-behaviors.mjs`.

3. src/cargo-manifest-kit.js

Status: `implemented-and-proven` | SHA-256 at `S`: `96bc115eb1e25cc89a8280b6e9c6cf1be08c69666185783a58d348eb813957e3` | Lineage: `K` -> `S` -> `X`

Historical factories: `createCargoManifestKit`. Historical helpers: `queryNearestCargo`. Historical resources: `CargoManifestState`. Historical events: `CargoPickedUp`, `CargoDeposited`, `CargoQuotaCompleted`, `CargoConditionChanged`, `CargoConditionDepleted`. Historical engine APIs: `engine.cargoManifest`.

Import Cutover

Before:

```
import { createCargoManifestKit } from "nexusengine";
```

After:

```
import { createCargoManifestKit } from "nexusengine/domains/simulation/economy/cargo";

engine.n.cargoManifest.pickUp({ operationId: "cargo:pickup:1", itemId, carrierId })
```

Required Changes

- Configuration: Use finite nonnegative item values, quantities, weights, capacity, and quota.
- Snapshot: Map items, carriers, deposited value, and quota state into the validated portable snapshot.
- Events: Use pickup/deposit operation receipts; quota completion is an accepted transition event.
- Optional adapters: None.
- Consumer status: `not-started-outside-restoration-wave`.

Corrected Defects

- Non-finite or negative values and quantities were accepted.
- Nested item metadata escaped by reference.
- Pickup and deposit lacked exact-once command identity.

Proof: `src/core-domains/simulation/tests/restored-economy-operations.mjs`.

4. src/character-camera-kit.js

Status: `implemented-and-proven` | SHA-256 at `S`: `15ef74380bbb6b1d575b1ca6d7a2c486f7dbfe7944903cdd7e95d1d35e6f2891` | Lineage: `I` -> `K` -> `S` -> `X`

Historical factories: `createCameraKit`, `createCharacterCameraKit`. Historical helpers: None. Historical resources: None. Historical events: None. Historical engine APIs: None.

Import Cutover

Before:

```
import { createCameraKit, createCharacterCameraKit } from "nexusengine";
```

After:

```
import { createThirdPersonCameraKit } from "nexusengine/domains/camera/third-person";

engine.n.thirdPersonCamera.describe({ operationId: "camera:1", delta: 1 / 60 })
```

Required Changes

- Configuration: Provide characterId and install public Character, Motion, and Camera capabilities before this Kit.
- Snapshot: Map yaw, pitch, sequence, and renderer-neutral descriptor; host camera objects are never snapshot data.
- Events: describe() returns a receipt containing the camera descriptor; host rendering remains external.
- Optional adapters: `camera-world-occlusion-adapter-kit`.
- Consumer status: `not-started-outside-restoration-wave`.

Corrected Defects

- Camera behavior read private movement state.
- Required bindings failed only during the first update.
- World occlusion was coupled to camera ownership.

Proof: `src/core-domains/presentation/tests/third-person-camera-restoration.mjs`, `src/core-domains/simulation/tests/restored-integration-adapters.mjs`.

5. src/economy-kit.js

Status: `implemented-and-proven` | SHA-256 at `S`: `c503e016dee6bda434349c2231080c735744ca478c5a199be9f6316fd7c4fc7a` | Lineage: `K` -> `S` -> `X`

Historical factories: `createEconomyKit`. Historical helpers: None. Historical resources: `EconomyState`. Historical events: `EconomyTransactionRequest`, `EconomyTransactionCompleted`, `EconomyTransactionRejected`. Historical engine APIs: `engine.economy`.

Import Cutover

Before:

```
import { createEconomyKit } from "nexusengine";
```

After:

```
import { createEconomyAccountKit } from "nexusengine/domains/simulation/economy/accounts";

engine.n.economy.transact({ operationId: "economy:1", account: "cash", amount: 10 })
```

Required Changes

- Configuration: Rename initial balance data to accounts and install the Runtime Transaction ledger before Economy Accounts.
- Snapshot: Map accounts and optional ledger records into the validated snapshot; old event queues are discarded.

- Events: transact() returns exact-once completed or rejected receipts and records them in Runtime Transaction.
- Optional adapters: None.
- Consumer status: `not-started-outside-restoration-wave`.

Corrected Defects

- Transactions depended on an internal engine.tick(0).
- ledgerLimit zero was mishandled.
- Non-finite balances and escaped metadata were possible.

Proof: `src/core-domains/simulation/tests/restored-economy-operations.mjs`.

6. src/environmental-affordance-kit.js

Status: `implemented-and-proven` | SHA-256 at `S`: `4f2e25cce64efff5b250cae303e568f1e90b0e8eb821fbf3f9701d3e16745a2c` | Lineage: `K` -> `S` -> `X`

Historical factories: `createEnvironmentalAffordanceKit`. Historical helpers: `queryNearbyAffordances`. Historical resources: `EnvironmentalAffordanceState`. Historical events: `AffordanceEntered`, `AffordanceActivated`, `AffordanceCompleted`. Historical engine APIs: `engine.environmentalAffordances`.

Import Cutover

Before:

```
import { createEnvironmentalAffordanceKit } from "./src/environmental-affordance-kit.js";
```

After:

```
import { createEnvironmentalAffordanceKit } from "nexusengine/domains/interaction/environmental-
affordance";

engine.n.environmentalAffordances.activate({ operationId: "affordance:1", affordanceId })
```

Required Changes

- Configuration: Keep portable affordance position, radius, requirements, and activation policy.
- Snapshot: Map affordances and counts; query diagnostics are not semantic state.
- Events: Queries are silent; activate() emits only accepted transitions through receipts.
- Optional adapters: None.
- Consumer status: `not-started-outside-restoration-wave`.

Corrected Defects

- Queries mutated active state and emitted events.
- Unknown activation could mutate before rejection.
- Initial completion counts were wrong.

Proof: `src/core-domains/interaction/tests/restored-interaction-behaviors.mjs`.

7. src/facility-operations-kit.js

Status: `implemented-and-proven` | SHA-256 at `S`: `f2e287cea085c259825c7321c70784c26f9f7a8084e74ed6bdc00bd99e419712` | Lineage: `K` -> `S` -> `X`

Historical factories: `createFacilityOperationsKit`. Historical helpers: None. Historical resources: `FacilityOperationsState`. Historical events: `FacilityOutputProduced`, `FacilityConditionChanged`. Historical engine APIs: `engine.facilityOperations`.

Import Cutover

Before:

```
import { createFacilityOperationsKit } from "nexusengine";
```

After:

```
import { createFacilityOperationsKit } from "nexusengine/domains/simulation/operations/facility";

engine.n.facilityOperations.advance({ operationId: "facility:step:1", delta: 1 })
```

Required Changes

- Configuration: Use finite output/upkeep descriptors and explicit intervals, capacity, condition, and status.
- Snapshot: Map facility cycle and condition state into portable records.
- Events: advance() returns portable operation outputs; install Facility Economy only when account effects are desired.
- Optional adapters: `facility-economy-adapter-kit`.
- Consumer status: `not-started-outside-restoration-wave`.

Corrected Defects

- Non-finite output values were accepted.
- Nested configuration escaped by reference.
- Economy side effects were coupled to facility state.

Proof: `src/core-domains/simulation/tests/restored-economy-operations.mjs`, `src/core-domains/simulation/tests/restored-integration-adapters.mjs`.

8. src/hazard-field-kit.js

Status: `implemented-and-proven` | SHA-256 at `S`: `555f7b16007f8adb7fc1e58c6143524e9f8ff37f18c0ca7bfa087921d77e0895` | Lineage: `K` -> `S` -> `X`

Historical factories: `createHazardFieldKit`. Historical helpers: `queryHazardCircle`. Historical resources: `HazardFieldState`. Historical events: `HazardFieldSpawned`, `HazardFieldCollision`. Historical engine APIs: `engine.hazardField`.

Import Cutover

Before:

```
import { createHazardFieldKit } from "nexusengine";
```

After:

```
import { createHazardFieldKit } from "nexusengine/domains/simulation/hazard-field";

engine.n.hazardField.advance({ operationId: "hazard:step:1", delta: 1 / 60 })
```

Required Changes

- Configuration: Use finite bounded hazard descriptors and stable IDs.
- Snapshot: Map hazards, sequences, and elapsed state; collision query results are not stored by querying.
- Events: Spawn and advance mutations emit accepted events; queryCircle() is read-only.
- Optional adapters: None.
- Consumer status: `not-started-outside-restoration-wave`.

Corrected Defects

- Generated IDs could collide.
- Bounds were not fully validated.
- Collision queries exposed references or mutated query state.

Proof: `src/core-domains/simulation/tests/restored-progression-pressure.mjs`.

9. src/landmark-guidance-kit.js

Status: `implemented-and-proven` | SHA-256 at `S`: `2233a9b147af597c62472152736a133c797b6ca554fdba0c5f8d43760f06894b` | Lineage: `K` -> `S` -> `X`

Historical factories: `createLandmarkGuidanceKit`. Historical helpers: `queryNearestLandmark`. Historical resources: `LandmarkGuidanceState`. Historical events: `LandmarkDiscovered`, `LandmarkReached`, `LandmarkActivated`. Historical engine APIs: `engine.landmarkGuidance`.

Import Cutover

Before:

```
import { createLandmarkGuidanceKit } from "./src/landmark-guidance-kit.js";
```

After:

```
import { createLandmarkGuidanceKit } from "nexusengine/domains/world/navigation/landmark-guidance";

engine.n.landmarkGuidance.reach({ operationId: "landmark:reach:1", landmarkId })
```

Required Changes

- Configuration: Keep stable landmark IDs, positions, radii, status, and metadata.
- Snapshot: Map landmarks and counts; discard legacy lastQuery mutation state.
- Events: Use exact discover/reach/activate receipts; nearest() is a deterministic read-only query.
- Optional adapters: None.
- Consumer status: `not-started-outside-restoration-wave`.

Corrected Defects

- Inactive or completed landmarks could transition again.
- Initial counts and nested metadata were unsafe.
- Nearest ties were unstable and queries mutated state.

Proof: `src/core-domains/world/tests/restored-world-behaviors.mjs`.

10. src/lifecycle-progression-kit.js

Status: `implemented-and-proven` | SHA-256 at `S`: `8e42b8135cb5d18351d7bc5589d94cfd84e248bdd9223bd0d4e609d2cfdb6da8` | Lineage: `K` -> `S` -> `X`

Historical factories: `createLifecycleProgressionKit`. Historical helpers: None. Historical resources: `LifecycleProgressionState`. Historical events: `LifecycleProgressionStart`, `LifecycleProgressionCompleted`. Historical engine APIs: `engine.lifecycleProgression`.

Import Cutover

Before:

```
import { createLifecycleProgressionKit } from "nexusengine";
```

After:

```
import { createLifecycleProgressionKit } from "nexusengine/domains/simulation/progression/lifecycle";

engine.n.lifecycleProgression.start({ operationId: "lifecycle:start:1", itemId })
```

Required Changes

- Configuration: Use items with stable IDs, prerequisites, finite duration, optional cost descriptor, and portable effects.
- Snapshot: Map item status, elapsed time, and completed IDs. Old queued events are not loaded.
- Events: start()/advance() return effect descriptors; Economy and Facility effects require explicit adapters.
- Optional adapters: `lifecycle-economy-adapter-kit`, `lifecycle-facility-adapter-kit`.
- Consumer status: `not-started-outside-restoration-wave`.

Corrected Defects

- Costs or events could occur before prerequisites passed.
- Payloads leaked references.
- Economy and Facility mutations were coupled to progression.

Proof: `src/core-domains/simulation/tests/restored-progression-pressure.mjs`, `src/core-domains/simulation/tests/restored-integration-adapters.mjs`.

11. src/navmesh-kit.js

Status: `implemented-and-proven` | SHA-256 at `S`: `6b655c26562901c360c9152a01b22e0adc8d20352e900f8bcab301b9c856de3a` | Lineage: `N` -> `S` -> `X`

Historical factories: `createNavMeshKit`. Historical helpers: `createNavMeshFromWalkability`, `create3DNavigationGraph`. Historical resources: None. Historical events: None. Historical engine APIs: `engine.navmesh`.

Import Cutover

Before:

```
import { createNavMeshKit } from "nexusengine";
```

After:

```
import { createNavMeshKit } from "nexusengine/domains/world/navigation/navmesh";

engine.n.navmesh.build({ operationId: "navmesh:build:1", walkability })
```

Required Changes

- Configuration: Provide normalized walkability cells with unique stable IDs and finite blocked-cost policy.
- Snapshot: Map portable nodes, edges, and source hash; Maps, Sets, and Infinity are rejected.
- Events: build() returns an exact operation receipt; pure graph builders remain exported from the subpath.
- Optional adapters: None.
- Consumer status: `not-started-outside-restoration-wave`.

Corrected Defects

- Duplicate coordinates or keys silently collapsed IDs.
- Blocked costs used non-portable Infinity.

Proof: `src/core-domains/world/tests/restored-world-behaviors.mjs`.

12. src/occupant-flow-kit.js

Status: `implemented-and-proven` | SHA-256 at `S`: `c07eb5bc1d6e862b4462324cbf707f442da08afd86c9ca87af1e0f2b19077d29` | Lineage: `K` -> `S` -> `X`

Historical factories: `createOccupantFlowKit`. Historical helpers: None. Historical resources: `OccupantFlowState`. Historical events: `OccupantSpawn`, `OccupantNeedCreated`, `OccupantServed`, `OccupantAbandoned`. Historical engine APIs: `engine.occupantFlow`.

Import Cutover

Before:

```
import { createOccupantFlowKit } from "nexusengine";
```

After:

```
import { createOccupantFlowKit } from "nexusengine/domains/simulation/operations/occupant-flow";

engine.n.occupantFlow.advance({ operationId: "occupants:step:1", delta: 1 })
```

Required Changes

- Configuration: Use portable occupants and deterministic spawn rules. Request policy is not part of this config.
- Snapshot: Map occupants, rules, elapsed time, and next sequence; reset restores the original normalized rules.
- Events: spawn/serve/advance return receipts; install Occupant Request to create queue records.
- Optional adapters: `occupant-request-adapter-kit`.
- Consumer status: `not-started-outside-restoration-wave`.

Corrected Defects

- Reset reused mutated spawn configuration.
- Generated IDs could collide.
- Request creation and internal engine.tick(0) coupled other domains.

Proof: `src/core-domains/simulation/tests/restored-economy-operations.mjs`, `src/core-domains/simulation/tests/restored-integration-adapters.mjs`.

13. src/pathfinding-kit.js

Status: `implemented-and-proven` | SHA-256 at `S`: `54fcc95c2e9a2dd8afdcf44629627de7b712de6f558608a948b7cb00e5d37b7b` | Lineage: `N` -> `S` -> `X`

Historical factories: `createPathfindingKit`. Historical helpers: `navigationHeuristics`, `createAStarPathfinder`, `navigationAdapters`, `createNavigationQuery`. Historical resources: None. Historical events: None. Historical engine APIs: `engine.navigation`.

Import Cutover

Before:

```
import { createPathfindingKit } from "nexusengine";
```

After:

```
import { createPathfindingKit } from "nexusengine/domains/world/navigation/pathfinding";

engine.n.pathfinding.findPath({ operationId: "path:1", start, goal })
```

Required Changes

- Configuration: Select an explicit public navigation adapter and finite heuristic/cost policy.
- Snapshot: Map portable request and result records; provider functions are never snapshot data.
- Events: findPath() returns resolved/failed receipts; pure A* helpers remain available at the subpath.
- Optional adapters: None.
- Consumer status: `not-started-outside-restoration-wave`.

Corrected Defects

- Negative or non-finite edge costs were accepted.
- Requests, paths, and results leaked references.

Proof: `src/core-domains/world/tests/restored-world-behaviors.mjs`.

14. src/procedural-kit.js

Status: `implemented-and-proven` | SHA-256 at `S`: `26a313962334fe915b21d079d0f93a4c99962cbf19bcc7d90a94ca506d312962` | Lineage: `N` -> `S` -> `X`

Historical factories: `createProceduralKit`. Historical helpers: `proceduralAlgorithms`, `createProceduralSnapshot`, `createProceduralQuery`. Historical resources: None. Historical events: None. Historical engine APIs: `engine.procedural`.

Import Cutover

Before:

```
import { createProceduralKit } from "nexusengine";
```

After:

```
import { createProceduralGenerationKit } from "nexusengine/domains/world/generation";

engine.n.proceduralGeneration.generate({ operationId: "generation:1" })
```

Required Changes

- Configuration: Provide seed, algorithm IDs, version, and neutral region/connector/point configuration. Authored meanings remain recipe data.
- Snapshot: Regenerate from the SHA-256 identity or map only portable generated descriptors; live providers are excluded.
- Events: generate() returns an exact receipt and snapshot identity; pure query helpers remain at the subpath.
- Optional adapters: None.
- Consumer status: `not-started-outside-restoration-wave`.

Corrected Defects

- Identity did not hash the complete normalized generation contract.
- Game-shaped meanings and live state leaked through the generic facade.

Proof: `src/core-domains/world/tests/restored-world-behaviors.mjs`.

15. src/pursuit-pressure-kit.js

Status: `implemented-and-proven` | SHA-256 at `S`: `3925ee176302ba2936cd2fd767e760b8dcc52bc7f661b4f6ee356876ea669915` | Lineage: `K` -> `S` -> `X`

Historical factories: `createPursuitPressureKit`. Historical helpers: `pursuitBand`. Historical resources: `PursuitPressureState`. Historical events: `PursuitPressureChanged`, `PursuitPressureCaught`, `PursuitPressureRecovered`. Historical engine APIs: `engine.pursuitPressure`.

Import Cutover

Before:

```
import { createPursuitPressureKit } from "./src/pursuit-pressure-kit.js";
```

After:

```
import { createPursuitPressureKit } from "nexusengine/domains/simulation/pursuit-pressure";

engine.n.pursuitPressure.setDistance({ operationId: "pursuit:1", distance })
```

Required Changes

- Configuration: Use finite distance thresholds and explicit caught/recovery rules.
- Snapshot: Map distance, band, caught state, and transition history into portable state.
- Events: setDistance()/adjust() return one receipt per accepted transition.
- Optional adapters: None.
- Consumer status: `not-started-outside-restoration-wave`.

Corrected Defects

- Initial caught state could disagree with distance.
- Accepted band transitions were not all recorded.
- Recovery and caught transitions could be incoherent.

Proof: `src/core-domains/simulation/tests/restored-progression-pressure.mjs`.

16. src/request-fulfillment-kit.js

Status: `implemented-and-proven` | SHA-256 at `S`: `d491b00e3c4e6f3b0762bea676340ba12e1a1ce30ef9beaace0c6dc6f610f0b8` | Lineage: `K` -> `S` -> `X`

Historical factories: `createRequestFulfillmentKit`. Historical helpers: `queryNearestOpenRequest`. Historical resources: `RequestFulfillmentState`. Historical events: `RequestFulfillmentCreated`, `RequestFulfillmentCompleted`, `RequestFulfillmentExpired`. Historical engine APIs: `engine.requestFulfillment`.

Import Cutover

Before:

```
import { createRequestFulfillmentKit } from "./src/request-fulfillment-kit.js";
```

After:

```
import { createRequestFulfillmentKit } from "nexusengine/domains/interaction/request/fulfillment";

engine.n.requestFulfillment.complete({ operationId: "request:complete:1", requestId })
```

Required Changes

- Configuration: Use stable request IDs, portable positions, finite radii/timeouts, and explicit rewards as data.
- Snapshot: Map request status and counters; old resource identities are rejected.
- Events: create/complete/advance mutations return exact receipts; nearest() is read-only.
- Optional adapters: None.
- Consumer status: `not-started-outside-restoration-wave`.

Corrected Defects

- Generated IDs could collide.
- Metadata and nearest results leaked references.
- Completion was not exact-once.

Proof: `src/core-domains/interaction/tests/restored-interaction-behaviors.mjs`.

17. src/request-queue-kit.js

Status: `implemented-and-proven` | SHA-256 at `S`: `4f4471547e57c848054f18a81fdd296472a74171703961c44c90a80e37aa5fc3` | Lineage: `K` -> `S` -> `X`

Historical factories: `createRequestQueueKit`. Historical helpers: None. Historical resources: `RequestQueueState`. Historical events: `RequestQueueAdd`, `RequestQueueFulfill`, `RequestQueueExpired`, `RequestQueueFulfilled`. Historical engine APIs: `engine.requestQueue`.

Import Cutover

Before:

```
import { createRequestQueueKit } from "nexusengine";
```

After:

```
import { createRequestQueueKit } from "nexusengine/domains/interaction/request/queue";

engine.n.requestQueue.add({ operationId: "request:add:1", request })
```

Required Changes

- Configuration: Normalize default reward/penalty once and keep effects descriptive.
- Snapshot: Map portable queue records, counts, and last outcome; no event queue is loaded.
- Events: add/fulfill/advance return exact receipts; install Request Economy to apply account effects.
- Optional adapters: `request-economy-adapter-kit`.
- Consumer status: `not-started-outside-restoration-wave`.

Corrected Defects

- Commands called engine.tick(0) internally.
- Default rewards were repeatedly normalized.
- Reward effects depended on install order.

Proof: `src/core-domains/interaction/tests/restored-interaction-behaviors.mjs`, `src/core-domains/simulation/tests/restored-integration-adapters.mjs`.

18. src/route-field-kit.js

Status: `implemented-and-proven` | SHA-256 at `S`: `fcbbf44196b5f8dc1ab3b705f84ee0fe20d1b80ec6b02533da6c3a47e49a571a` | Lineage: `K` -> `S` -> `X`

Historical factories: `createRouteFieldKit`. Historical helpers: `queryNearestRouteMarker`. Historical resources: `RouteFieldState`. Historical events: None. Historical engine APIs: `engine.routeField`.

Import Cutover

Before:

```
import { createRouteFieldKit } from "nexusengine";
```

After:

```
import { createRouteFieldKit } from "nexusengine/domains/world/navigation/route-field";

engine.n.routeField.nearest(point, { filter })
```

Required Changes

- Configuration: Use stable marker IDs, finite positions, and portable marker metadata.
- Snapshot: Map marker data only; legacy query diagnostics are discarded.
- Events: nearest() is a deterministic read-only query; mutations require operation IDs.
- Optional adapters: None.
- Consumer status: `not-started-outside-restoration-wave`.

Corrected Defects

- Nearest-marker queries mutated lastQuery.
- Nested marker data leaked references.
- Equal-distance ordering was unstable.

Proof: `src/core-domains/world/tests/restored-world-behaviors.mjs`.

19. src/schedule-kit.js

Status: `implemented-and-proven` | SHA-256 at `S`: `5f5b0c2af888fa1db633627ce04bcf5d742357f9a4e693fc9ac644d8d7058dbc` | Lineage: `K` -> `S` -> `X`

Historical factories: `createScheduleKit`. Historical helpers: None. Historical resources: `ScheduleState`. Historical events: `ScheduleCycle`. Historical engine APIs: `engine.schedule`.

Import Cutover

Before:

```
import { createScheduleKit } from "nexusengine";
```

After:

```
import { createScheduleKit } from "nexusengine/domains/runtime/sequence/schedule";

engine.n.schedule.advance({ operationId: "schedule:step:1", delta })
```

Required Changes

- Configuration: Use finite cycle intervals, firstAt, repeat, active, and time scale fields.
- Snapshot: Map elapsed time, scale, and nextAt; one-shot completion uses active false and nextAt null instead of Infinity.
- Events: advance() returns every occurrence crossed by the delta in deterministic order.
- Optional adapters: None.
- Consumer status: `not-started-outside-restoration-wave`.

Corrected Defects

- Non-finite time scales were accepted.
- Large deltas lost cycles or residual time.

Proof: `src/core-domains/simulation/tests/restored-economy-operations.mjs`.

20. src/spatial-scale-kit.js

Status: `implemented-and-proven` | SHA-256 at `S`: `34dd18654c89f523e211beddc00438ed5e1e7cbca52be81c1c0f19365b8c602c` | Lineage: `K` -> `S` -> `X`

Historical factories: `createSpatialScaleKit`. Historical helpers: `queryNearestScaleAnchor`. Historical resources: `SpatialScaleState`. Historical events: `SpatialScaleBandChanged`, `SpatialScaleAnchorEntered`. Historical engine APIs: `engine.spatialScale`.

Import Cutover

Before:

```
import { createSpatialScaleKit } from "./src/spatial-scale-kit.js";
```

After:

```
import { createSpatialScaleKit } from "nexusengine/domains/spatial/scale";

engine.n.spatialScale.update({ operationId: "scale:1", subject: point })
```

Required Changes

- Configuration: Use stable anchors with finite position, radius, and band data.
- Snapshot: Map subject, active band, and active anchor; nearest-query diagnostics are not required.
- Events: update() evaluates every anchor before emitting one accepted band/enter transition; nearest() is read-only.
- Optional adapters: None.
- Consumer status: `not-started-outside-restoration-wave`.

Corrected Defects

- An outside nearest anchor could suppress a valid enter transition from another anchor.
- Queries stored diagnostics in semantic state.

Proof: `src/core-domains/spatial/tests/spatial-scale-restoration.mjs`.

21. src/terrain-kit.js

Status: `implemented-and-proven` | SHA-256 at `S`: `cf81275942d6e019a31b0a060f162f67eeadc3e424d88f78d87eb4578c4106b5` | Lineage: `I` -> `K` -> `T` -> `R` -> `S` -> `X`

Historical factories: `createTerrainKit`. Historical helpers: `terrainLayers`, `createTerrainQuery`. Historical resources: None. Historical events: None. Historical engine APIs: None.

Import Cutover

Before:

```
import { createTerrainKit } from "nexusengine";
```

After:

```
import { createTerrainKit } from "nexusengine/domains/world/terrain";

engine.n.terrain.sample({ operationId: "terrain:sample:1", points })
```

Required Changes

- Configuration: Keep only neutral terrain layers, finite bounds, spline data, cell policy, and surface descriptors. Move authored presets and palettes to recipe/game data.
- Snapshot: Map portable sampled cells and query version; old Map/Set caches and renderer data are rejected.
- Events: sample/build/release operations return receipts; pure terrain query helpers remain at the subpath.
- Optional adapters: `camera-world-occlusion-adapter-kit`.
- Consumer status: `not-started-outside-restoration-wave`.

Corrected Defects

- Hidden cells were retained indefinitely.
- Spline bounds were incompletely evaluated.
- Core defaults contained cozy-beach presets, palettes, camera volumes, routes, and product fall data.

Proof: `src/core-domains/world/tests/restored-world-behaviors.mjs`.

22. src/transfer-zone-kit.js

Status: `implemented-and-proven` | SHA-256 at `S`: `b1fc29d1665490f3b68abe9380cedd405273e246a492cd546681e65f3d7de4c7` | Lineage: `K` -> `S` -> `X`

Historical factories: `createTransferZoneKit`. Historical helpers: None. Historical resources: `TransferZoneState`. Historical events: `TransferZoneCompleted`. Historical engine APIs: `engine.transferZones`.

Import Cutover

Before:

```
import { createTransferZoneKit } from "./src/transfer-zone-kit.js";
```

After:

```
import { createTransferZoneKit } from "nexusengine/domains/interaction/transfer-zone";

engine.n.transferZones.transfer({ operationId: "transfer:1", zoneId, candidate })
```

Required Changes

- Configuration: Use explicit zone shape, acceptedTypes, dwellSeconds, capacity, and portable metadata.
- Snapshot: Map occupancy, dwell, and completed transfer records into validated state.
- Events: enter/advance/transfer mutations emit only after every constraint passes.
- Optional adapters: None.
- Consumer status: `not-started-outside-restoration-wave`.

Corrected Defects

- Accepted types, dwell time, and capacity were described but not enforced before transfer.

Proof: `src/core-domains/interaction/tests/restored-interaction-behaviors.mjs`.

23. src/transport-route-kit.js

Status: `implemented-and-proven` | SHA-256 at `S`: `a49cc513571ffdb9b9bf723507f8f7392108a222fdc52749e1771475468ee51b` | Lineage: `K` -> `S` -> `X`

Historical factories: `createTransportRouteKit`. Historical helpers: None. Historical resources: `TransportRouteState`. Historical events: `TransportRouteCall`, `TransportRouteArrived`. Historical engine APIs: `engine.transportRoutes`.

Import Cutover

Before:

```
import { createTransportRouteKit } from "nexusengine";
```

After:

```
import { createTransportRouteKit } from "nexusengine/domains/simulation/operations/transport-route";

engine.n.transportRoutes.advance({ operationId: "transport:step:1", delta })
```

Required Changes

- Configuration: Use stable stops, finite carrier speed/capacity, and portable call metadata.
- Snapshot: Map carriers, waiting calls, progress, and arrival receipts.
- Events: call/advance return exact receipts; install Transport Request to fulfill queue entries.
- Optional adapters: `transport-request-adapter-kit`.
- Consumer status: `not-started-outside-restoration-wave`.

Corrected Defects

- Large deltas dropped route legs and residual progress.
- Caller metadata escaped by reference.
- Request integration and internal engine.tick(0) were coupled.

Proof: `src/core-domains/simulation/tests/restored-economy-operations.mjs`, `src/core-domains/simulation/tests/restored-integration-adapters.mjs`.

24. src/vehicle-dynamics-kit.js

Status: `implemented-and-proven` | SHA-256 at `S`: `07cd227201a85f0aa6a2ba202332ffac87e69a98e3b6893fb7f1b8cdc19db7e6` | Lineage: `K` -> `S` -> `X`

Historical factories: `createVehicleDynamicsKit`. Historical helpers: None. Historical resources: `VehicleDynamicsState`. Historical events: `VehicleDynamicsInput`, `VehicleImpact`. Historical engine APIs: `engine.vehicleDynamics`.

Import Cutover

Before:

```
import { createVehicleDynamicsKit } from "nexusengine";
```

After:

```
import { createVehicleDynamicsKit } from "nexusengine/domains/simulation/motion/vehicle";

engine.n.vehicleDynamics.step({ operationId: "vehicle:step:1", delta: 1 / 60, input })
```

Required Changes

- Configuration: Use finite dynamics, boost, bounds, and initial motion values. Do not include water policy.
- Snapshot: Map position, velocity, heading, boost, bounds, and last frame.
- Events: step() returns a receipt and emits impact only when accepted; water response requires an adapter.
- Optional adapters: `vehicle-water-response-adapter-kit`.
- Consumer status: `not-started-outside-restoration-wave`.

Corrected Defects

- boost.max zero could still create a boosted frame.
- Dynamics values were incompletely validated.
- Water response was coupled to vehicle state.

Proof: `src/core-domains/simulation/tests/restored-motion-behaviors.mjs`, `src/core-domains/simulation/tests/restored-integration-adapters.mjs`.

25. src/water-surface-kit.js

Status: `implemented-and-proven` | SHA-256 at `S`: `d278cedbd3d1903148ec5fd3bd1b64b79972364890ece485260822bd266518bf` | Lineage: `K` -> `S` -> `X`

Historical factories: `createWaterSurfaceKit`. Historical helpers: `queryWaterSurface`. Historical resources: `WaterSurfaceState`. Historical events: None. Historical engine APIs: `engine.waterSurface`.

Import Cutover

Before:

```
import { createWaterSurfaceKit } from "nexusengine";
```

After:

```
import { createWaterSurfaceKit } from "nexusengine/domains/world/water-surface";

engine.n.waterSurface.query(point)
```

Required Changes

- Configuration: Use renderer-neutral water zones, depth, drag, current, wave, hazard, and metadata descriptors.
- Snapshot: Map zones and elapsed wave state; renderer objects and query diagnostics are rejected.
- Events: query() is read-only; advance() is the only elapsed-state mutation.
- Optional adapters: `vehicle-water-response-adapter-kit`.
- Consumer status: `not-started-outside-restoration-wave`.

Corrected Defects

- Queries mutated lastQuery.
- Nested zone metadata and query results escaped by reference.
- Renderer-shaped surface data was mixed with world queries.

Proof: `src/core-domains/world/tests/restored-world-behaviors.mjs`, `src/core-domains/simulation/tests/restored-integration-adapters.mjs`.

26. src/world-physics-kit.js

Status: `implemented-and-proven` | SHA-256 at `S`: `ad6f7324b3e9ae4c2048198ada08458b5a5d25f320599cb0fe04ad89841fe84a` | Lineage: `I` -> `K` -> `S` -> `X`

Historical factories: `createPhysicsKit`, `createWorldPhysicsKit`. Historical helpers: None. Historical resources: None. Historical events: None. Historical engine APIs: None.

Import Cutover

Before:

```
import { createPhysicsKit, createWorldPhysicsKit } from "nexusengine";
```

After:

```
import { createWorldContactKit } from "nexusengine/domains/simulation/physics/world-contact";
import { createSoftRespawnKit } from "nexusengine/domains/simulation/recovery/soft-respawn";

engine.n.worldContact.resolve({ operationId: "contact:1", body, sample });
engine.n.softRespawn.recover({ operationId: "respawn:1", subjectId, reason })
```

Required Changes

- Configuration: Split contact bounds/slope/impact config from recovery points and recovery policy. The current createPhysicsKit is only the provider-neutral Physics contract and is not this legacy behavior.
- Snapshot: Split old state into World Contact and Soft Respawn snapshots. No old world-physics snapshot is accepted directly.
- Events: resolve() emits contact/correction/recovery-required data; recover() commits one exact-once coherent recovery receipt.
- Optional adapters: `locomotion-contact-response-adapter-kit`.
- Consumer status: `not-started-outside-restoration-wave`.

Corrected Defects

- One module owned contact, stability, fall policy, and respawn state.
- Respawn events and contact state could become incoherent.
- The historical createPhysicsKit name now conflicts semantically with the provider-neutral Physics contract Kit.

Proof: `src/core-domains/simulation/tests/restored-motion-behaviors.mjs`, `src/core-domains/simulation/tests/restored-integration-adapters.mjs`.

Optional Integration Adapters

Adapters are independently installed. They import no private sibling and never auto-install either side of an integration.

Adapter	Import	Requires	Provides
`locomotion-contact-response-adapter-kit`	`nexusengine/domains/simulation/adapters/locomotion-contact-response`	`motion:locomotion-intent`, `physics:world-contact`	`motion:contact-response`
`camera-world-occlusion-adapter-kit`	`nexusengine/domains/camera/adapters/camera-world-occlusion`	`camera:third-person-descriptor`, `world:terrain-sampling`, `physics:query`	`camera:world-occlusion-response`
`vehicle-water-response-adapter-kit`	`nexusengine/domains/simulation/adapters/vehicle-water-response`	`motion:vehicle-dynamics`, `world:water-query`	`motion:vehicle-water-response`
`lifecycle-economy-adapter-kit`	`nexusengine/domains/simulation/adapters/lifecycle-economy`	`progression:lifecycle`, `economy:transaction`	`progression:lifecycle-economy-adapter`
`lifecycle-facility-adapter-kit`	`nexusengine/domains/simulation/adapters/lifecycle-facility`	`progression:lifecycle`, `operations:facility`	`progression:lifecycle-facility-adapter`
`facility-economy-adapter-kit`	`nexusengine/domains/simulation/adapters/facility-economy`	`operations:facility`, `economy:transaction`	`operations:facility-economy-adapter`
`occupant-request-adapter-kit`	`nexusengine/domains/interaction/adapters/occupant-request`	`operations:occupant-flow`, `interaction:request-queue`	`interaction:occupant-request-adapter`

Adapter	Import	Requires	Provides
`transport-request-adapter-kit`	`nexusengine/domains/interaction/adapters/transport-request`	`operations:transport-route`, `interaction:request-queue`	`interaction:transport-request-adapter`
`request-economy-adapter-kit`	`nexusengine/domains/interaction/adapters/request-economy`	`interaction:request-queue`, `economy:transaction`	`interaction:request-economy-adapter`

Generated Recipes

- `procedural-navigation`
- `terrain-character-traversal`
- `management-operations`
- `vehicle-rescue-logistics`
- `spatial-guidance`
- `hazard-pursuit`

Physics Name Warning

The historical `createPhysicsKit` from `src/world-physics-kit.js` and the current provider-neutral `createPhysicsKit` are not equivalent. Replace historical world behavior with `createWorldContactKit` plus `createSoftRespawnKit`; use the current `createPhysicsKit` only for Physics contracts and providers.

GENERATED REFERENCE

NexusEngine 0.0.4 Root Module Dispositions

Source commit: `a9adca5b3620f996f00860358c4864dd4bdfa6d9`

This is a hard cutover. Removed modules are not forwarded. Sources marked core-restored now use manifest-owned semantic subpaths; all other rows name their external owner or replacement.

Removed Exports From Retained Modules

Source	Removed exports	Disposition	Replacement
`src/core-domains/compute/model/kits/model-kit/index.js`	`createMockModelAdapter`	recipe-data	Inject createModelAdapterBoundary({ infer }) into createModelKit({ adapters }).
`src/core-domains/presentation/sky/kits/sky-kit/index.js`	`CORE_SKYBOX_PRESETS`	recipe-data	Pass caller-owned preset data to createSkyDescriptorKit({ presets }) or registerPreset().
`src/core-domains/presentation/speech/kits/speech-kit/index.js`	`createTinyTTSAssetManifest`, `createTinyTTSProvider`, `registerTinyTTSAssets`	external-kit	Install a model, network, and asset implementation through createSpeechKit().registerProvider().
`src/core-domains/presentation/speech/kits/speech-kit/pocket-tts-provider.js`	`createPocketTTSAssetManifest`, `createPocketTTSHttpAssetProvider`, `createPocketTTSProvider`, `registerPocketTTSAssets`	external-kit	Install PocketTTS through createSpeechKit().registerProvider().

Retired Source Modules

Source	Disposition	Owner	Core requirements
`bin/nexus-editor.mjs`	external-kit	NexusEngine-Editor	`n:composition`, `n:host`, `n:mcp`
`src/action-movement-kit.js`	core-restored	NexusEngine manifests: n:simulation:motion:locomotion	`action-locomotion-kit`
`src/assistance-target-kit.js`	core-restored	NexusEngine manifests: n:interaction:assistance-target	`assistance-target-kit`
`src/cargo-manifest-kit.js`	core-restored	NexusEngine manifests: n:simulation:economy:cargo	`cargo-manifest-kit`
`src/character-camera-kit.js`	core-restored	NexusEngine manifests: n:camera:third-person	`third-person-camera-kit`
`src/common-game-definitions.js`	recipe-data	NexusEngine-Kits recipes	`n:actor`, `n:interaction`, `n:simulation`
`src/core-domains/compute/model/kits/model-kit/mock-model-adapter.js`	recipe-data	NexusEngine tests	`n:compute:model`
`src/core-domains/presentation/speech/kits/speech-kit/pocket-tts-provider.js`	external-kit	NexusEngine-Kits adapters	`n:asset`, `n:compute:model`, `n:presentation:speech`

Source	Disposition	Owner	Core requirements
`src/core-domains/world/scene/kits/scene-kit/hosts/headless-scene-host.js`	external-kit	NexusEngine-Editor adapters	`n:host`, `n:world:scene`
`src/core-domains/world/scene/kits/scene-kit/hosts/native-scene-host.js`	external-kit	NexusEngine-Editor adapters	`n:host`, `n:world:scene`
`src/core-domains/world/scene/kits/scene-kit/hosts/rust-native-scene-host.js`	external-kit	NexusEngine-Editor adapters	`n:host`, `n:world:scene`
`src/core-domains/world/scene/kits/scene-kit/hosts/web-html-scene-host.js`	external-kit	NexusEngine-Editor adapters	`n:host`, `n:world:scene`
`src/core-domains/world/scene/kits/scene-kit/hosts/web-scene-host.js`	external-kit	NexusEngine-Editor adapters	`n:host`, `n:world:scene`
`src/economy-kit.js`	core-restored	NexusEngine manifests: n:simulation:economy:accounts	`economy-account-kit`
`src/environmental-affordance-kit.js`	core-restored	NexusEngine manifests: n:interaction:environmental-affordance	`environmental-affordance-kit`
`src/facility-operations-kit.js`	core-restored	NexusEngine manifests: n:simulation:operations:facility	`facility-operations-kit`
`src/game-kit-composer.js`	duplicate	n:composition	`n:composition`, `n:policy`
`src/hazard-field-kit.js`	core-restored	NexusEngine manifests: n:simulation:hazard-field	`hazard-field-kit`
`src/host.js`	external-kit	NexusEngine-Editor	`n:composition`, `n:host`, `n:policy`
`src/hosts/browser/browser-indexeddb-asset-cache-adapter.js`	external-kit	NexusEngine-Editor adapters	`n:asset`, `n:host`, `n:presentation`
`src/hosts/browser/browser-presentation-surface-adapter.js`	external-kit	NexusEngine-Editor adapters	`n:asset`, `n:host`, `n:presentation`
`src/hosts/browser/browser-startup-presentation-adapter.js`	external-kit	NexusEngine-Editor adapters	`n:asset`, `n:host`, `n:presentation`
`src/landmark-guidance-kit.js`	core-restored	NexusEngine manifests: n:world:navigation:landmark-guidance	`landmark-guidance-kit`
`src/lifecycle-progression-kit.js`	core-restored	NexusEngine manifests: n:simulation:progression:lifecycle	`lifecycle-progression-kit`
`src/modules/nexus-diffusion/backend/index.js`	external-kit	NexusEngine-Kits	`n:compute`, `n:compute:model`, `n:runtime:data`
`src/modules/nexus-diffusion/checkpoint/index.js`	external-kit	NexusEngine-Kits	`n:compute`, `n:compute:model`, `n:runtime:data`

Source	Disposition	Owner	Core requirements
`src/modules/nexus-diffusion/dataset/index.js`	external-kit	NexusEngine-Kits	`n:compute`, `n:compute:model`, `n:runtime:data`
`src/modules/nexus-diffusion/index.js`	external-kit	NexusEngine-Kits	`n:compute`, `n:compute:model`, `n:runtime:data`
`src/modules/nexus-diffusion/model/index.js`	external-kit	NexusEngine-Kits	`n:compute`, `n:compute:model`, `n:runtime:data`
`src/modules/nexus-diffusion/nexus-diffusion-domain.js`	external-kit	NexusEngine-Kits	`n:compute`, `n:compute:model`, `n:runtime:data`
`src/modules/nexus-diffusion/noise/index.js`	external-kit	NexusEngine-Kits	`n:compute`, `n:compute:model`, `n:runtime:data`
`src/modules/nexus-diffusion/preview/index.js`	external-kit	NexusEngine-Kits	`n:compute`, `n:compute:model`, `n:runtime:data`
`src/modules/nexus-diffusion/sampling/index.js`	external-kit	NexusEngine-Kits	`n:compute`, `n:compute:model`, `n:runtime:data`
`src/modules/nexus-diffusion/tensor/index.js`	external-kit	NexusEngine-Kits	`n:compute`, `n:compute:model`, `n:runtime:data`
`src/modules/nexus-diffusion/training/index.js`	external-kit	NexusEngine-Kits	`n:compute`, `n:compute:model`, `n:runtime:data`
`src/modules/nexus-diffusion/utils/seeded-random.js`	external-kit	NexusEngine-Kits	`n:compute`, `n:compute:model`, `n:runtime:data`
`src/navmesh-kit.js`	core-restored	NexusEngine manifests: n:world:navigation:navmesh	`navmesh-kit`
`src/occupant-flow-kit.js`	core-restored	NexusEngine manifests: n:simulation:operations:occupant-flow	`occupant-flow-kit`
`src/pathfinding-kit.js`	core-restored	NexusEngine manifests: n:world:navigation:pathfinding	`pathfinding-kit`
`src/procedural-kit.js`	core-restored	NexusEngine manifests: n:world:generation	`procedural-generation-kit`
`src/procedural-navigation-extensions.js`	duplicate	NexusEngine-Kits	`n:composition`
`src/pursuit-pressure-kit.js`	core-restored	NexusEngine manifests: n:simulation:pursuit-pressure	`pursuit-pressure-kit`
`src/renderers.js`	external-kit	NexusEngine-Kits adapters	`n:host`, `n:presentation:output`
`src/renderers/three/three-object-capture-provider.js`	external-kit	NexusEngine-Kits adapters	`n:host`, `n:object`, `n:presentation`
`src/renderers/three/three-presentation-output-adapter.js`	external-kit	NexusEngine-Kits adapters	`n:host`, `n:object`, `n:presentation`
`src/renderers/three/three-subject-bounds-adapter.js`	external-kit	NexusEngine-Kits adapters	`n:host`, `n:object`, `n:presentation`
`src/request-fulfillment-kit.js`	core-restored	NexusEngine manifests: n:interaction:request:fulfillment	`request-fulfillment-kit`
`src/request-queue-kit.js`	core-restored	NexusEngine manifests: n:interaction:request:queue	`request-queue-kit`

Source	Disposition	Owner	Core requirements
`src/route-field-kit.js`	core-restored	NexusEngine manifests: n:world:navigation:route-field	`route-field-kit`
`src/scenario-driver-kit.js`	game-owned	Experiments or game repository	`n:composition`, `n:runtime:sequence`, `n:simulation`
`src/scenario-duration-kit.js`	game-owned	Experiments or game repository	`n:runtime:sequence`, `n:simulation`
`src/schedule-kit.js`	core-restored	NexusEngine manifests: n:runtime:sequence:schedule	`schedule-kit`
`src/shaders.js`	external-kit	NexusEngine-Kits adapters	`n:host`, `n:render:shader`, `n:material`
`src/spatial-scale-kit.js`	core-restored	NexusEngine manifests: n:spatial:scale	`spatial-scale-kit`
`src/terrain-kit.js`	core-restored	NexusEngine manifests: n:world:terrain	`terrain-kit`
`src/transfer-zone-kit.js`	core-restored	NexusEngine manifests: n:interaction:transfer-zone	`transfer-zone-kit`
`src/transport-route-kit.js`	core-restored	NexusEngine manifests: n:simulation:operations:transport-route	`transport-route-kit`
`src/vehicle-dynamics-kit.js`	core-restored	NexusEngine manifests: n:simulation:motion:vehicle	`vehicle-dynamics-kit`
`src/water-surface-kit.js`	core-restored	NexusEngine manifests: n:world:water-surface	`water-surface-kit`
`src/world-physics-kit.js`	core-restored	NexusEngine manifests: n:simulation:physics:world-contact, n:simulation:recovery:soft-respawn	`world-contact-kit`, `soft-respawn-kit`
`tools/headless-editor/`	external-kit	NexusEngine-Editor	`n:composition`, `n:host`, `n:mcp`, `n:policy`

GENERATED REFERENCE

Frozen ProtoKit Extraction

Canonical source: `https://github.com/LuminaryLabs-Agents/NexusEngine-ProtoKits.git` at `0d102649267737230d618b30fe6f9465b198d234`

Coverage: **3698/3698 source items (100%)** across **530 ProtoKit folders**. Source code was parsed as text and never executed.

Dispositions

Disposition	Source items
core-reuse	119
core-composition	115
core-new-atom	0
external-kit	3096
recipe-data	113
game-owned	236
duplicate	0
rejected-unproven	19

Core Gap Set

No historical source item passed every Core-new-atom gate. The frozen Core gap set is empty; unproven generic-looking behavior remains external or rejected rather than being promoted by name.

Deterministic Batches

Batch	First folder	Last folder	Folders
batch-01	`2d-platformer-domain`	`blackwake-kit-registry`	50
batch-02	`blackwake-preset-kit`	`domain-taxonomy-domain-kit`	50
batch-03	`downhill-race-kit`	`generic-cargo-kit`	50
batch-04	`generic-cargo-transfer-kit`	`generic-replay-test-kit`	50
batch-05	`generic-resource-loop-kit`	`ground-contact-kit`	50
batch-06	`guard-domain-kit`	`navigation-knowledge-domain-kit`	50
batch-07	`negotiation-commitment-domain-kit`	`openxr-hand-adapter-dsk`	50
batch-08	`orchard-biome-kit`	`render-layer-kit`	50
batch-09	`render-quality-budget-kit`	`stream-backpressure-kit`	50
batch-10	`stream-channel-kit`	`water-stream-kit`	50
batch-11	`water-surface-kit`	`zone-field-kit`	30

Artifacts

- `source.json`: immutable source identity and complete snapshot hash.

- `inventory.json`: files, exports, state/lifecycle signals, dependencies, consumers, providers, and adapters by folder.
- `dispositions.json`: canonical export-level ownership and reconstruction ledger.
- `core-gap-set.json`: frozen result of the new-Core-atom gate.